

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

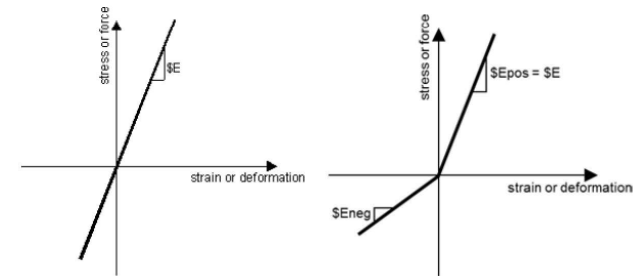
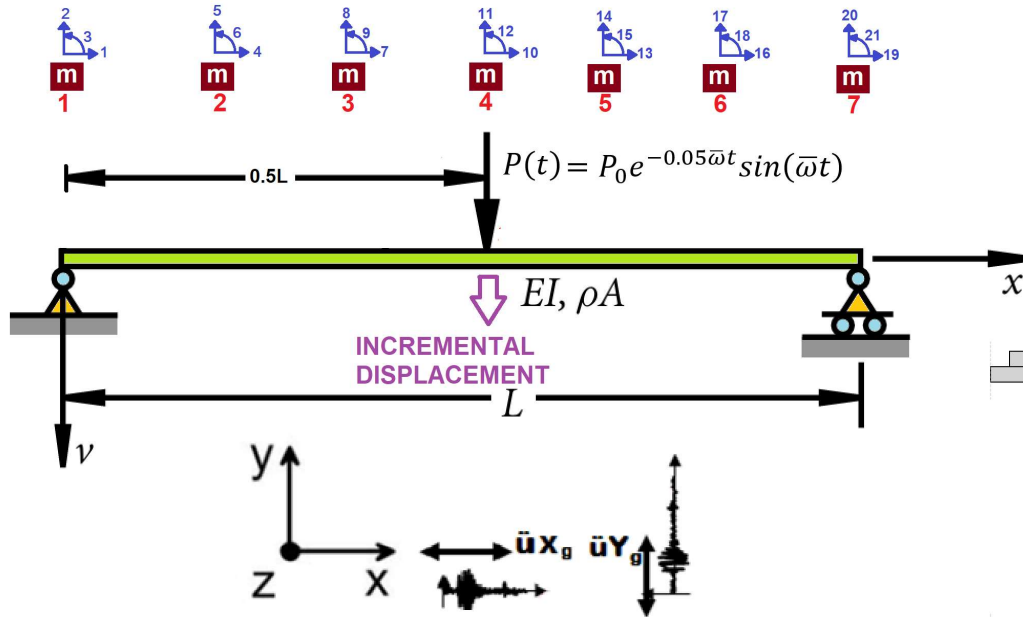
**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF STEEL A
SIMPLY SUPPORTED BEAM
ELEMENT: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

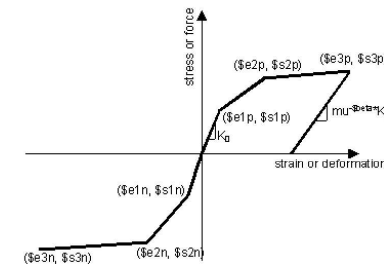
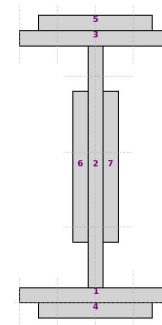
$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$



ELEMENT ELASTIC MATERIAL

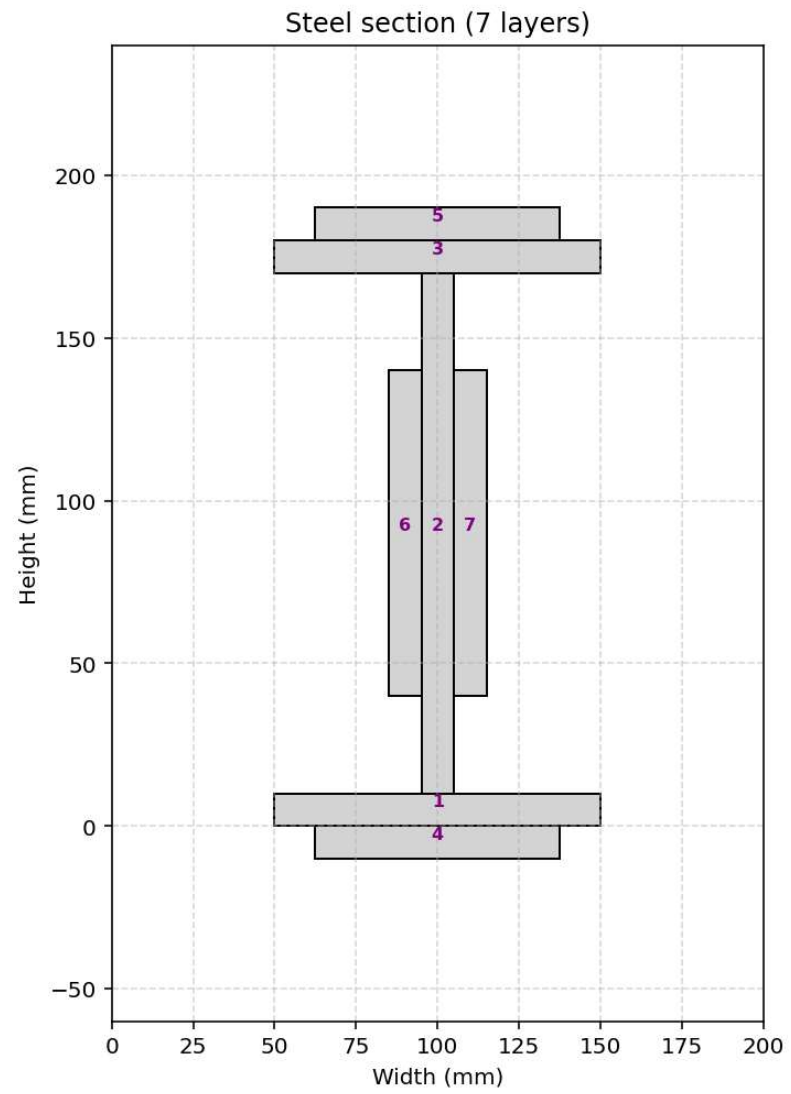


ELEMENT INELASTIC MATERIAL

$$\rho A \frac{\partial^2 v(x, t)}{\partial t^2} + EI \frac{\partial^4 v(x, t)}{\partial x^4} = P(t)$$

boundary conditions

$$v(0, t) = v(L, t) = 0, \quad \frac{\partial^2 v(x, t)}{\partial x^2} \Big|_{x=0} = \frac{\partial^2 v(x, t)}{\partial x^2} \Big|_{x=L} = 0.$$



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p(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

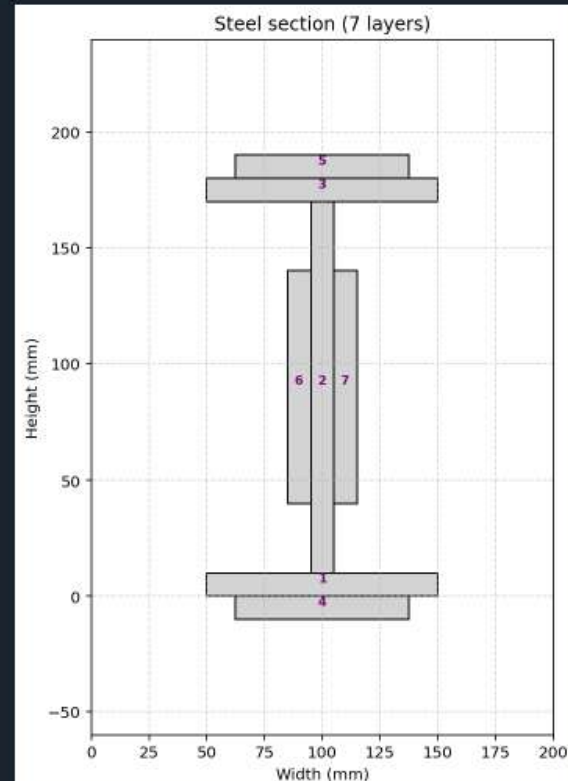
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1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A SIMPLY SUPPORTED BEAM ELEMENT : AN O
4 #FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HIST
5 #
6 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
7 #
8 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacemen}$ 
9 #
10 #  $P(t) = P_0 \sin(wt)$ 
11 #  $P(t) = P_0 \exp(-0.05wt) \sin(wt)$ 
12 #
13 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
14 # OF ENERGY DISSIPATION CAPACITY INDEX
15 #
16 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
17 # EMAIL: salar.d.ghashghaei@gmail.com
18 #####
19 """
20 Nonlinear Seismic Performance Assessment of a Beam Element:
21 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear
22
23 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Beam Elem
24 for performance-based earthquake engineering. The 2D model incorporates both material nonlin
25 (elastic-perfectly plastic or hysteretic steel with strain hardening)
26 and geometric nonlinearity to capture P-delta effects
27 and large displacements—essential for collapse assessment.
28
29 Eight analysis protocols are implemented:
30 (1) [PERIOD] : Structural Period
31 (2) [STATIC] : Gravity Load analysis establishing dead Load state
32 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
33 and plastic mechanism identification

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50 %



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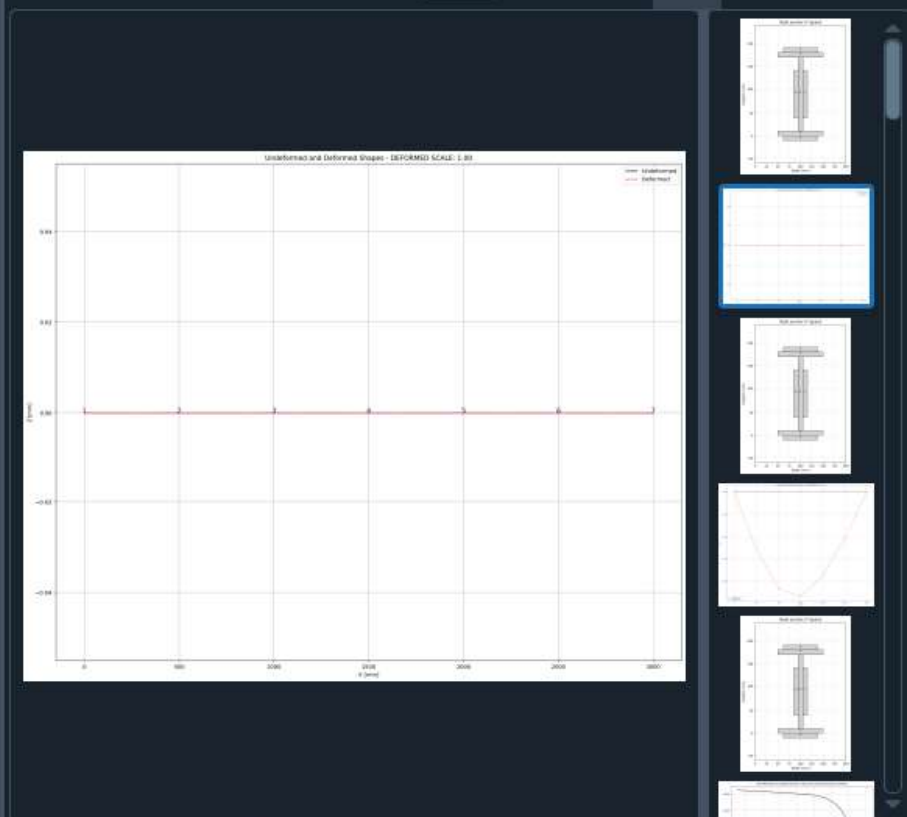
STATIC ANALYSIS

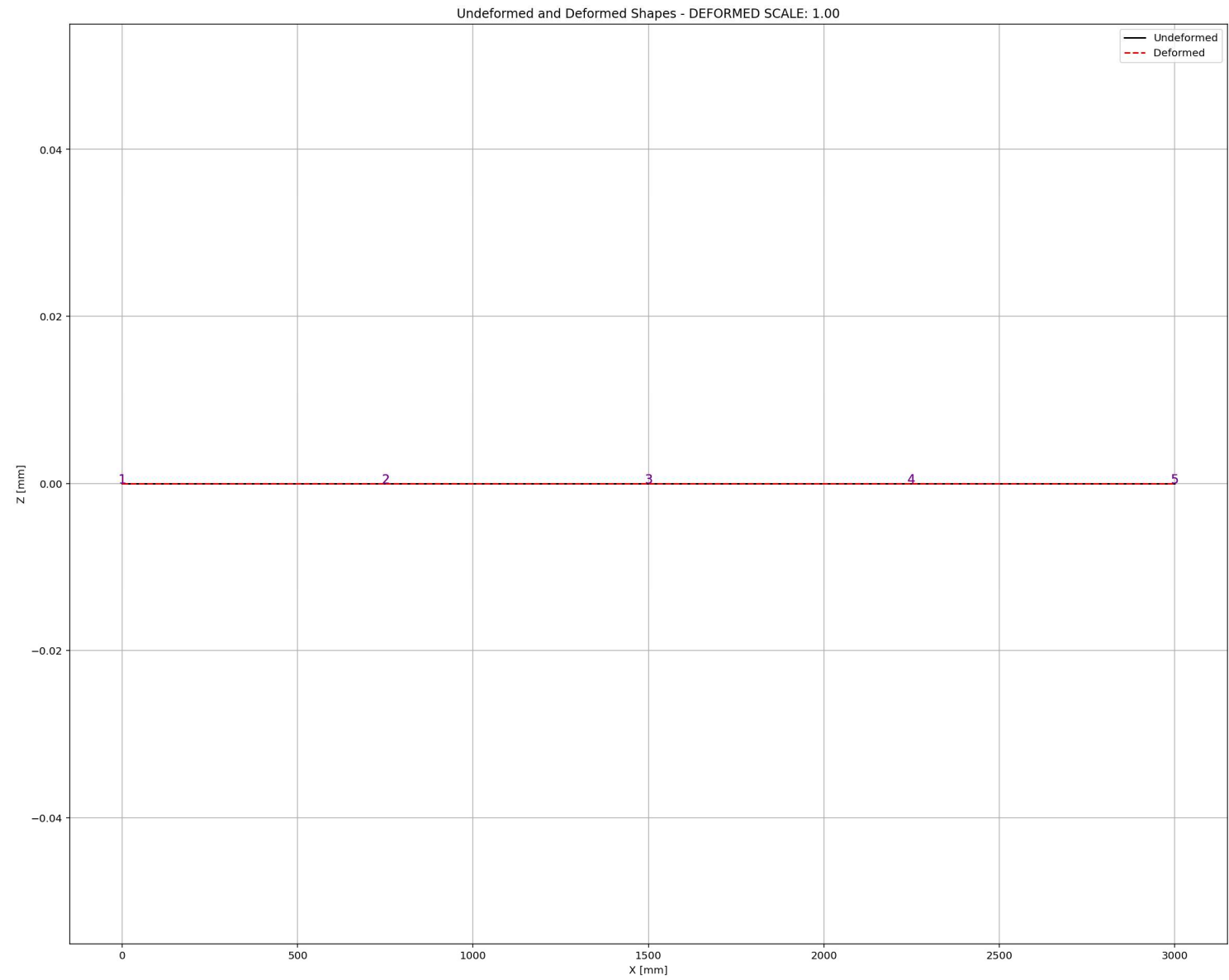
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P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

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1069 #%%-----
1070 # STATIC ANALYSIS
1071 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1072 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1073 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1074 ANAL_TYPE = 'STATIC'
1075
1076 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1077 (reaction, disp_mid,
1078  ele_axialforce, ele_shearforce, ele_momentforce,
1079  node_displacementsX, node_displacementsY,
1080  dispX, dispY) = DATA
1081
1082 S01.PLOT_2D FRAME(deformed_scale=1.0)
1083 #%%-----
1084 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1085 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1086 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1087 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1088 ANAL_TYPE = 'PUSHOVER'
1089
1090 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1091 (reaction_PUSH, disp_mid_PUSH,
1092  ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1093  node_displacementsX_PUSH, node_displacementsY_PUSH,
1094  dispX_PUSH, dispY_PUSH,
1095  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1096  STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) = DATA
1097
1098
1099 XDATA = disp_mid_PUSH
1100 YDATA = reaction_PUSH
1101 XLABEL = 'Displacement in Middle Span [mm]'
1102 YLABEL = 'Base Reaction [N]'
    
```





PUSHOVER ANALYSIS

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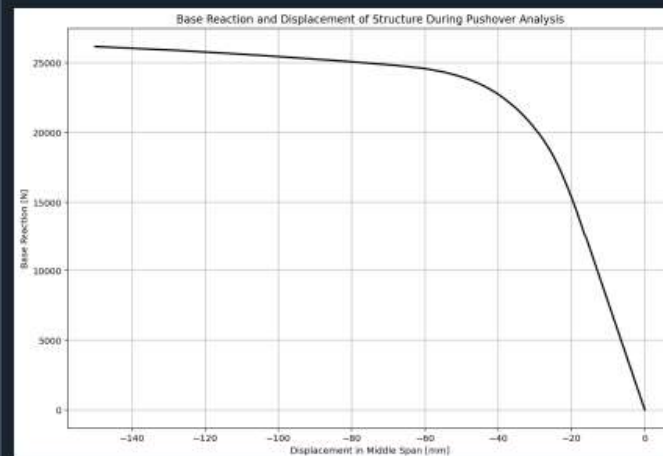
P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

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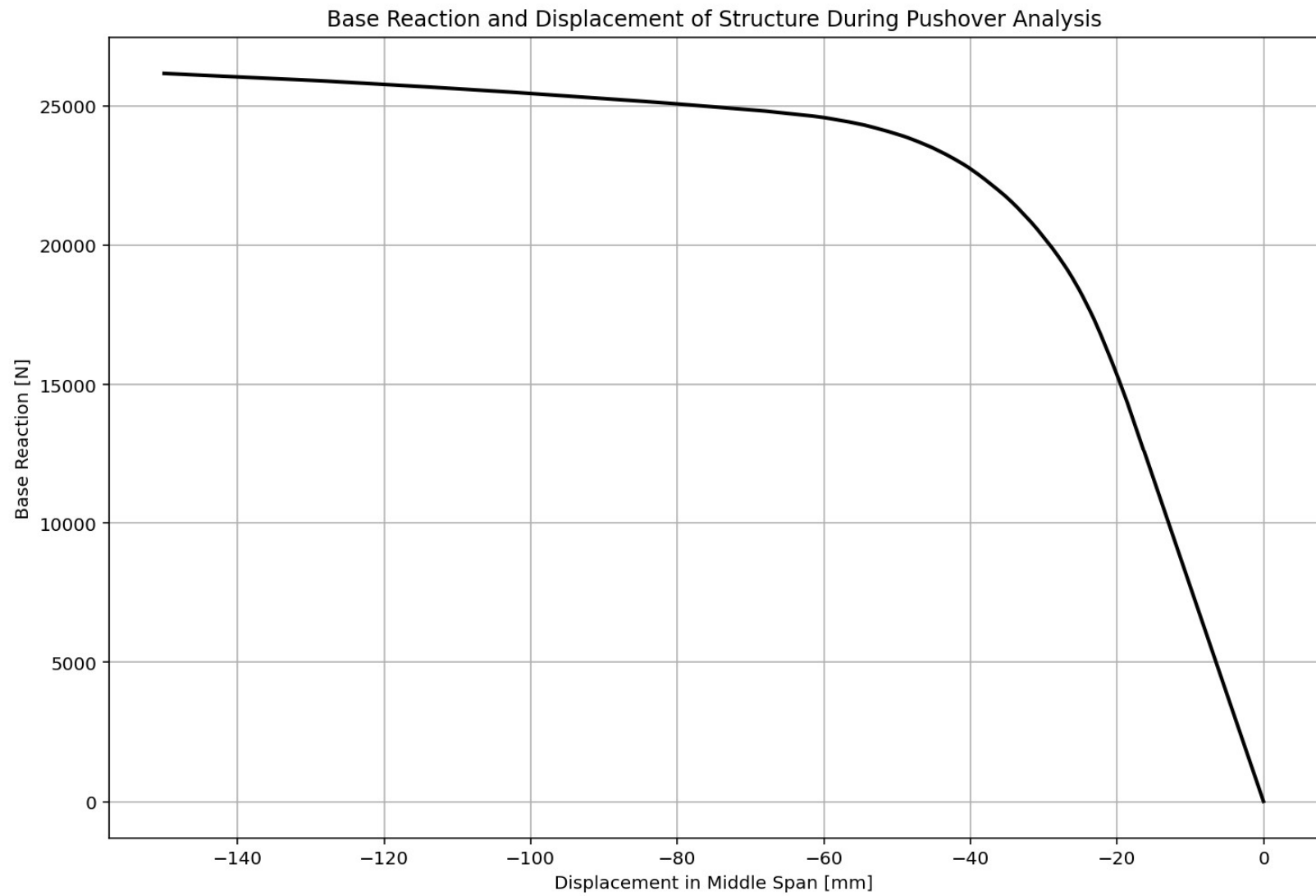
1084 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1085 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1086 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1087 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1088 ANAL_TYPE = 'PUSHOVER'
1089
1090 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1091 (reaction_PUSH, disp_mid_PUSH,
1092  ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1093  node_displacementsX_PUSH, node_displacementsY_PUSH,
1094  dispX_PUSH, dispY_PUSH,
1095  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1096  STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH) = DATA
1097
1098
1099 XDATA = disp_mid_PUSH
1100 YDATA = reaction_PUSH
1101 XLABEL = 'Displacement in Middle Span [mm]'
1102 YLABEL = 'Base Reaction [N]'
1103 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1104 COLOR = 'black'
1105 SEMILOGY = False
1106 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1107
1108 DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1109 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) =
1110
1111 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1112 plt.figure(0, figsize=(12, 8))
1113 plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1114 plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1115 plt.title('Period of Structure During Pushover Analysis')
1116 plt.ylabel('Structural Period [s]')
1117 plt.xlabel('Displacement [mm]')

```

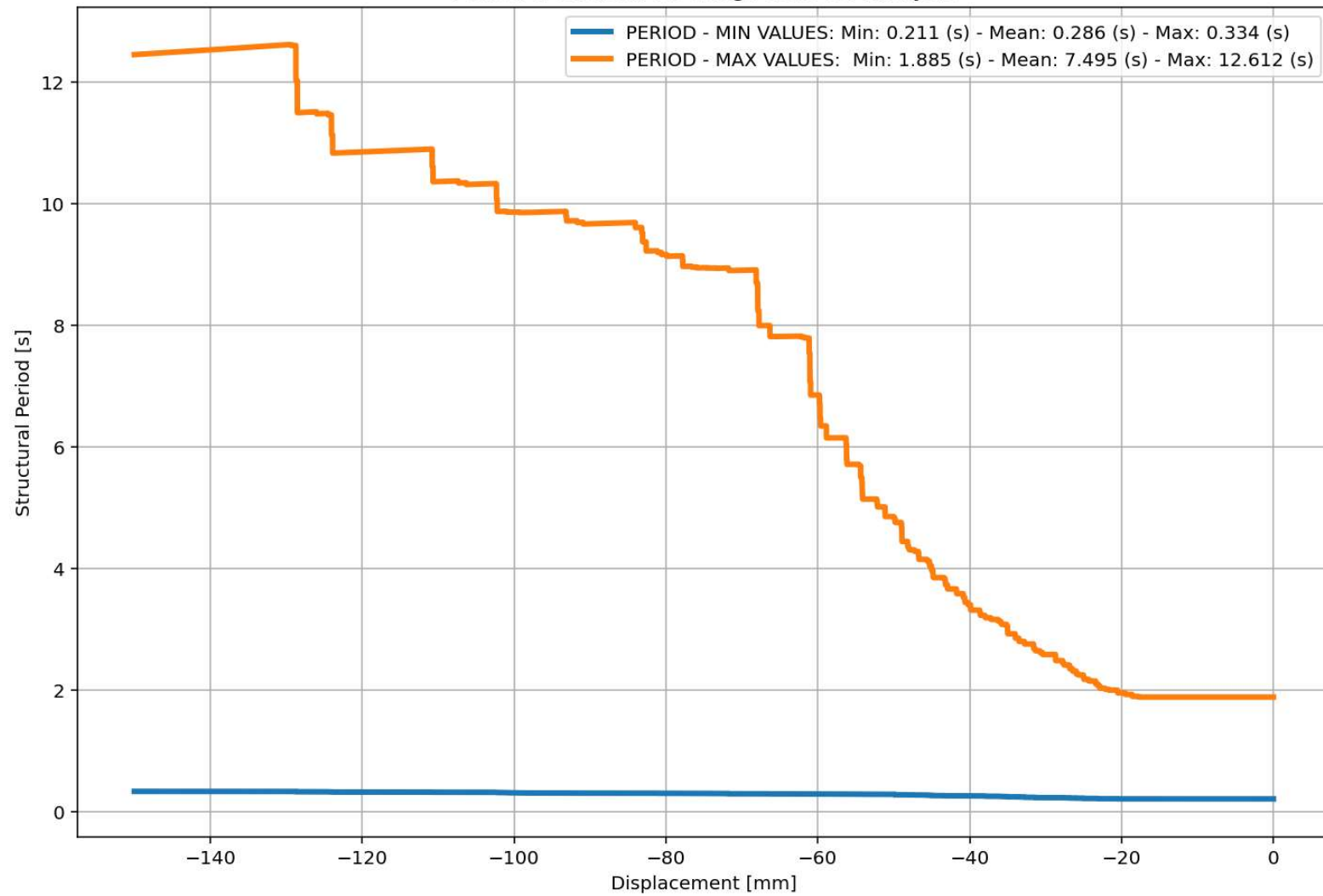
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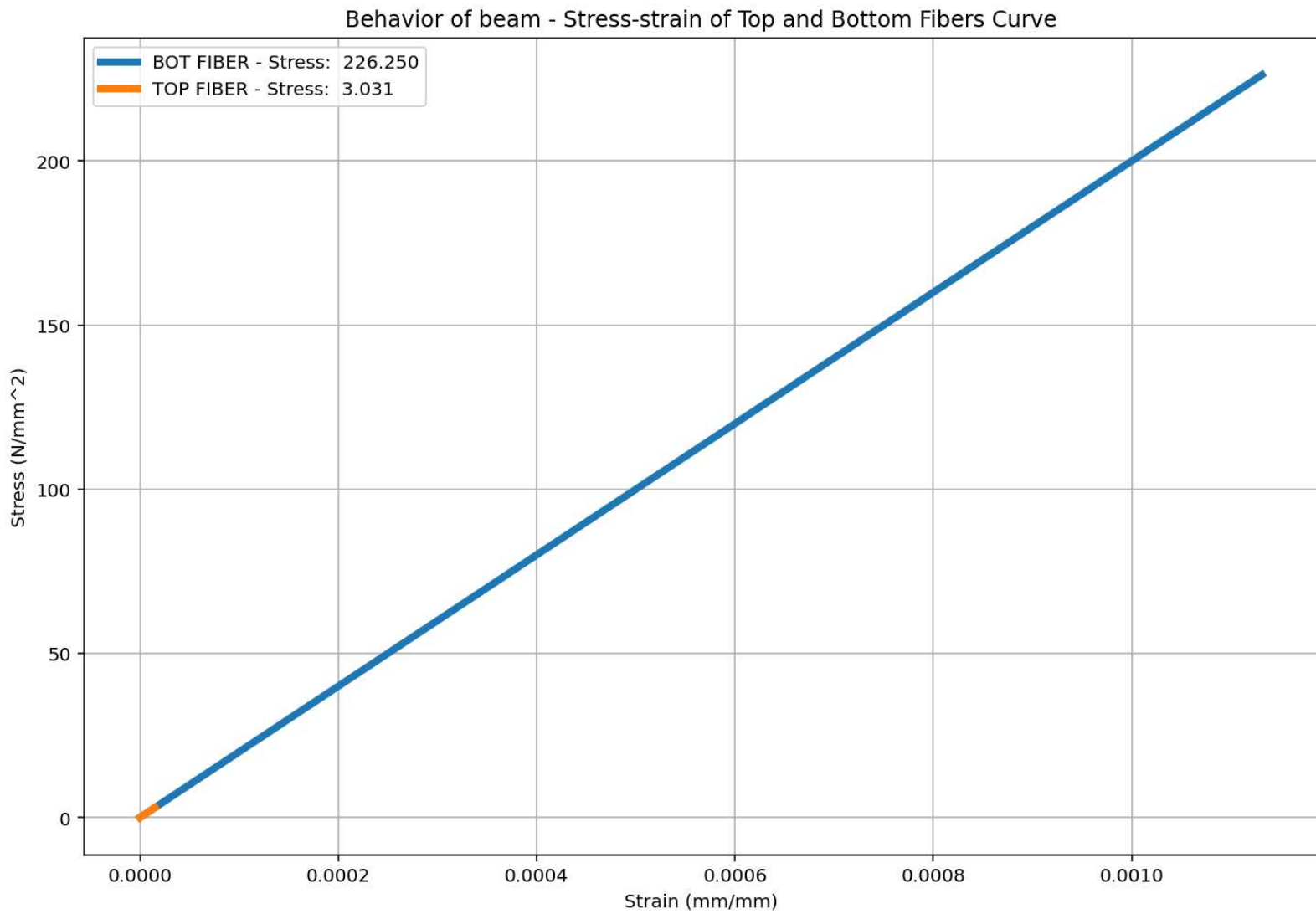


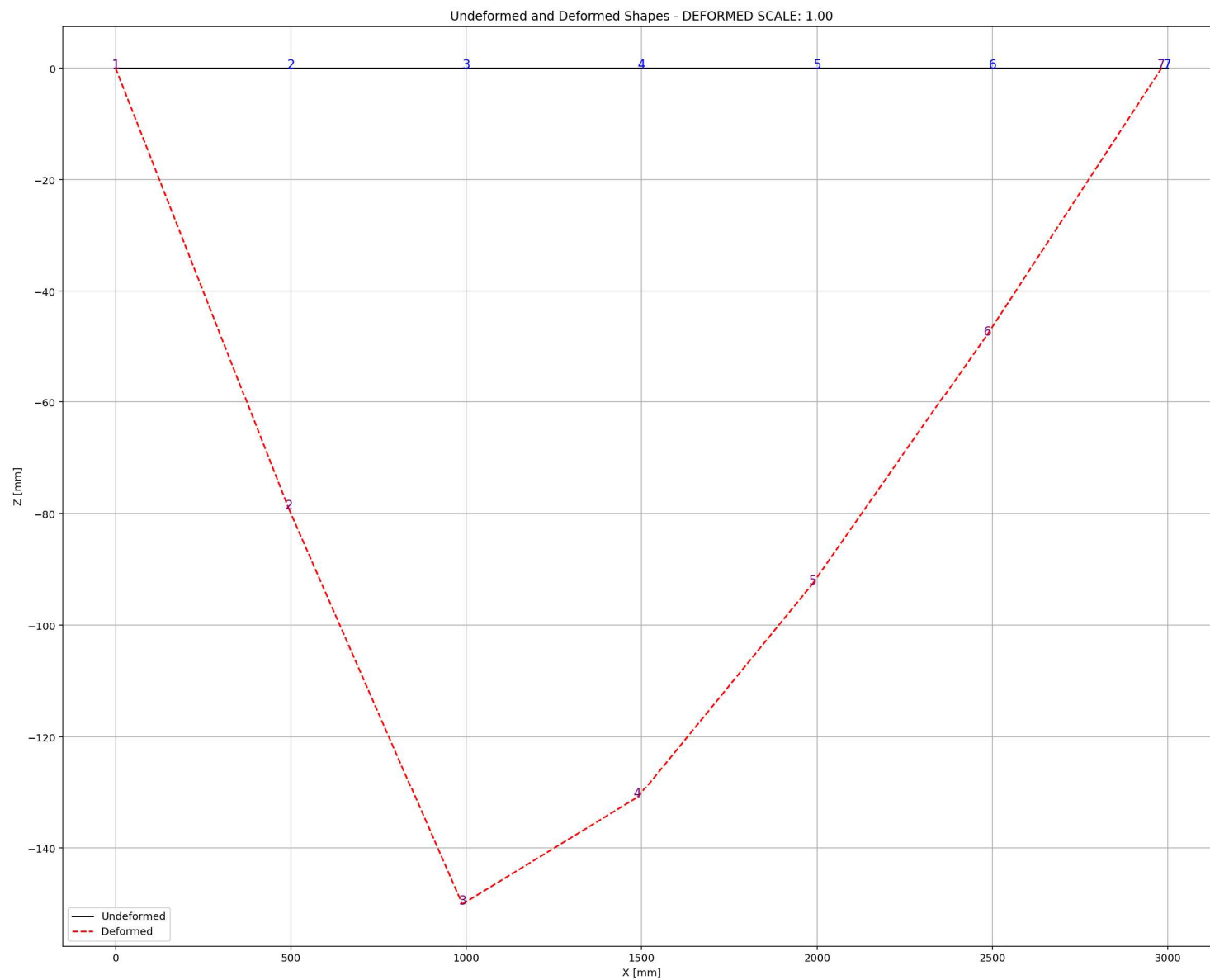
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Period of Structure During Pushover Analysis







CYCLIC DISPLACEMENT ANALYSIS

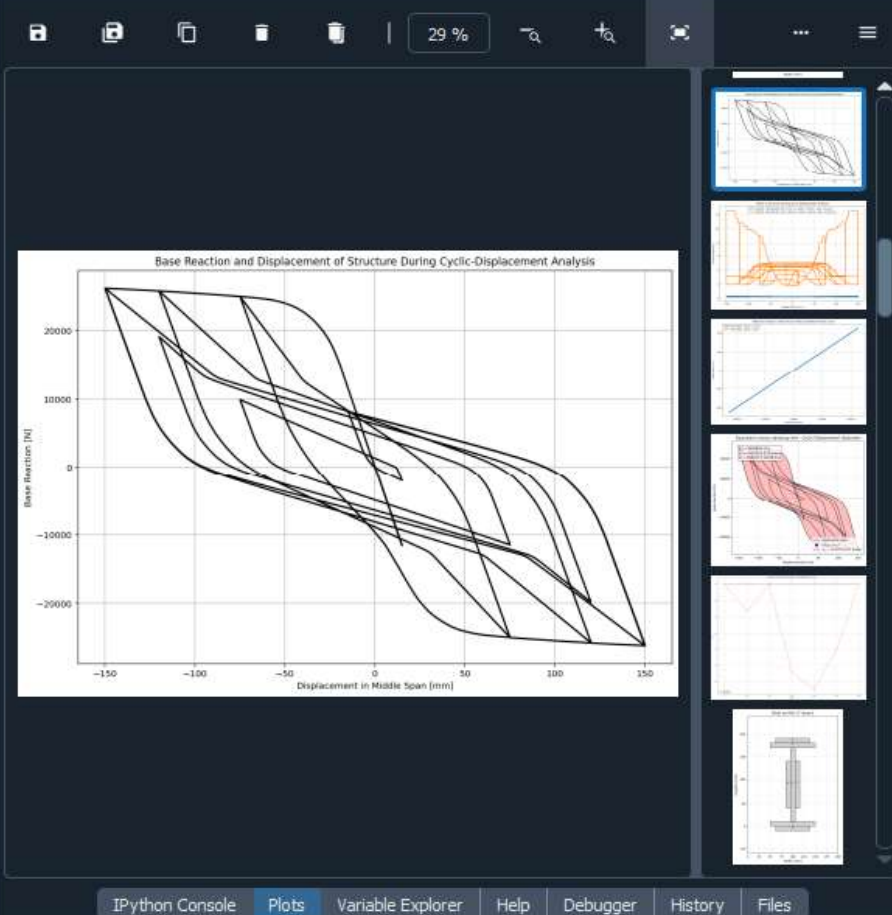
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P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

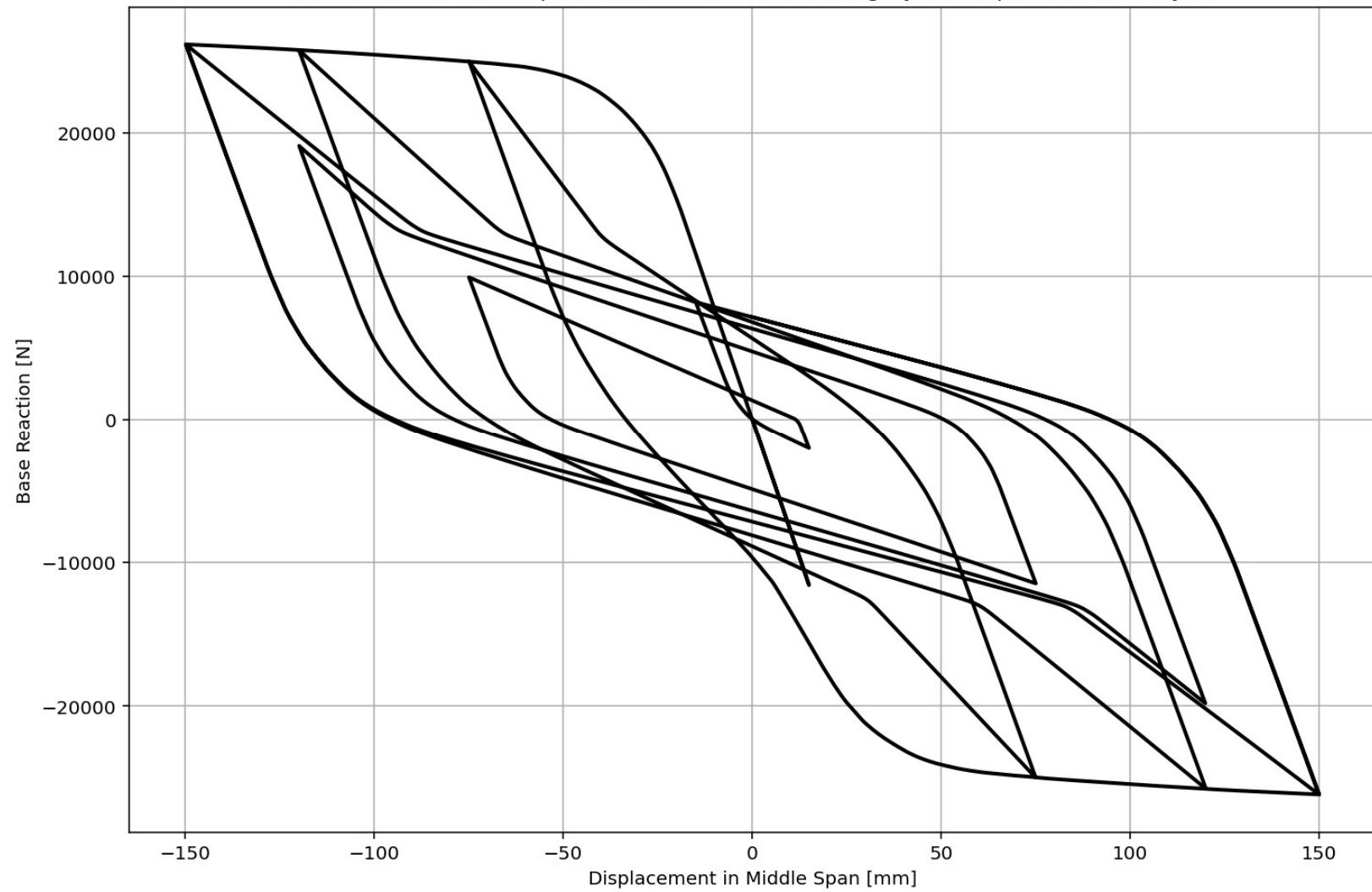
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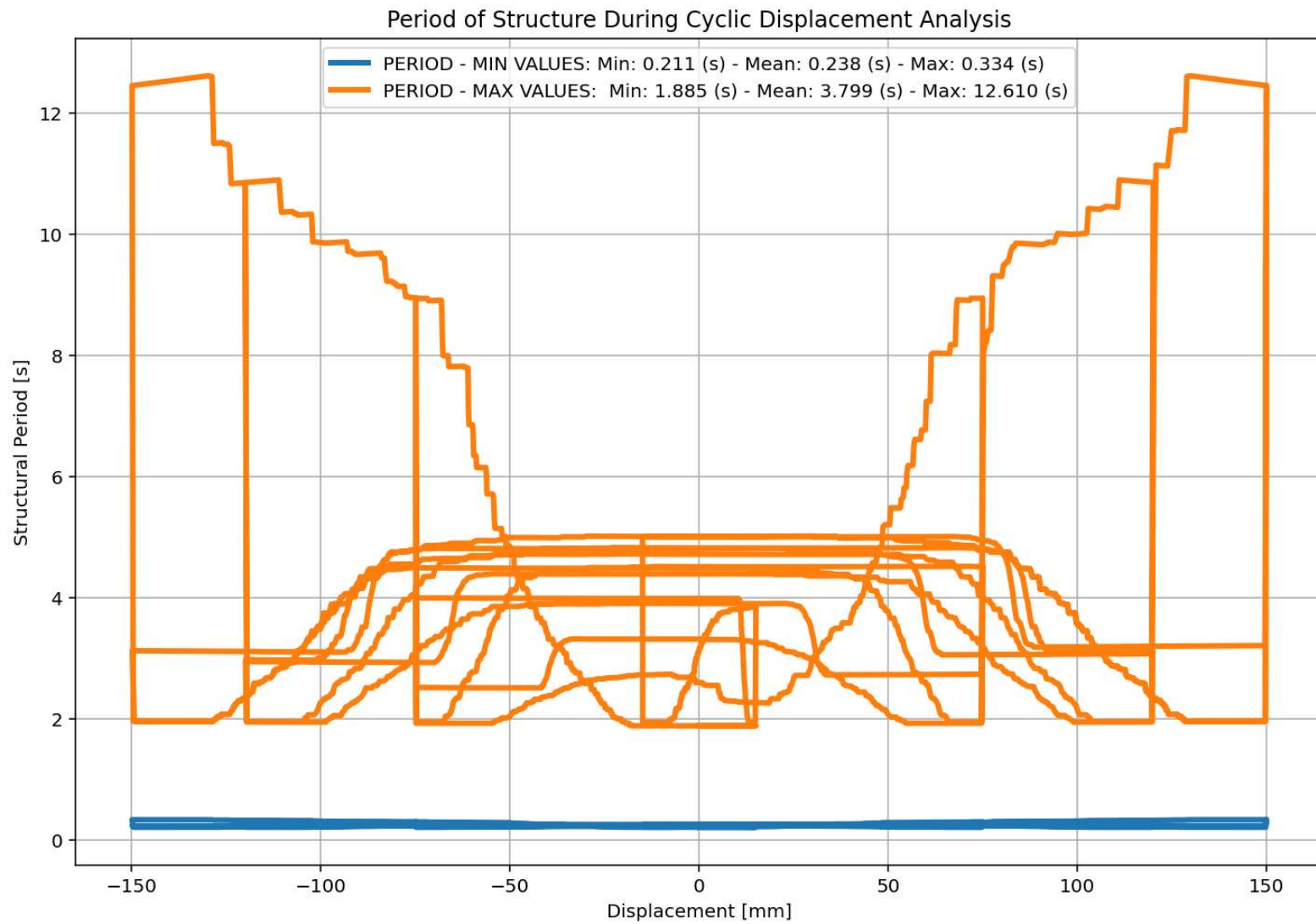
1138 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1139 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1140 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1141 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1142 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1143
1144 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1145 (reaction_CP, disp_mid_CP,
1146  ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1147  node_displacementsX_CP, node_displacementsY_CP,
1148  dispX_CP, dispY_CP,
1149  PERIOD_MIN_CP, PERIOD_MAX_CP,
1150  STRESSb_CP, STRAINb_CP, STRESSs_CP, STRAINt_CP, CURVATURE_CP) = DATA
1151
1152
1153 XDATA = disp_mid_CP
1154 YDATA = reaction_CP
1155 XLABEL = 'Displacement in Middle Span [mm]'
1156 YLABEL = 'Base Reaction [N]'
1157 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1158 COLOR = 'black'
1159 SEMILOGY = False
1160 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1161
1162 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1163 plt.figure(0, figsize=(12, 8))
1164 plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1165 plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1166 plt.title('Period of Structure During Cyclic Displacement Analysis')
1167 plt.ylabel('Structural Period [s]')
1168 plt.xlabel('Displacement [mm]')
1169 #plt.semilogy()
1170 plt.grid()
1171 plt.legend(['E' 'BERTON - MTN VALUES: Min: /mm min(BERTON MTN CP) - 3F1 / (c) - Mean: /mm mean(BERT

```

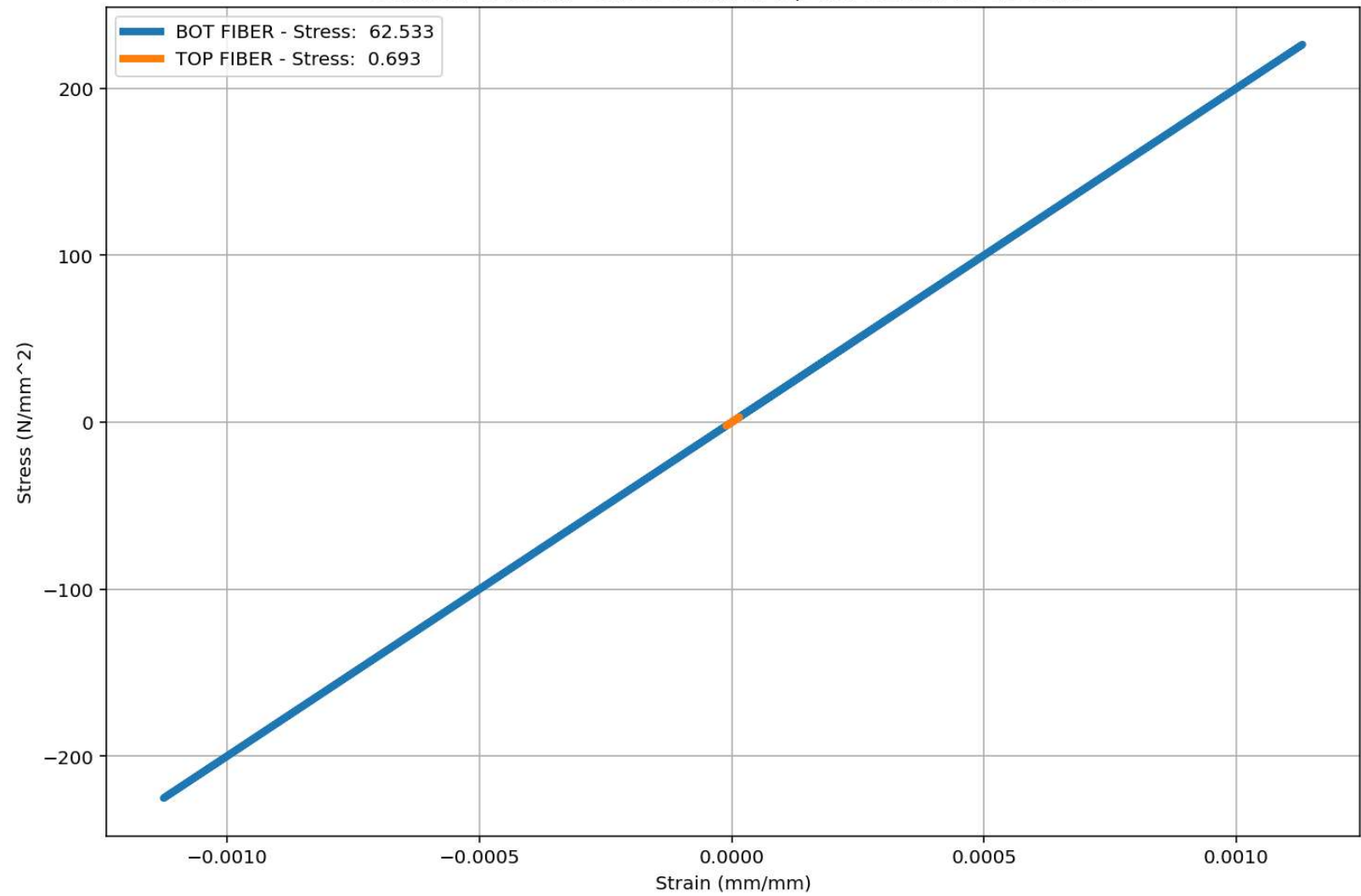


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

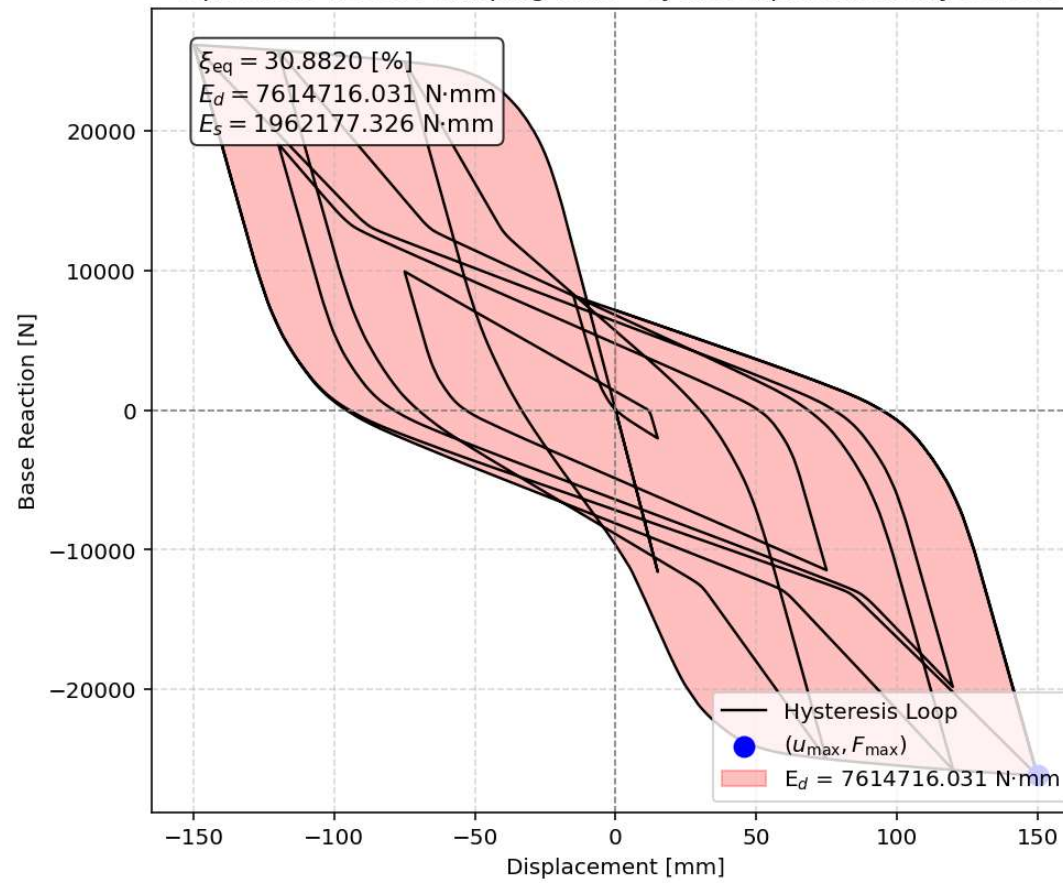


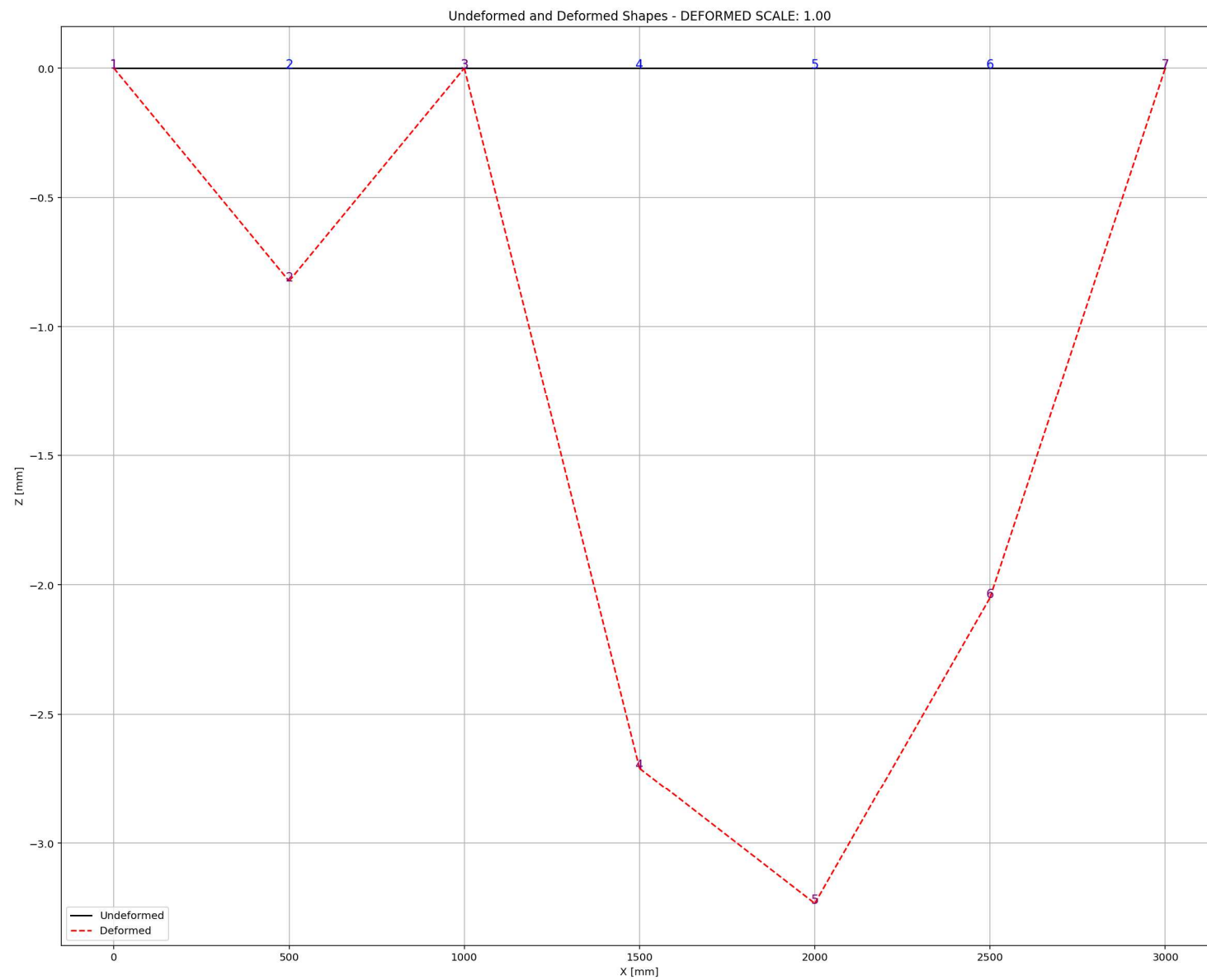


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis





STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

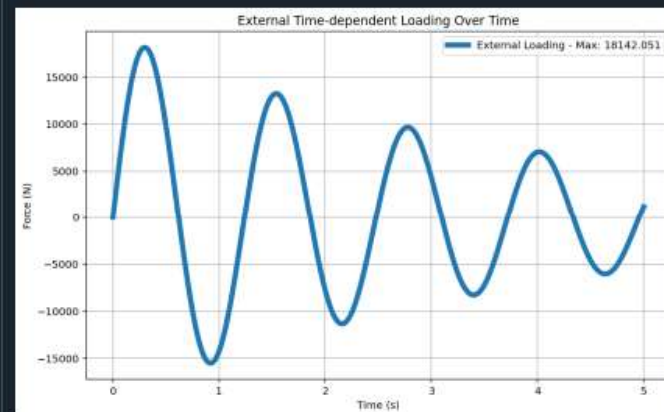
```

1196 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1197 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1198 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1199 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1200 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1201
1202 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1203
1204 (reaction_ETDLS, disp_mid_ETDLS,
1205  ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1206  node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1207  dispX_ETDLS, dispY_ETDLS,
1208  PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1209  STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINT_ETDLS, CURVATURE_ETDLS) = DATA
1210
1211
1212 XDATA = disp_mid_ETDLS
1213 YDATA = reaction_ETDLS
1214 XLABEL = 'Displacement in Middle Span [mm]'
1215 YLABEL = 'Base Reaction [N]'
1216 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Lo
1217 COLOR = 'black'
1218 SEMIOLOGY = False
1219 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1220
1221 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1222 plt.figure(0, figsize=(12, 8))
1223 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1224 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1225 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1226 plt.ylabel('Structural Period [s]')
1227 plt.xlabel('Displacement [mm]')
1228 #plt.semilogy()
1229 plt.grid()

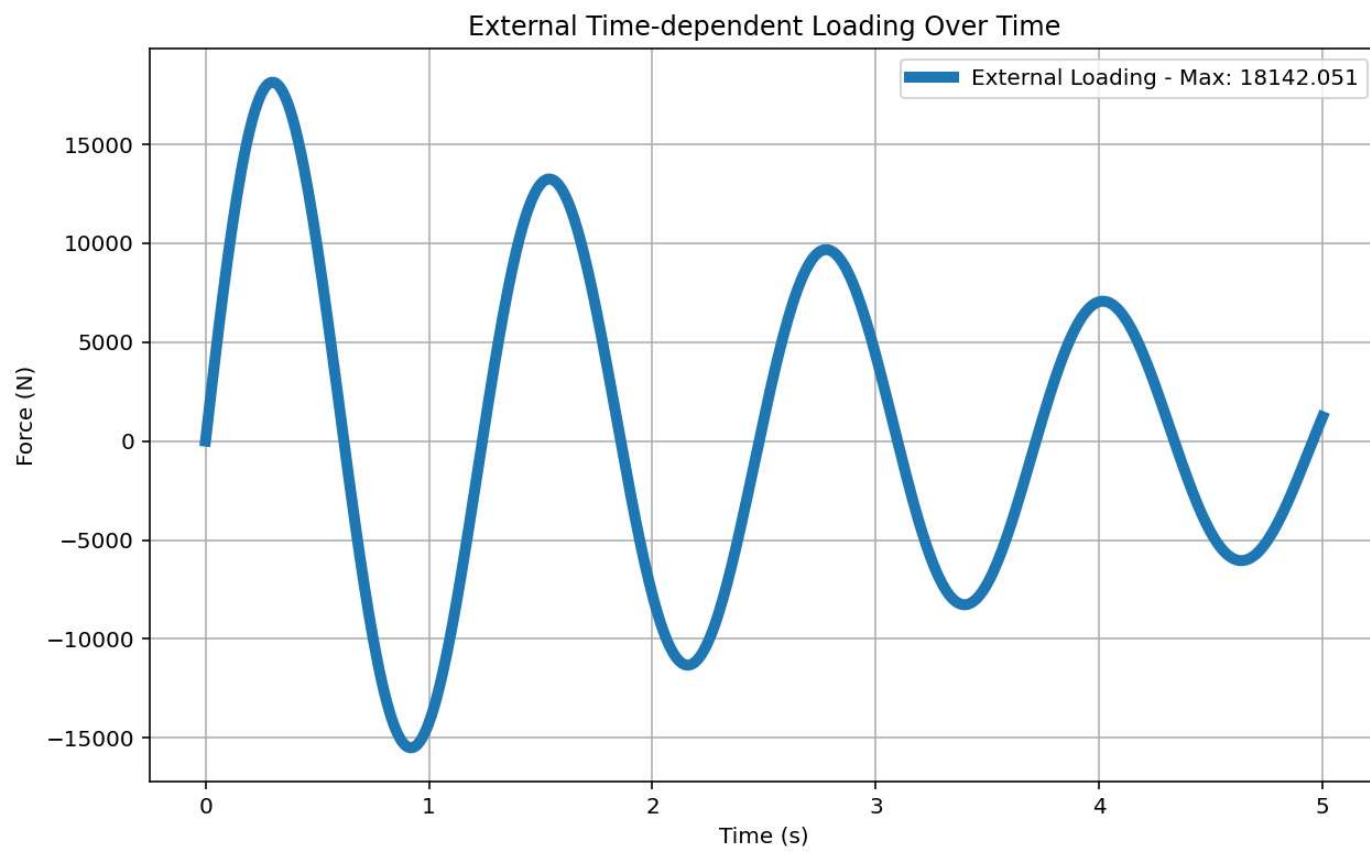
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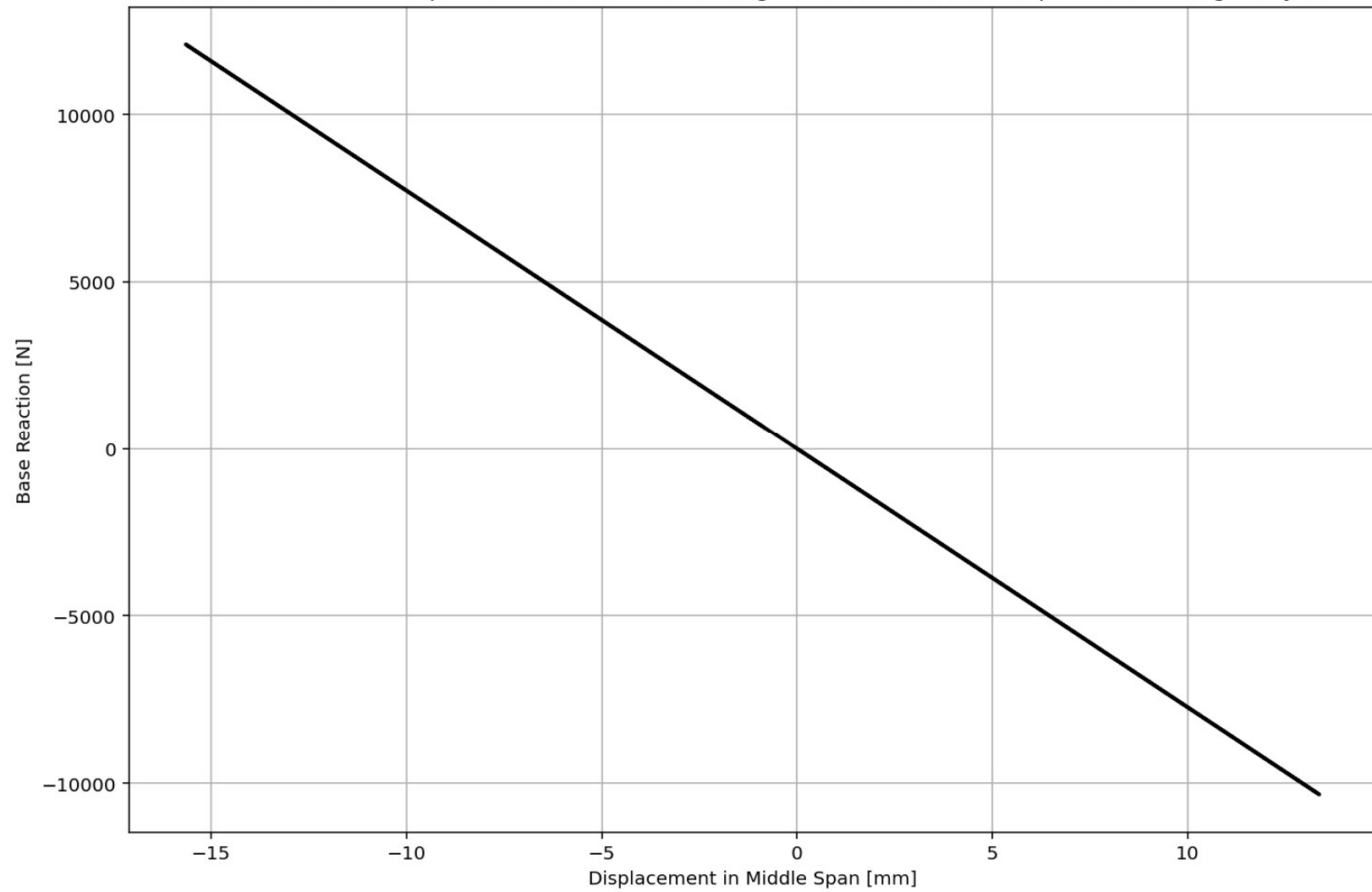
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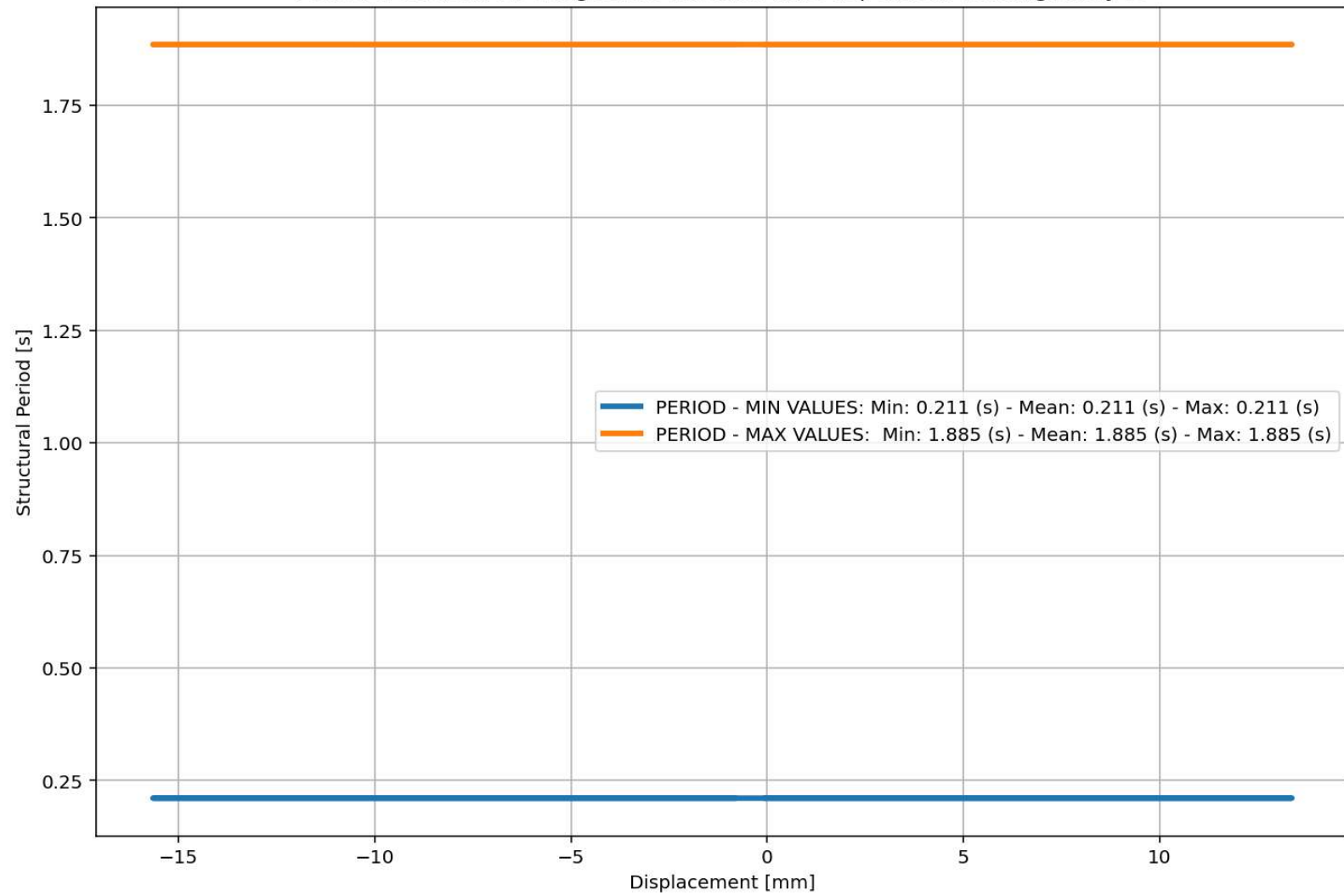
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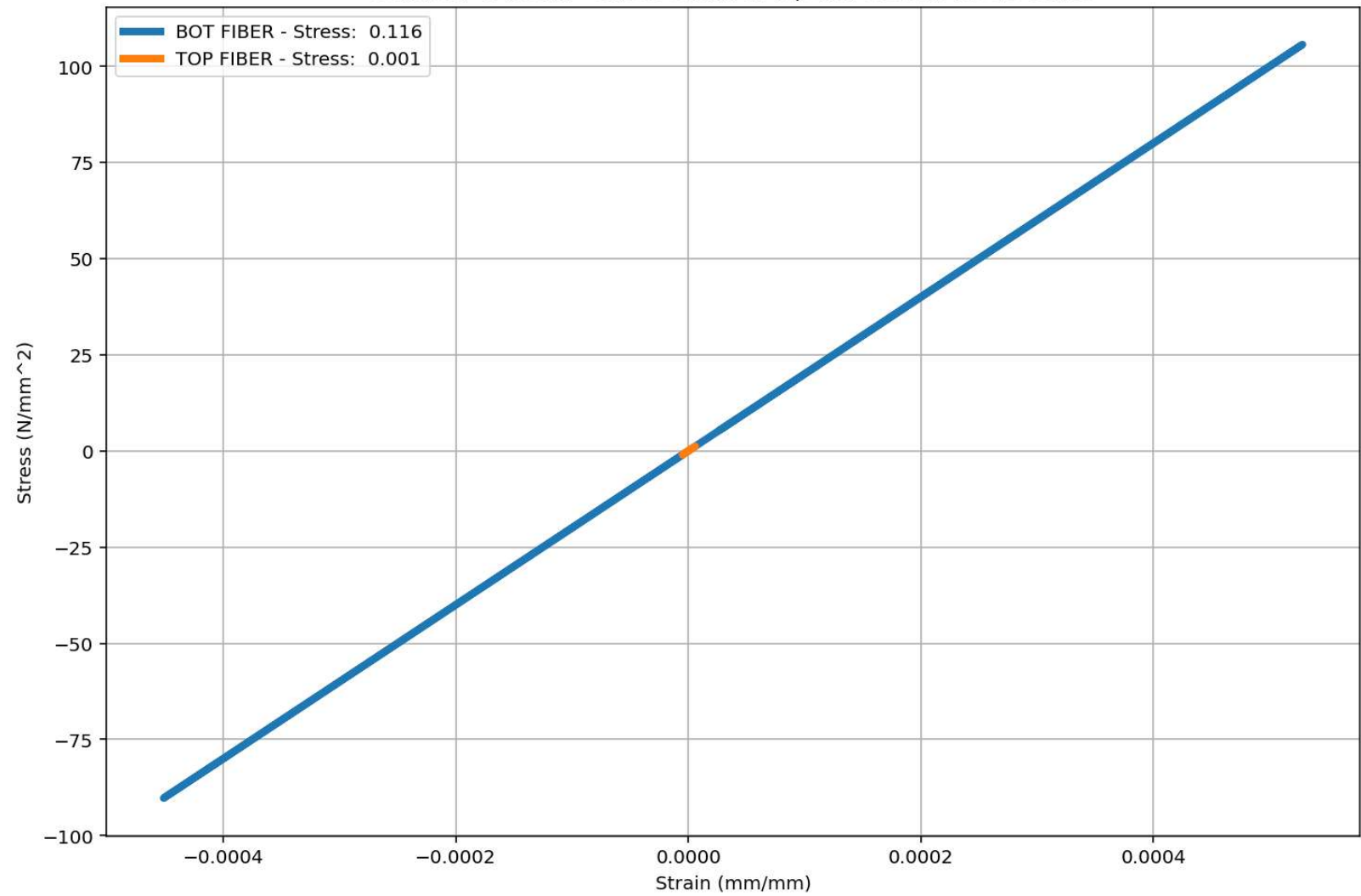
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

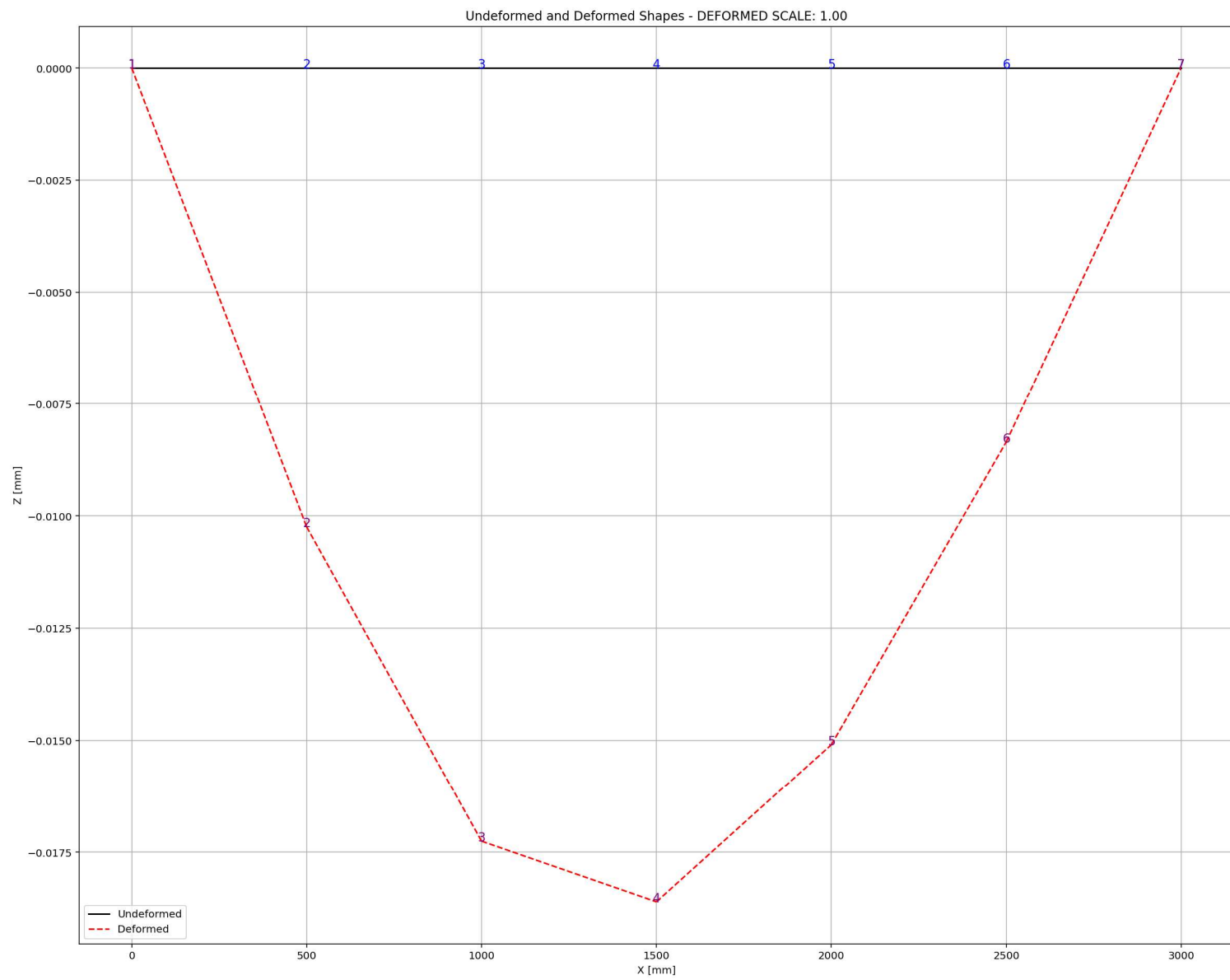


Period of Structure During Static External Time-dependent Loading Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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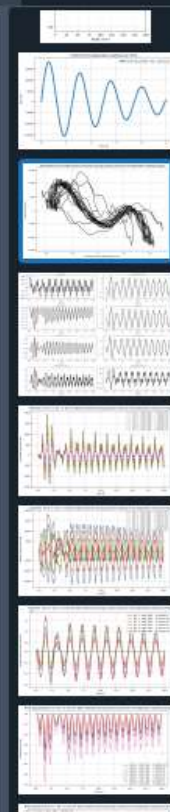
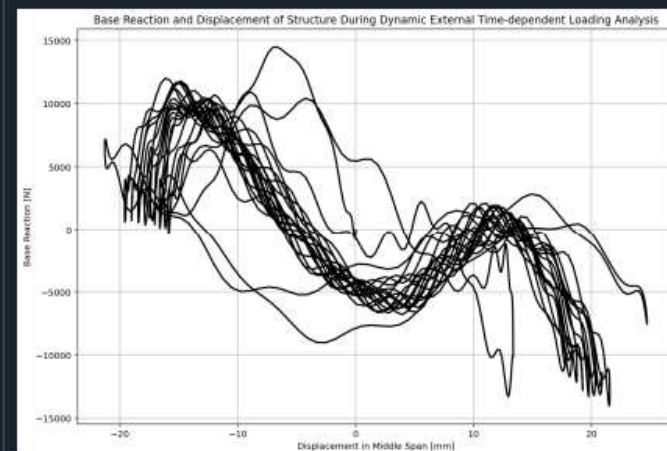
P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

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1248 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1249 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1250 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1251 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1252 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1253
1254 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1255
1256 (time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1257 ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
1258 node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1259 dispX_ETDLD, dispY_ETDLD,
1260 veloX_ETDLD, veloY_ETDLD,
1261 accX_ETDLD, accY_ETDLD,
1262 stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1263 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
1264 STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD) = DATA
1265
1266 XDATA = disp_mid_ETDLD
1267 YDATA = reaction_ETDLD
1268 XLABEL = 'Displacement in Middle Span [mm]'
1269 YLABEL = 'Base Reaction [N]'
1270 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent L
1271 COLOR = 'black'
1272 SEMILOGY = False
1273 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1274
1275 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1276 dispX_ETDLD, dispY_ETDLD,
1277 veloX_ETDLD, veloY_ETDLD,
1278 accX_ETDLD, accY_ETDLD)
1279
1280 # PLOT ELEMENTS AXIAL FORCE
1281

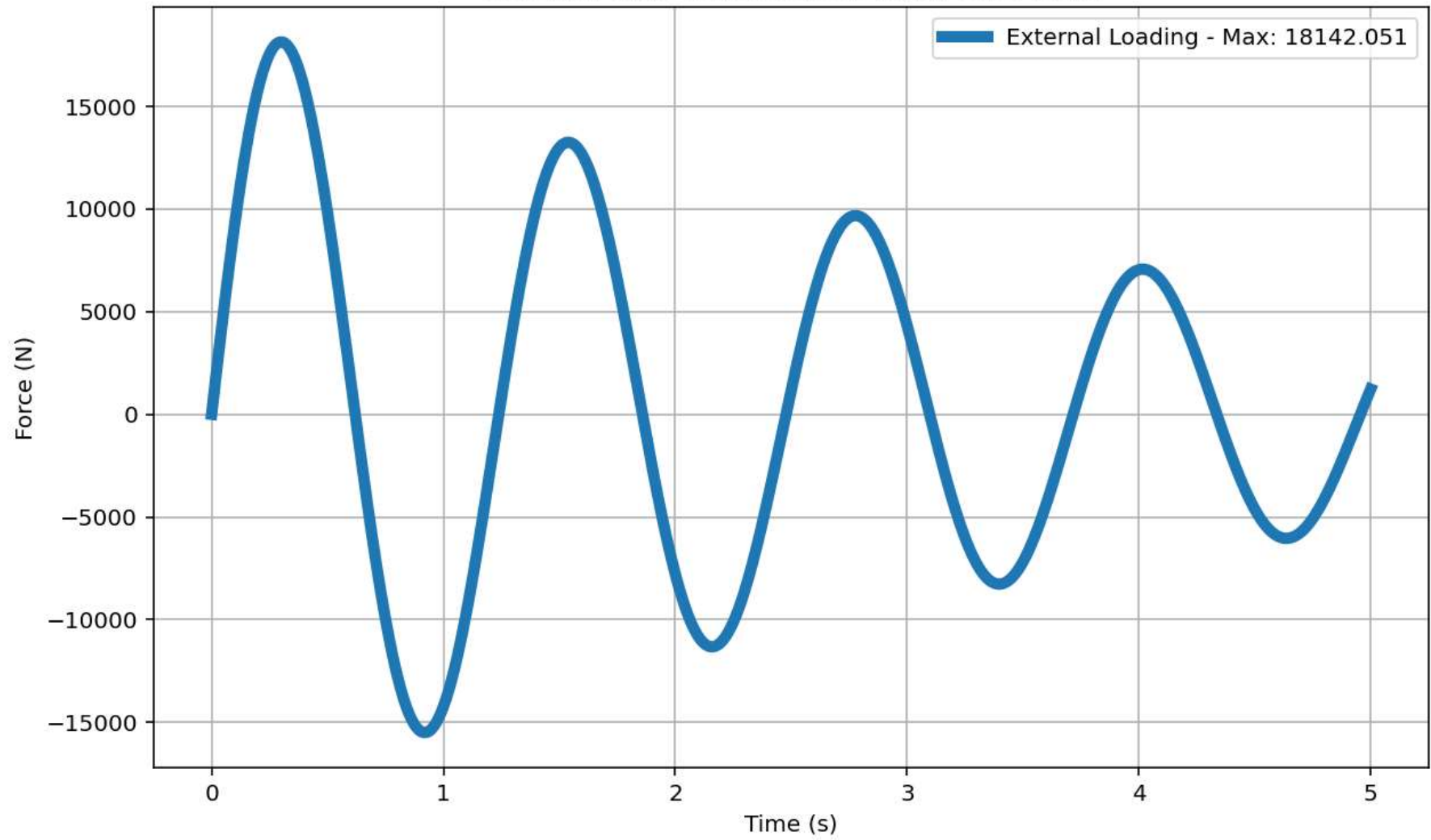
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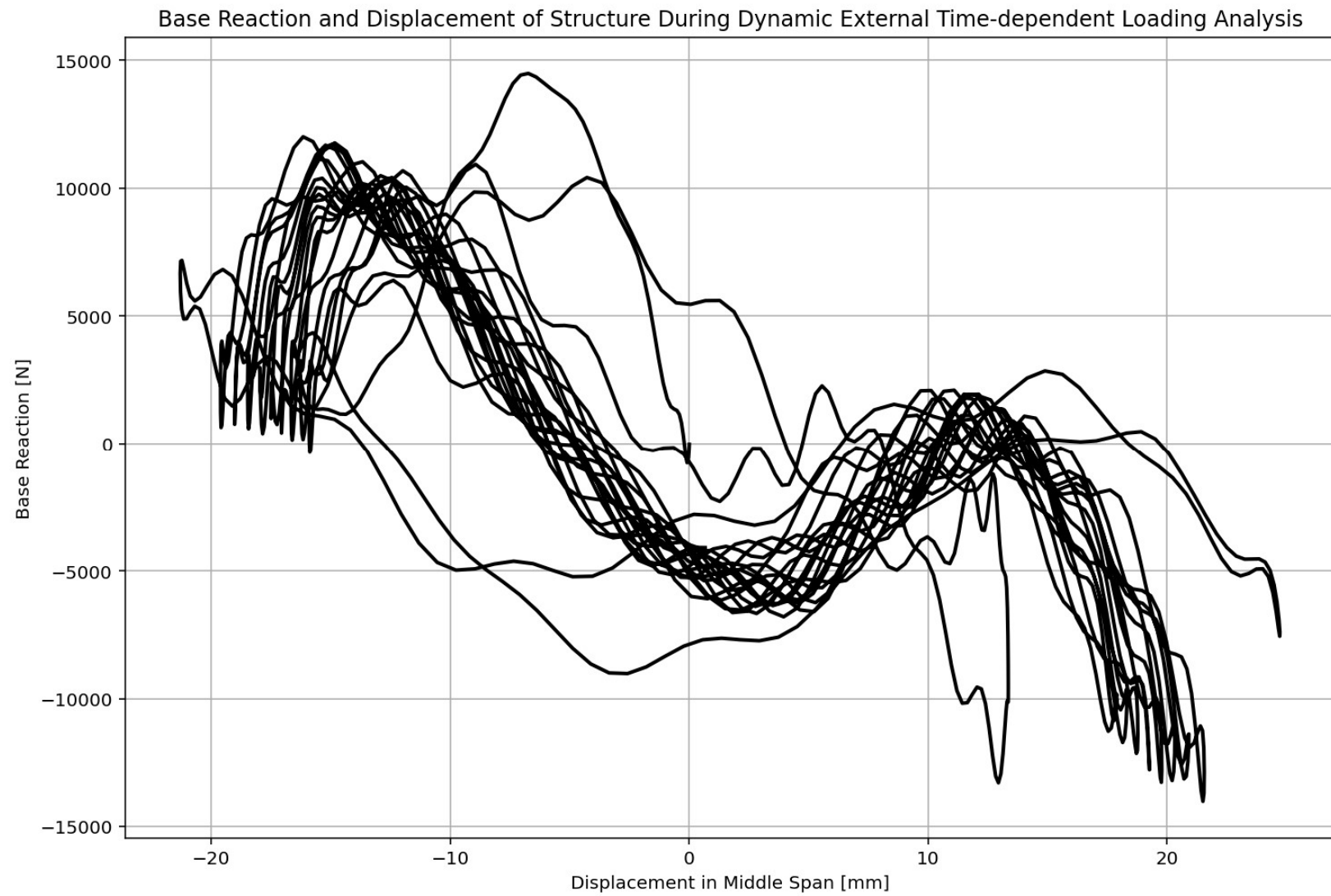
..PLY_SUPPORTED_8_ANA\P(t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED 29 %

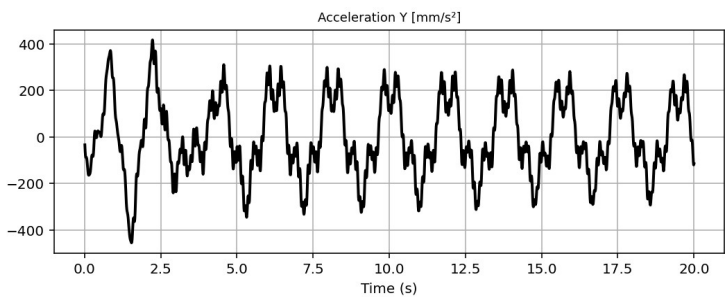
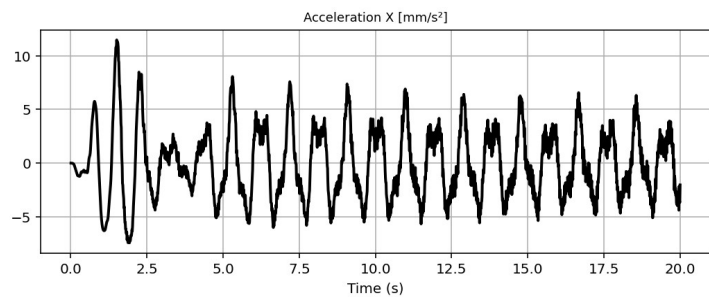
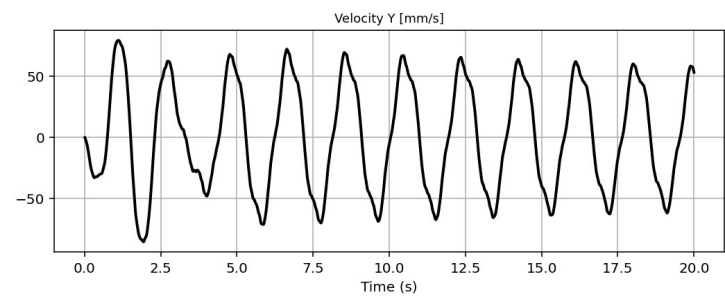
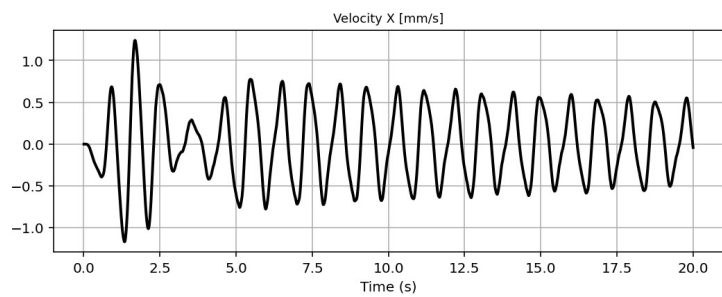
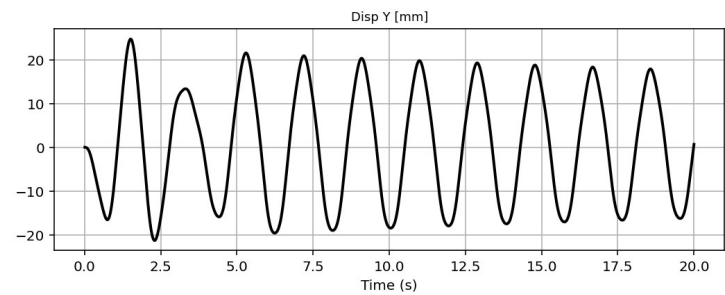
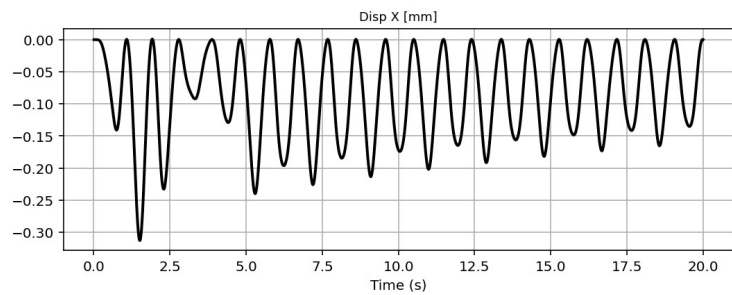
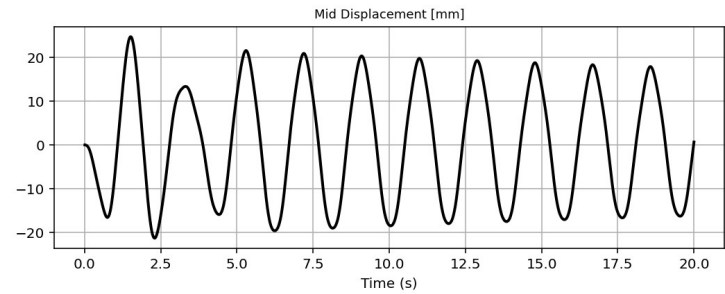
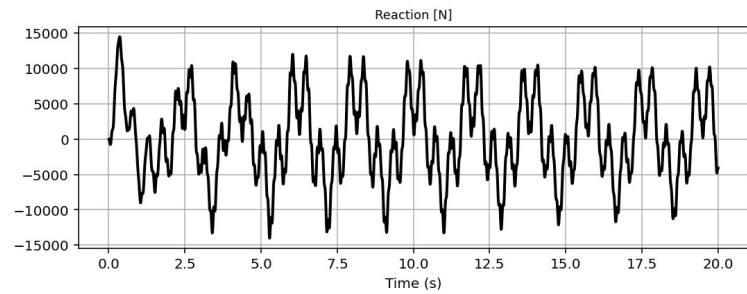


IPython Console Plots Variable Explorer Help Debugger History Files

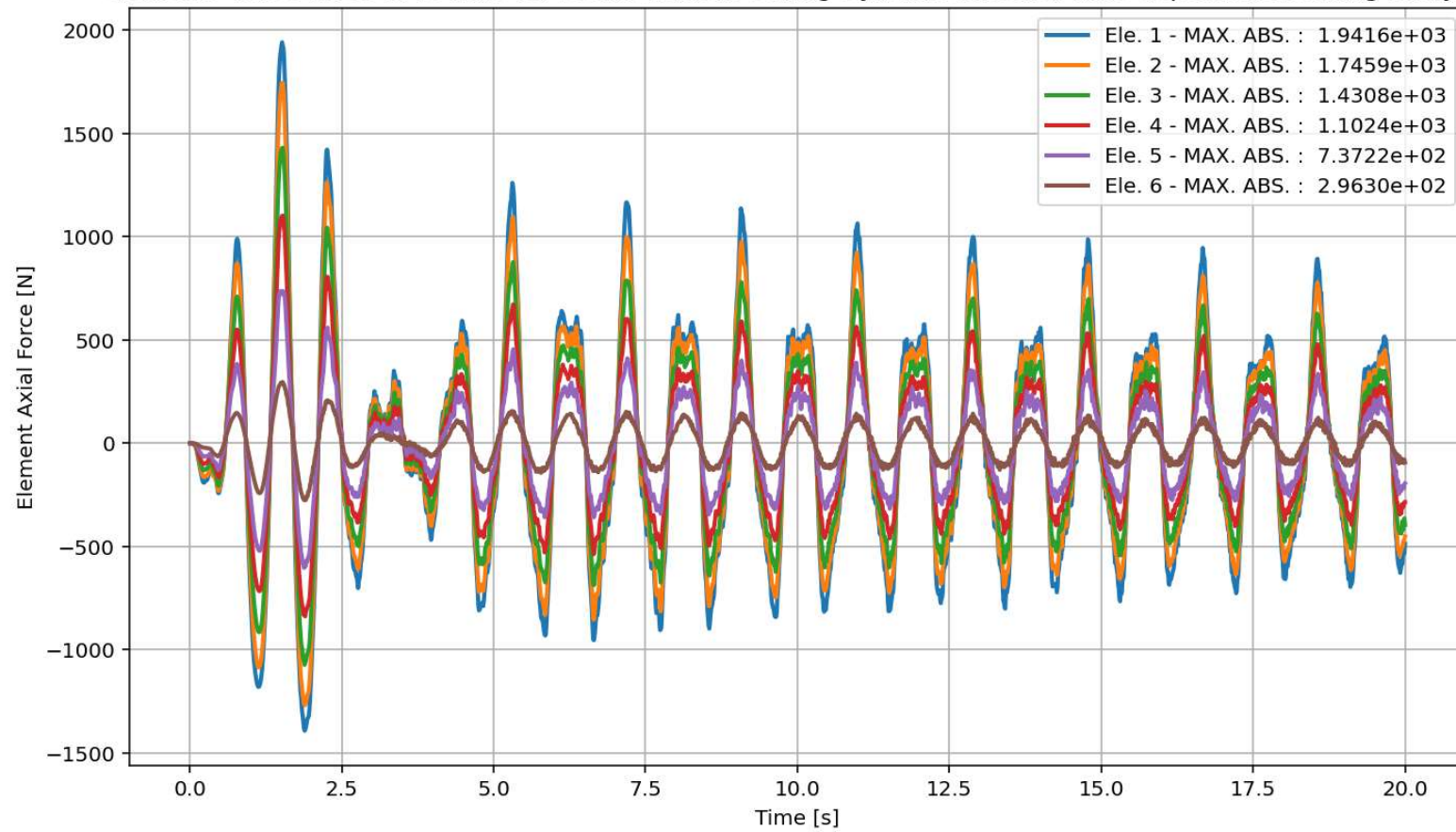
External Time-dependent Loading Over Time

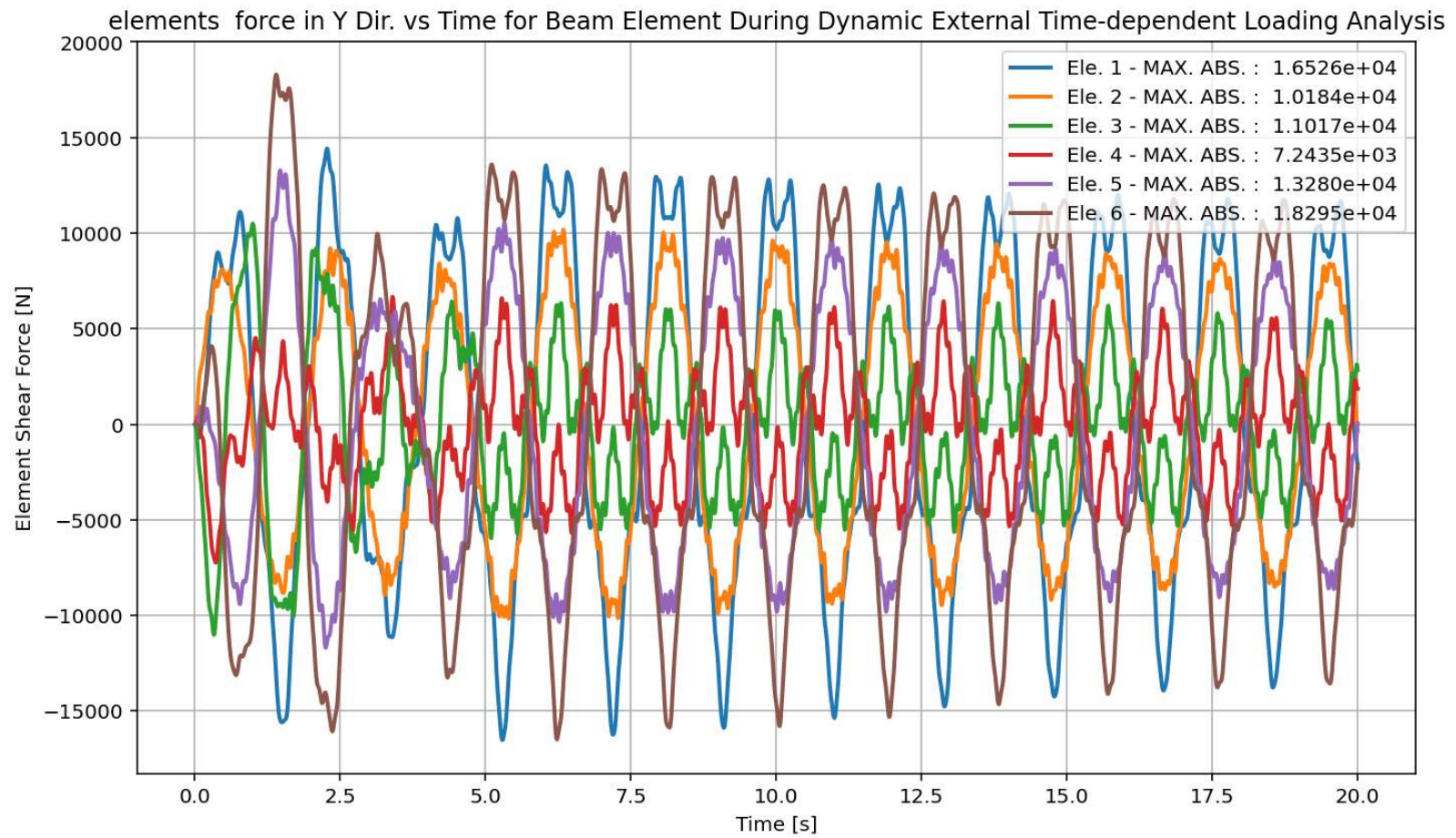




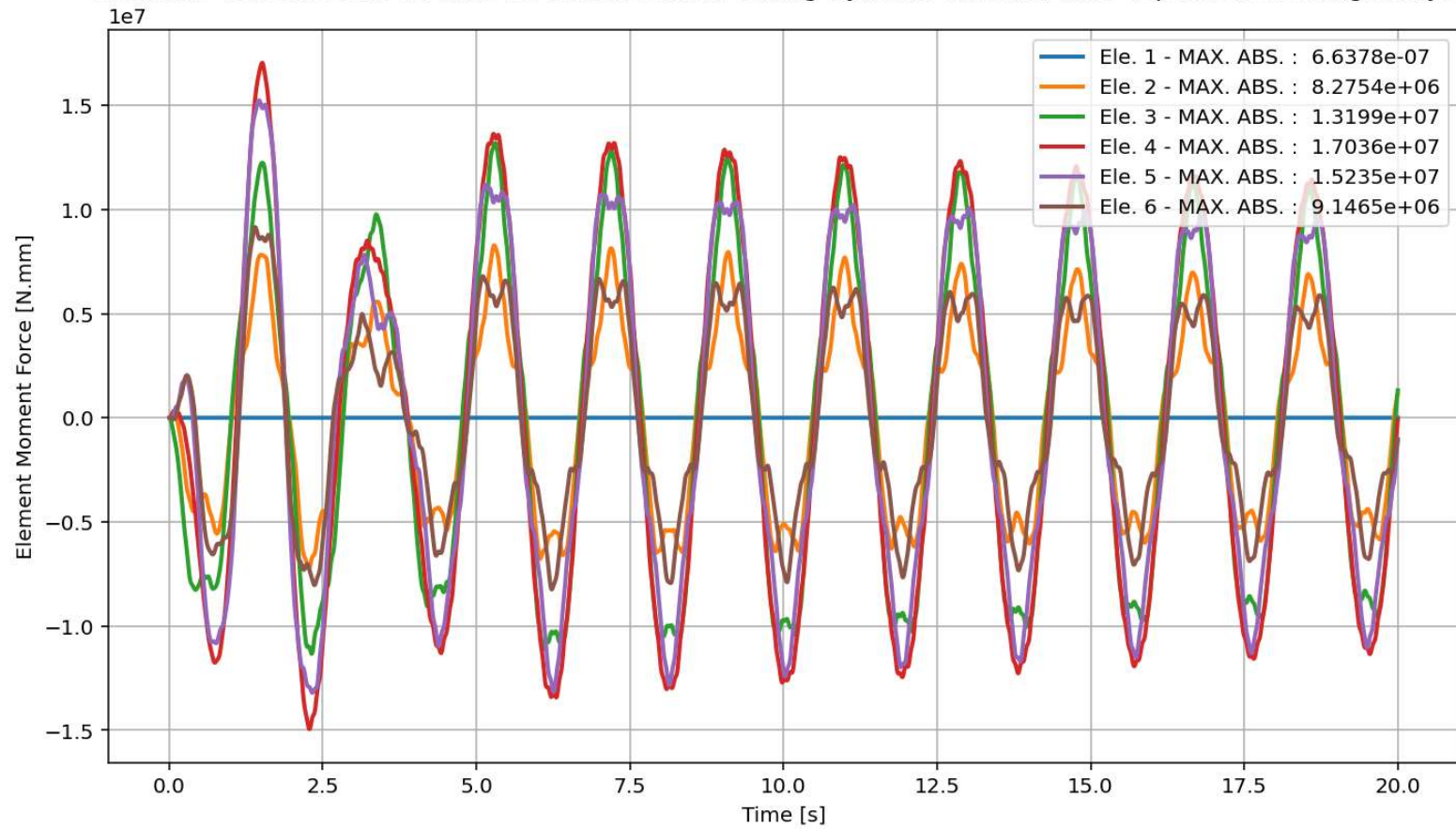


elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

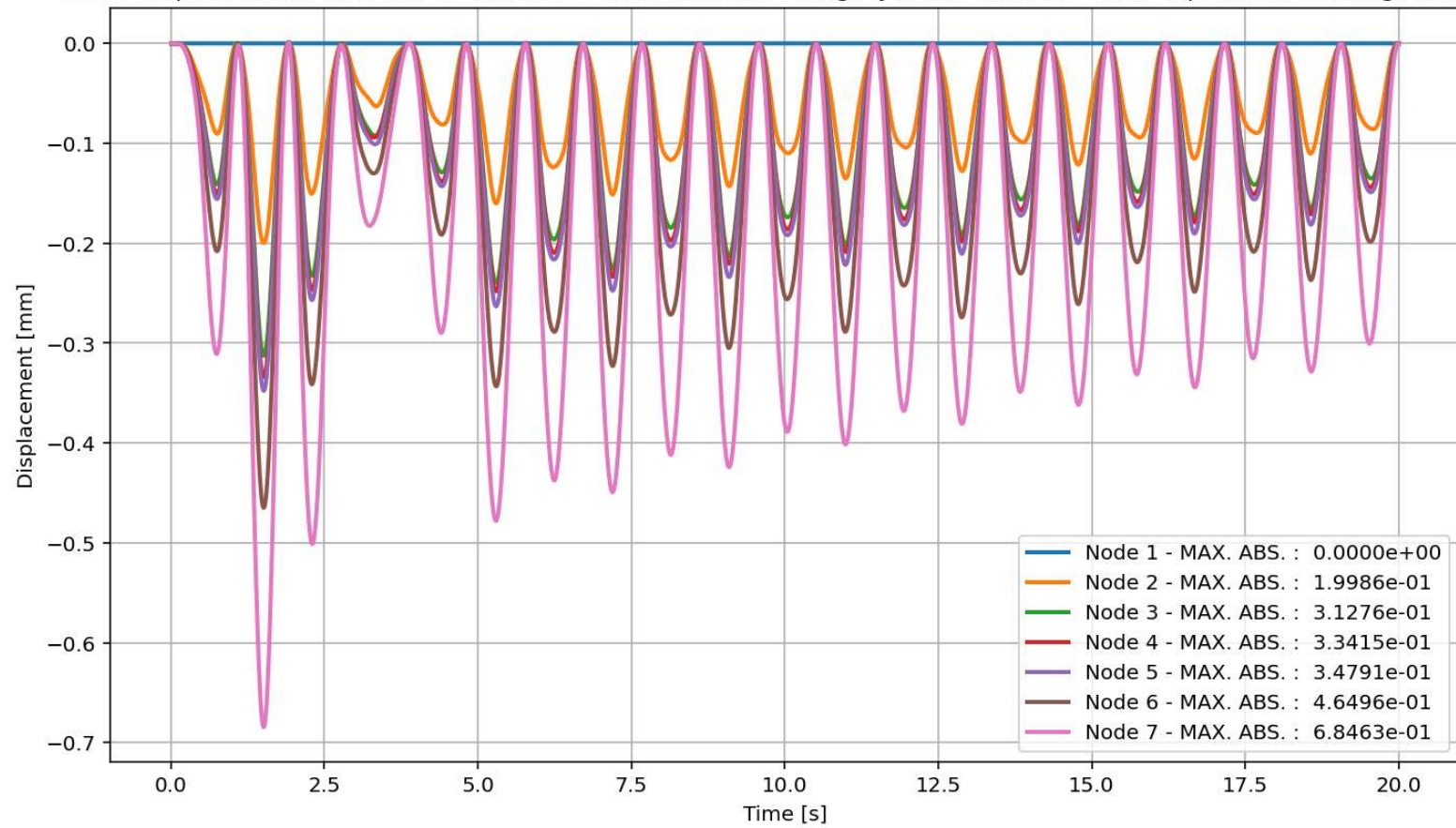




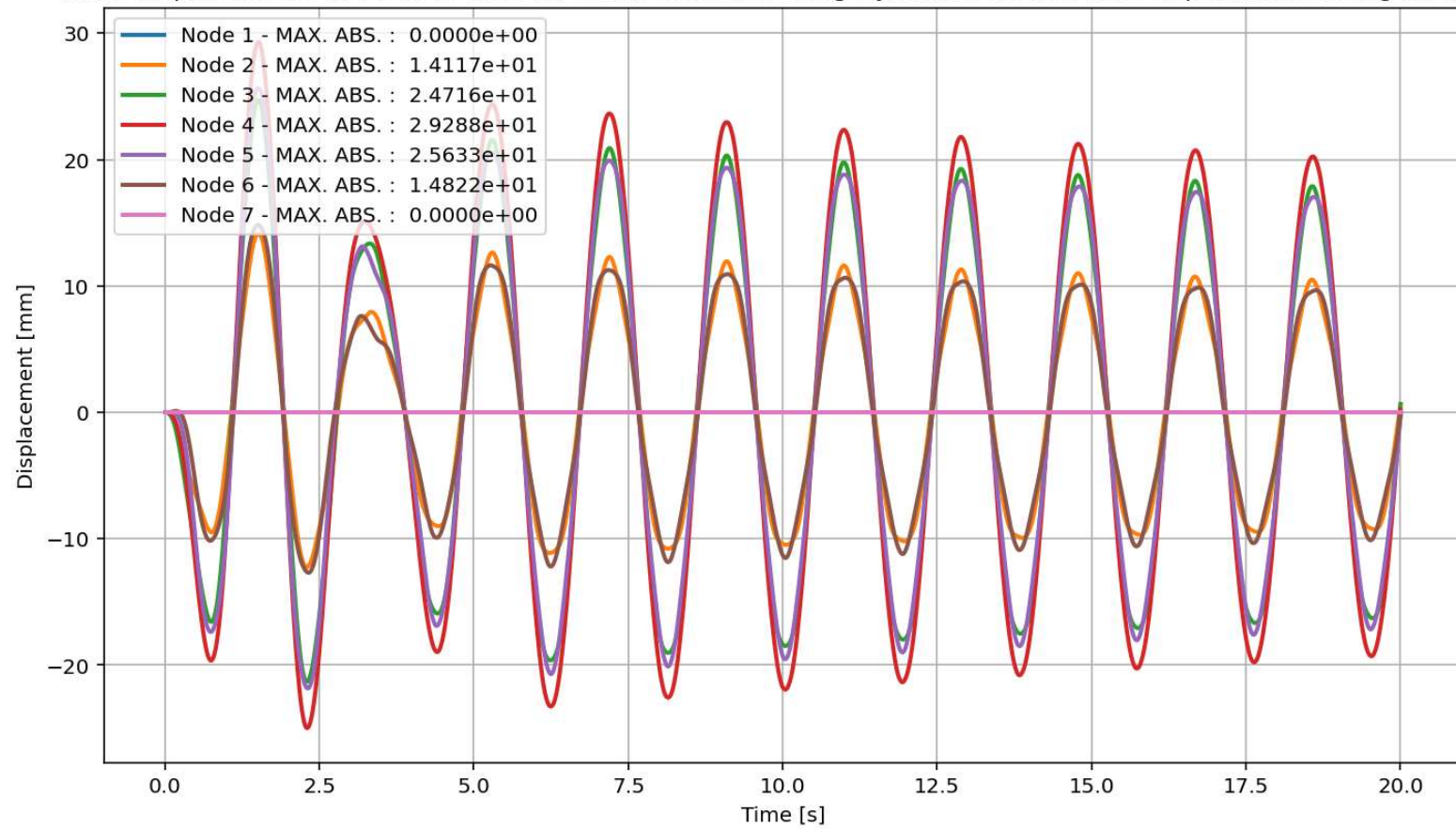
elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



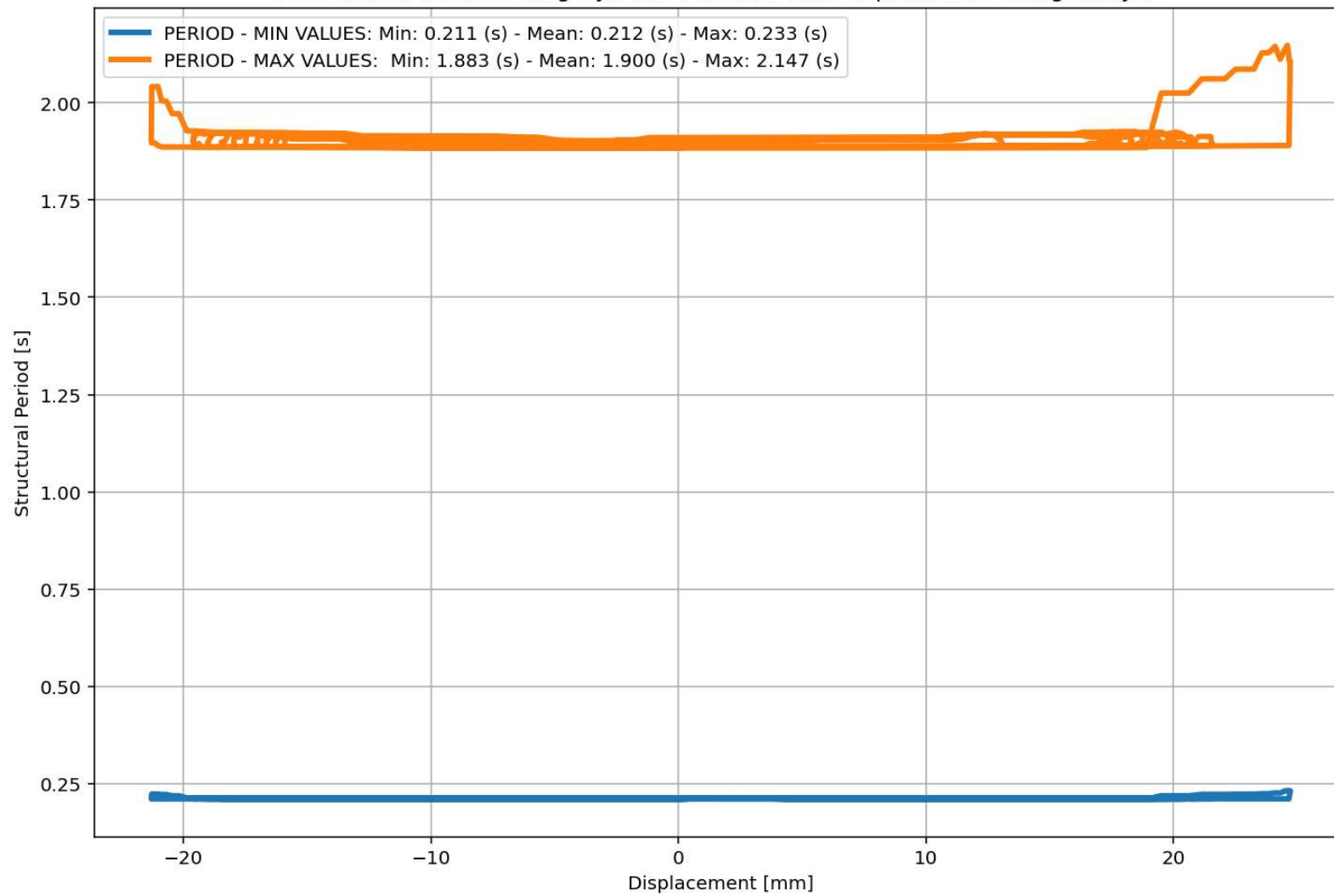
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



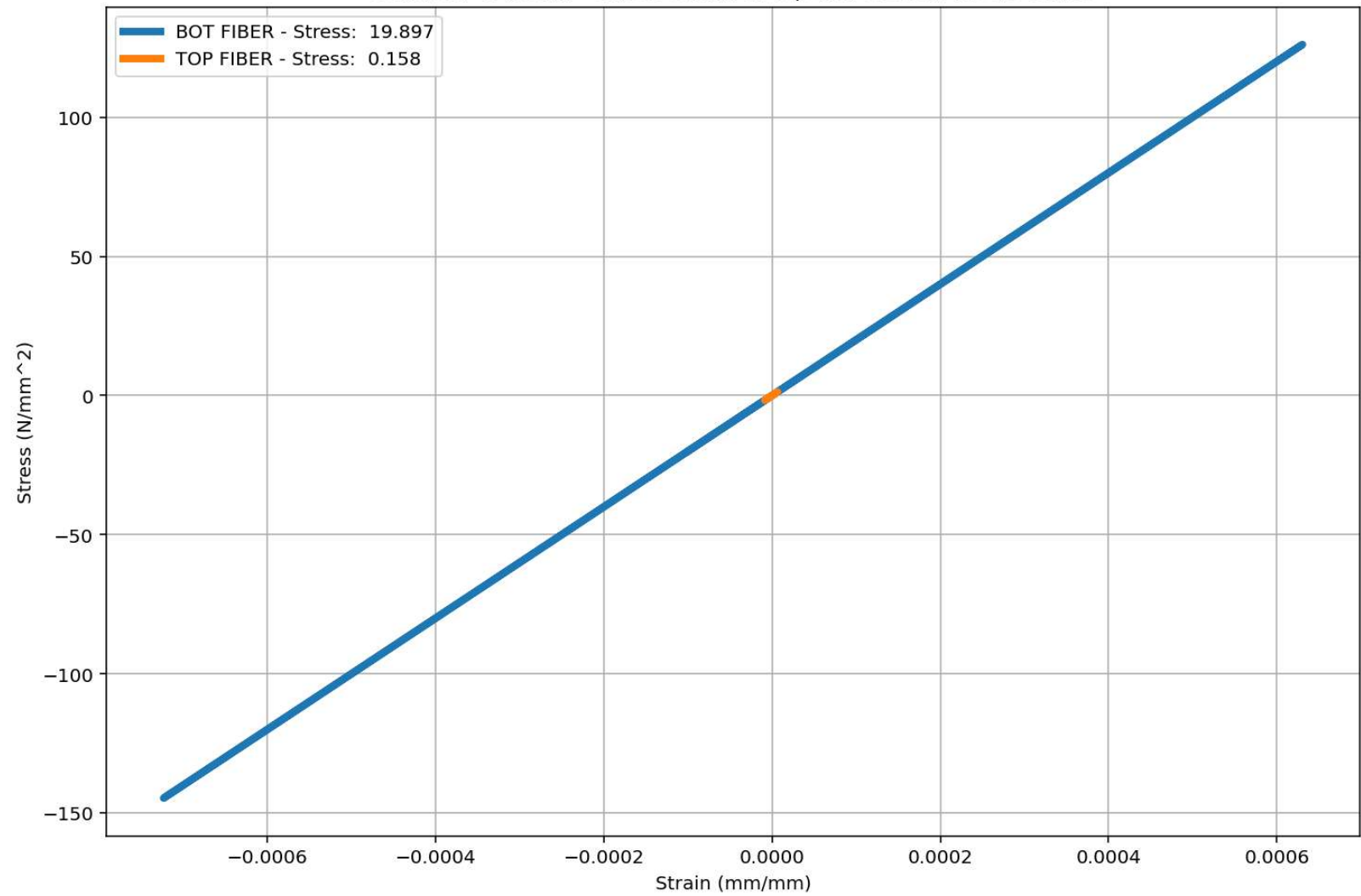
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

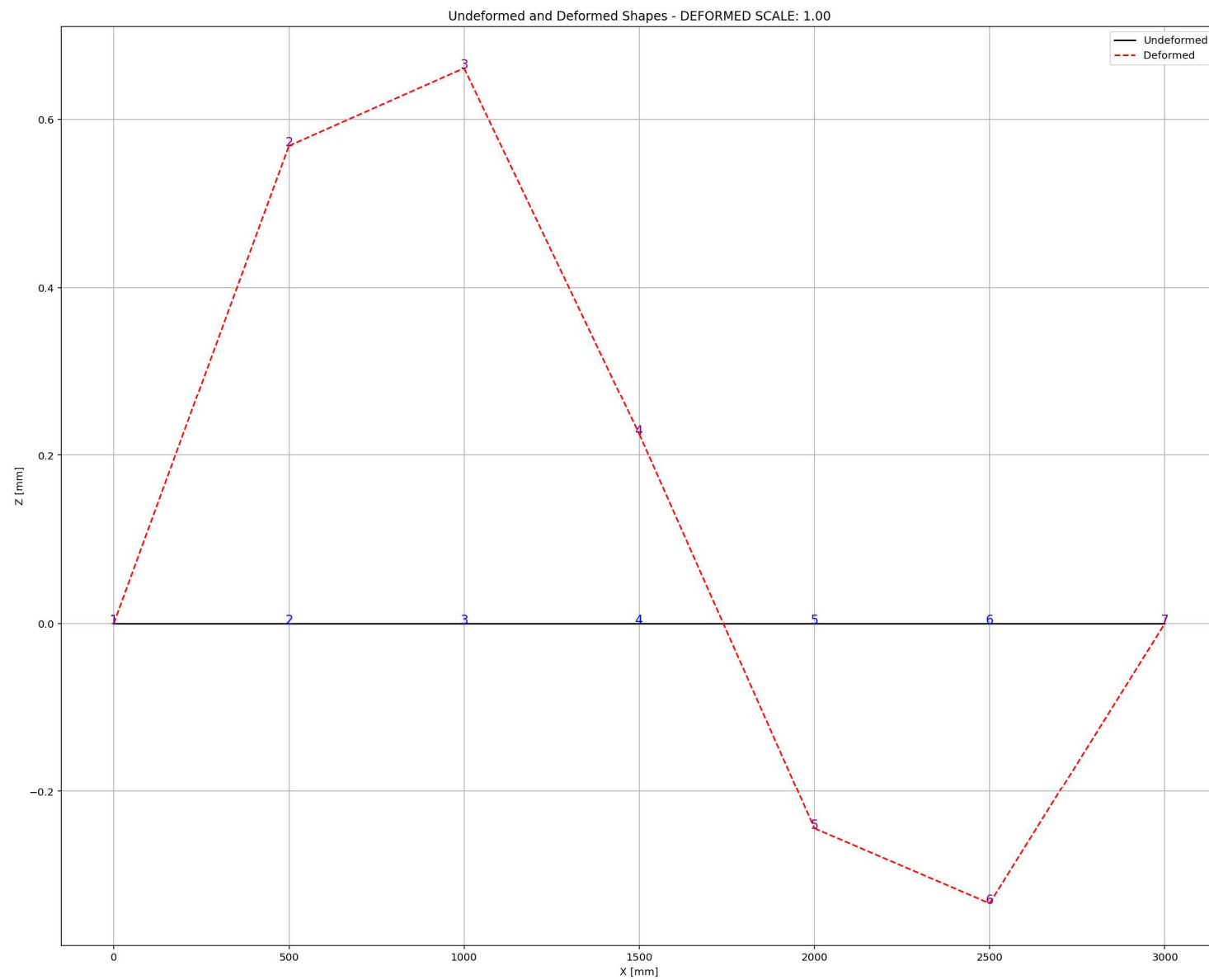


Period of Structure During Dynamic External Time-dependent Loading Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



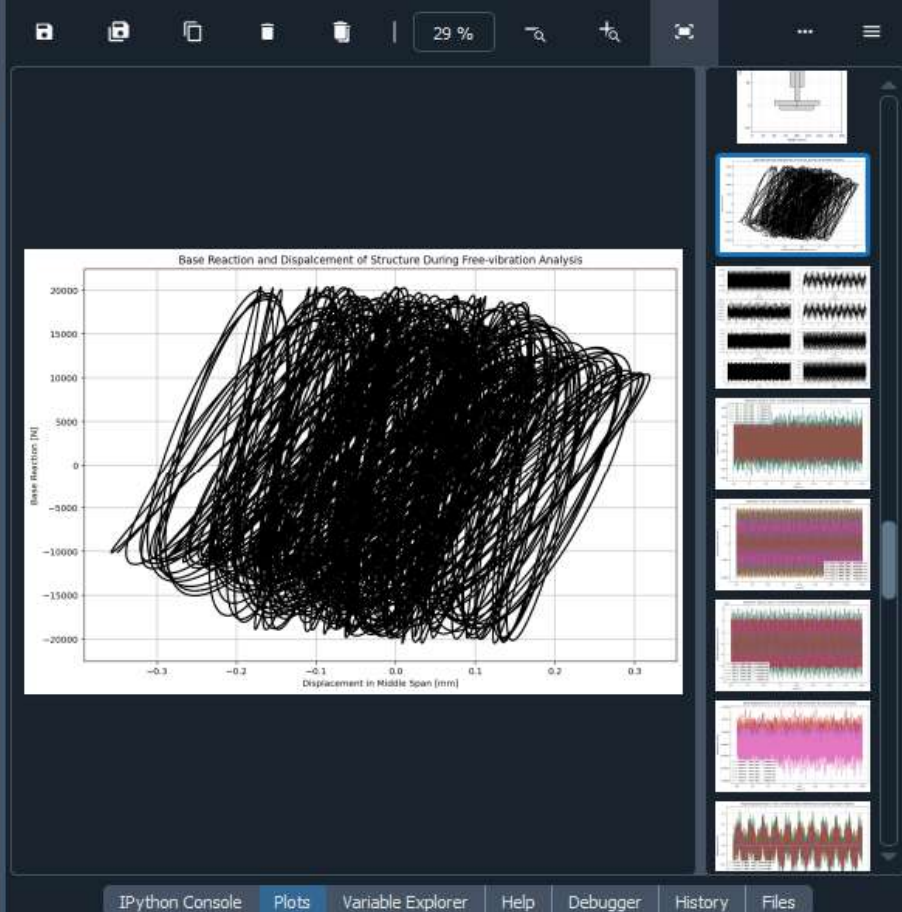


FREE-VIBRATION ANALYSIS

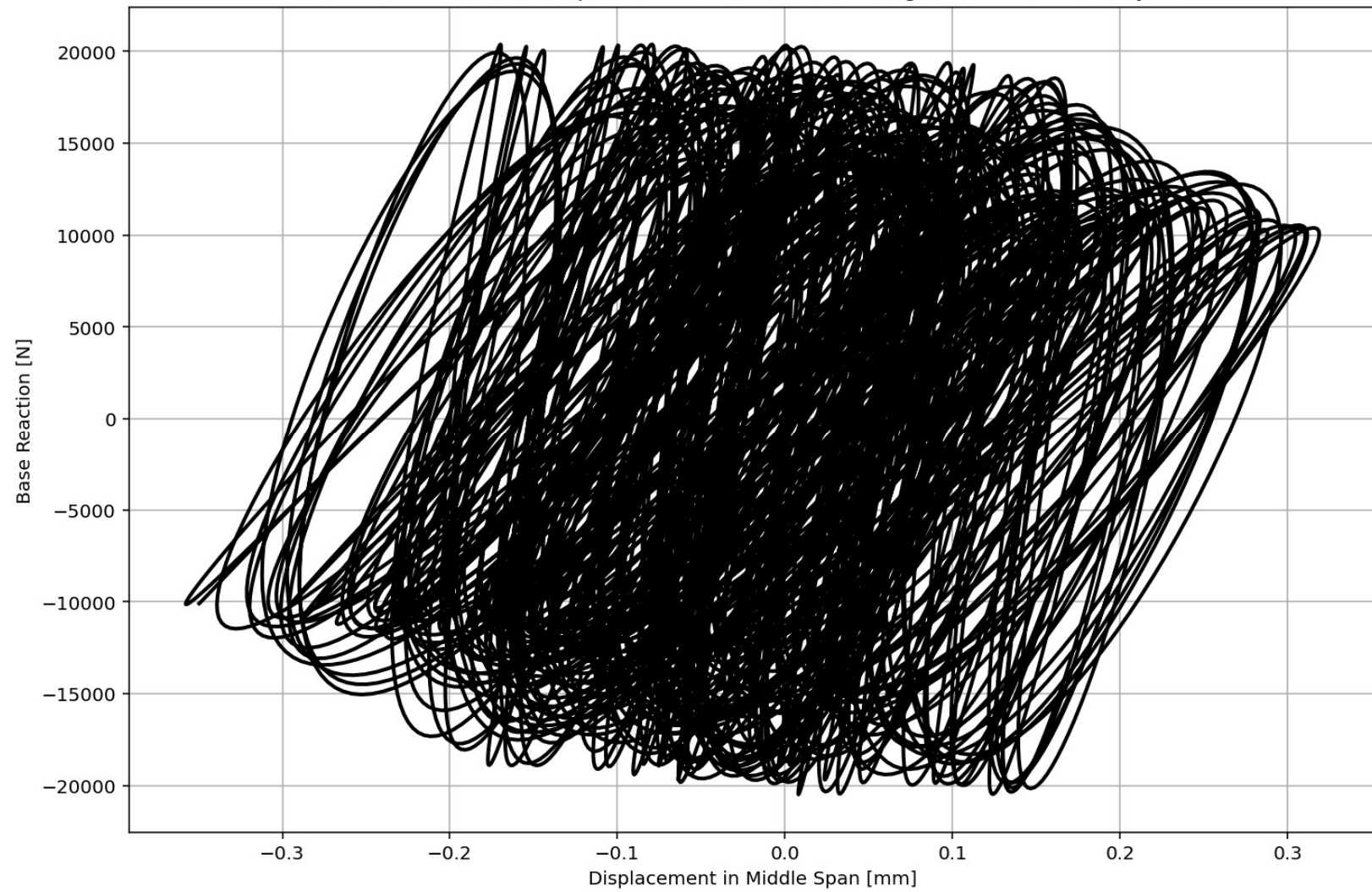
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...PPORTED\p(t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py

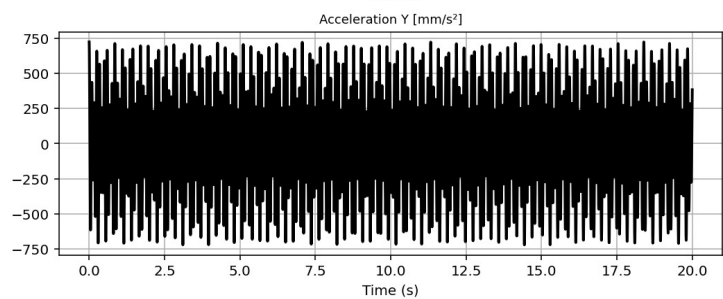
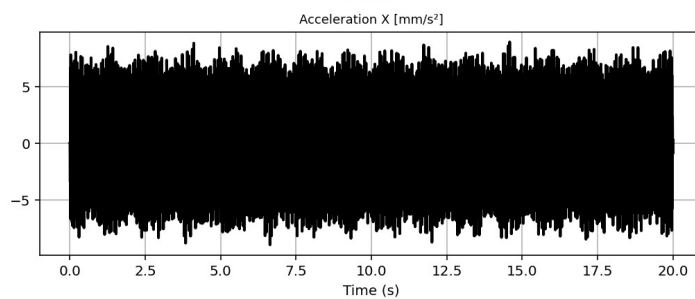
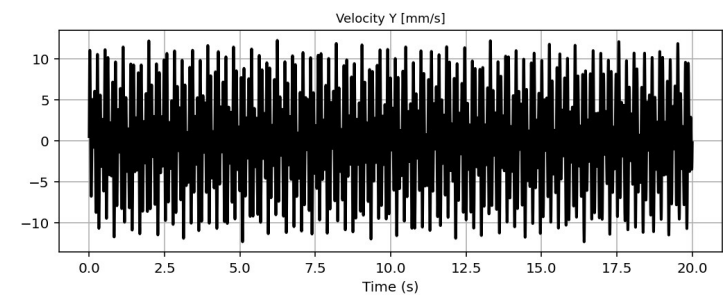
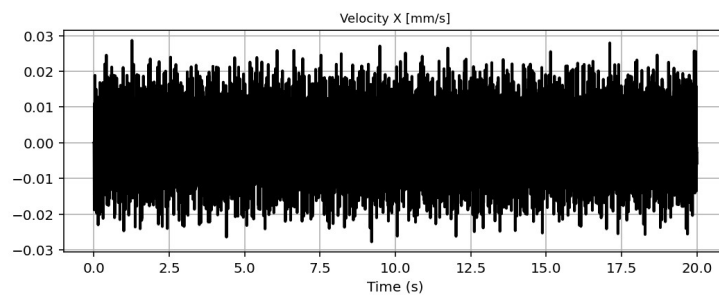
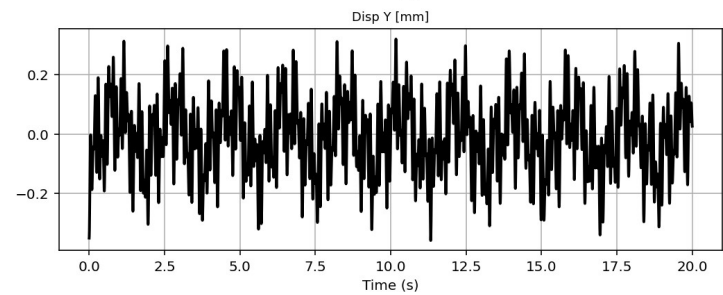
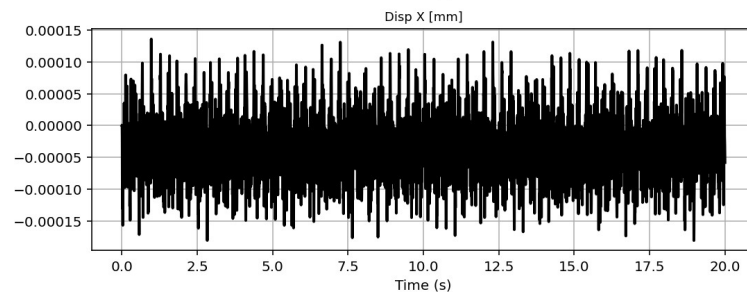
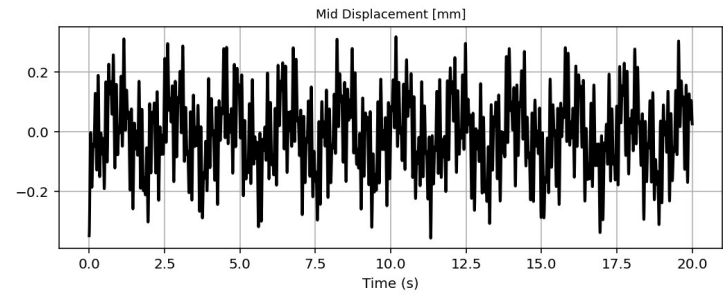
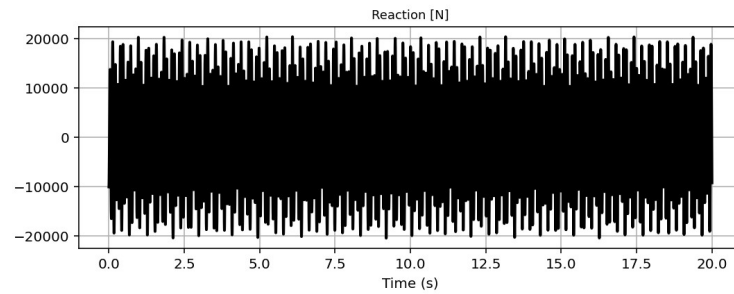
P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

```
1326 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1327 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1328 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1329 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1330 ANAL_TYPE = 'FREE-VIBRATION'
1331 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1332
1333 (time_FV, reaction_FV, disp_mid_FV,
1334 ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
1335 node_displacementsX_FV, node_displacementsY_FV,
1336 dispX_FV, dispY_FV,
1337 veloX_FV, veloY_FV,
1338 accX_FV, accY_FV,
1339 stiffness_FV, PERIOD_FV, damping_ratio_FV,
1340 PERIOD_MIN_FV, PERIOD_MAX_FV,
1341 STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
1342
1343 XDATA = disp_mid_FV
1344 YDATA = reaction_FV
1345 XLABEL = 'Displacement in Middle Span [mm]'
1346 YLABEL = 'Base Reaction [N]'
1347 TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
1348 COLOR = 'black'
1349 SEMILOGY = False
1350 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1351
1352 PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
1353                   dispX_FV, dispY_FV,
1354                   veloX_FV, veloY_FV,
1355                   accX_FV, accY_FV)
1356
1357 # PLOT ELEMENTS AXIAL FORCE
1358 YLABEL = 'Element Axial Force [N]'
1359 TITLE = "elements force in X Dis vs Time for Beam Element During Free-vibration Analysis"
```

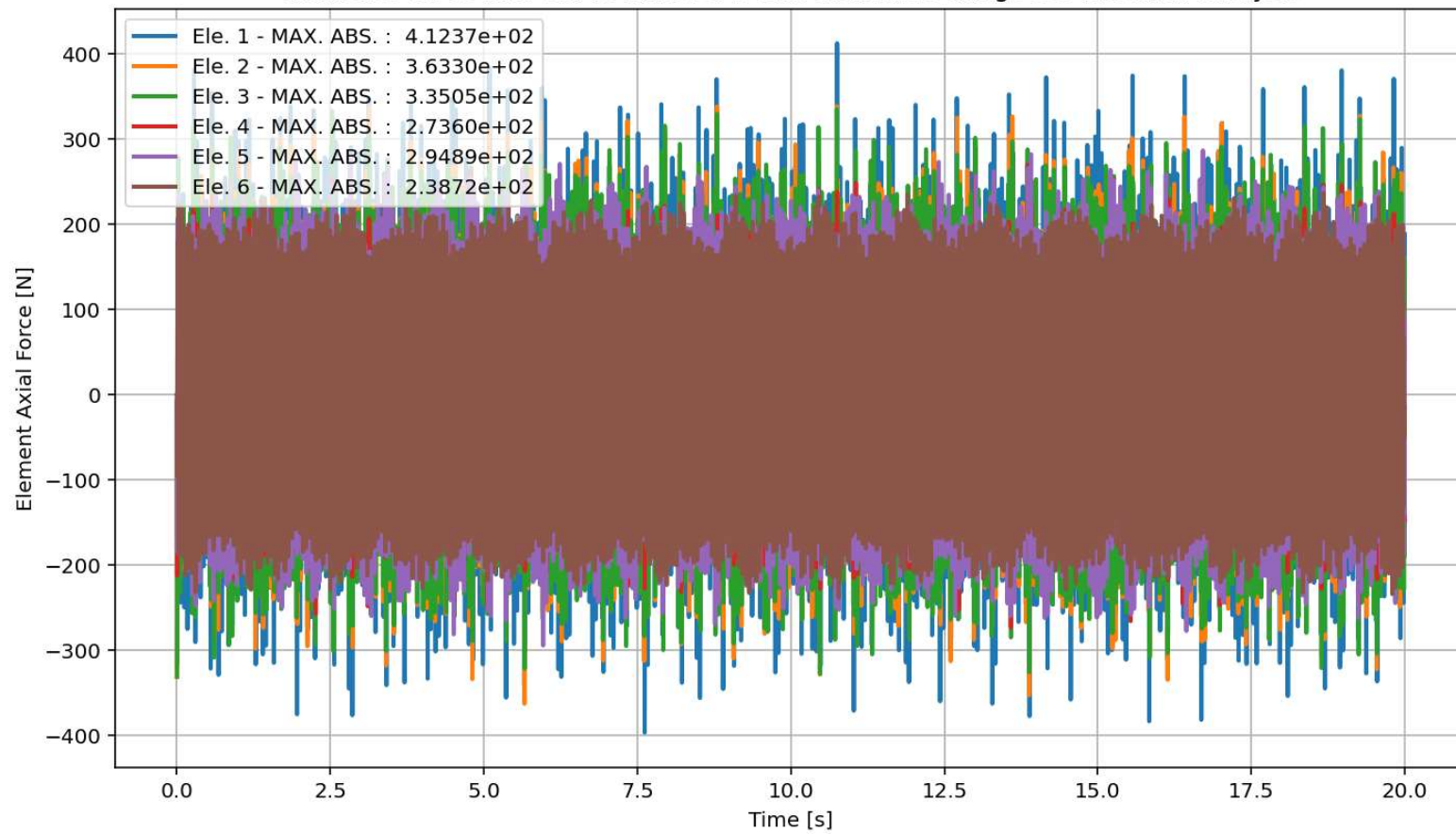


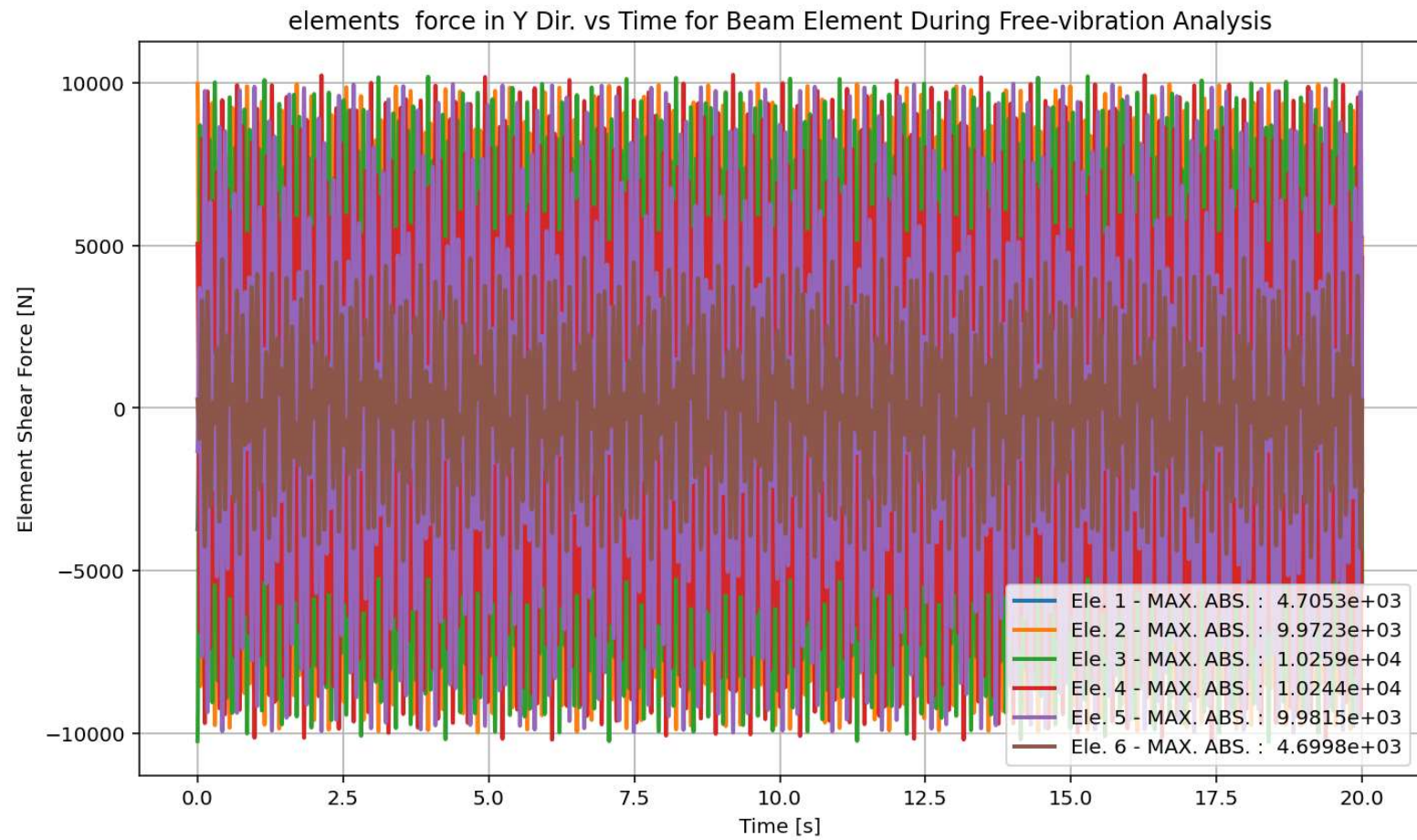
Base Reaction and Dispalcement of Structure During Free-vibration Analysis

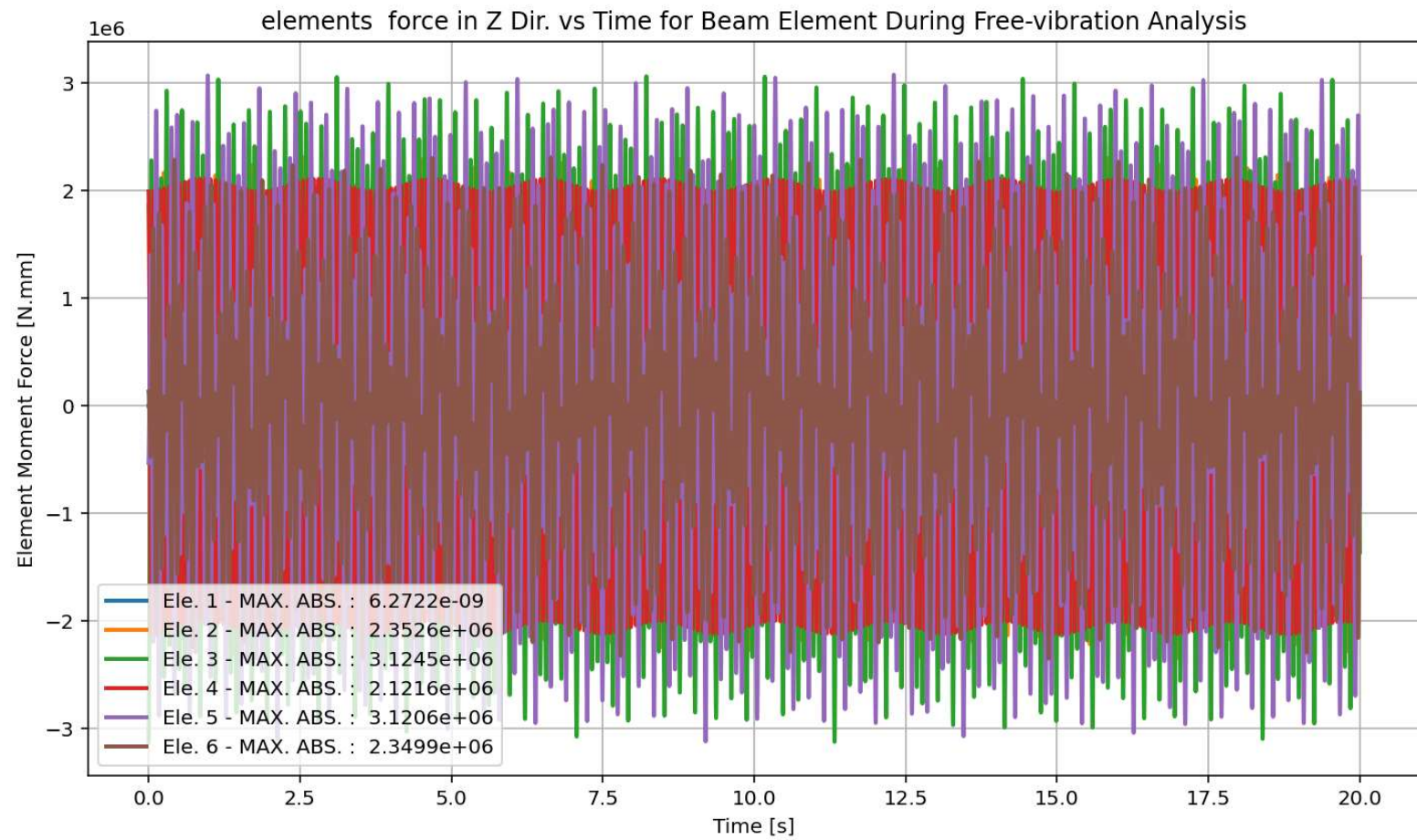


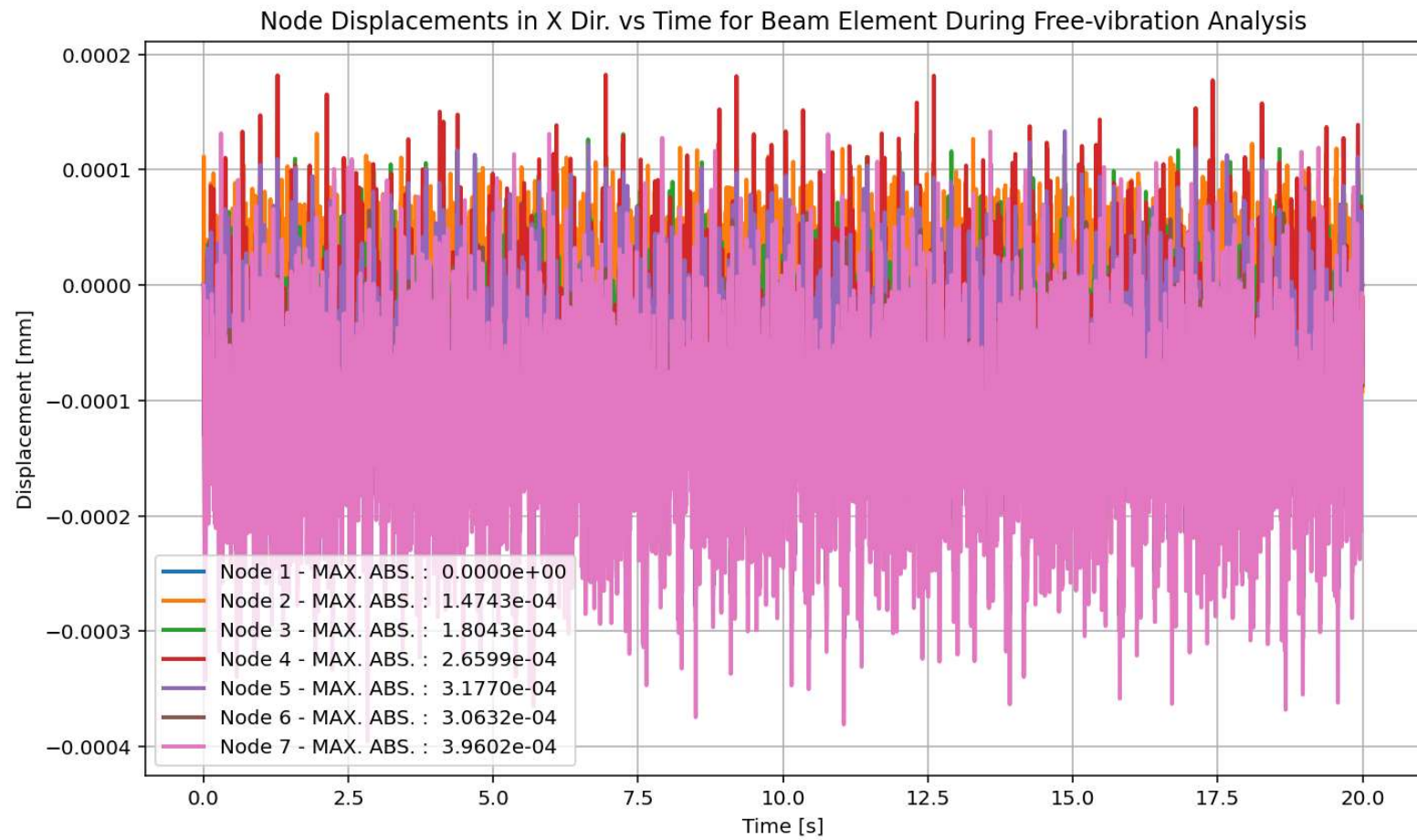


elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis

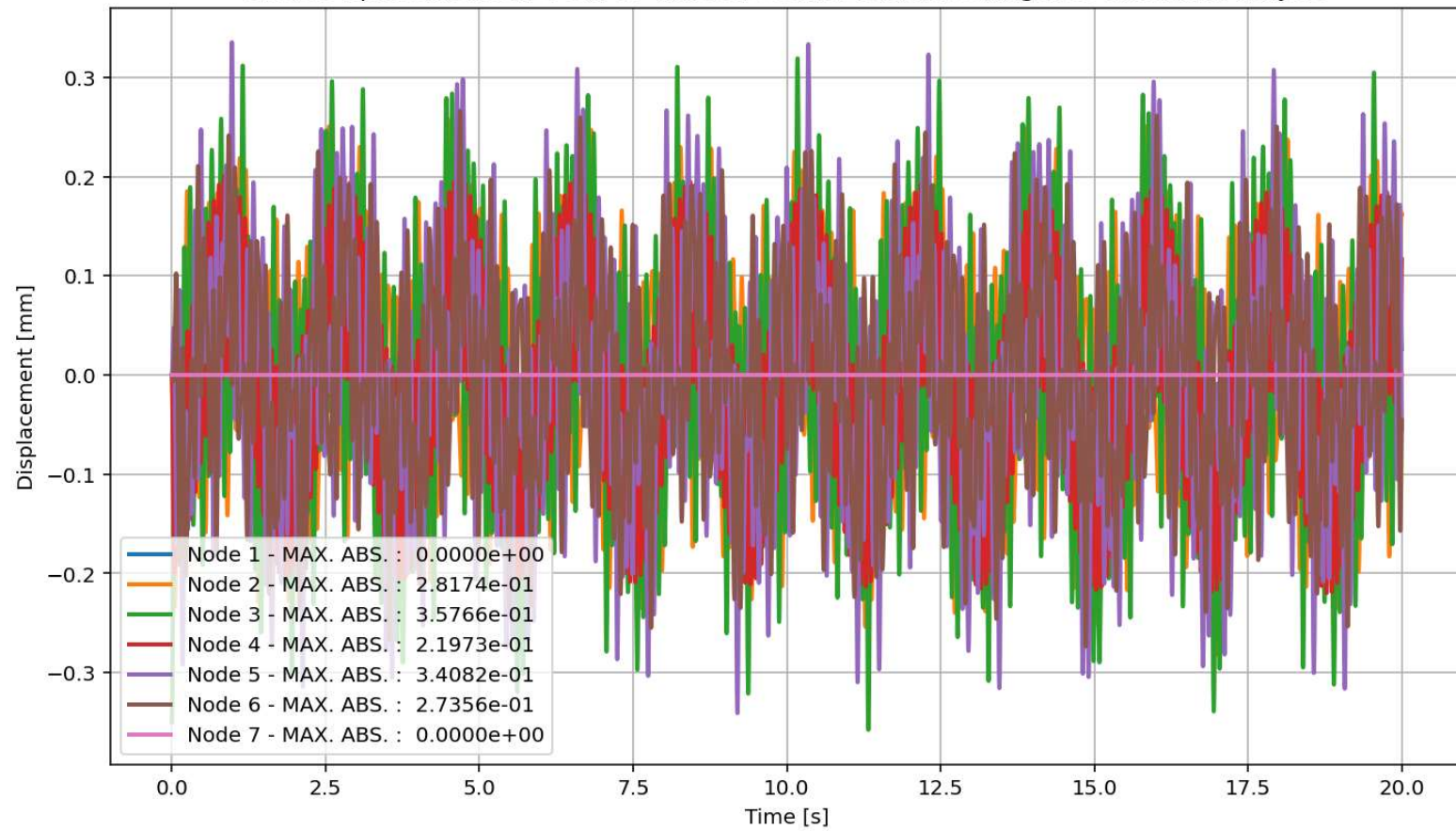




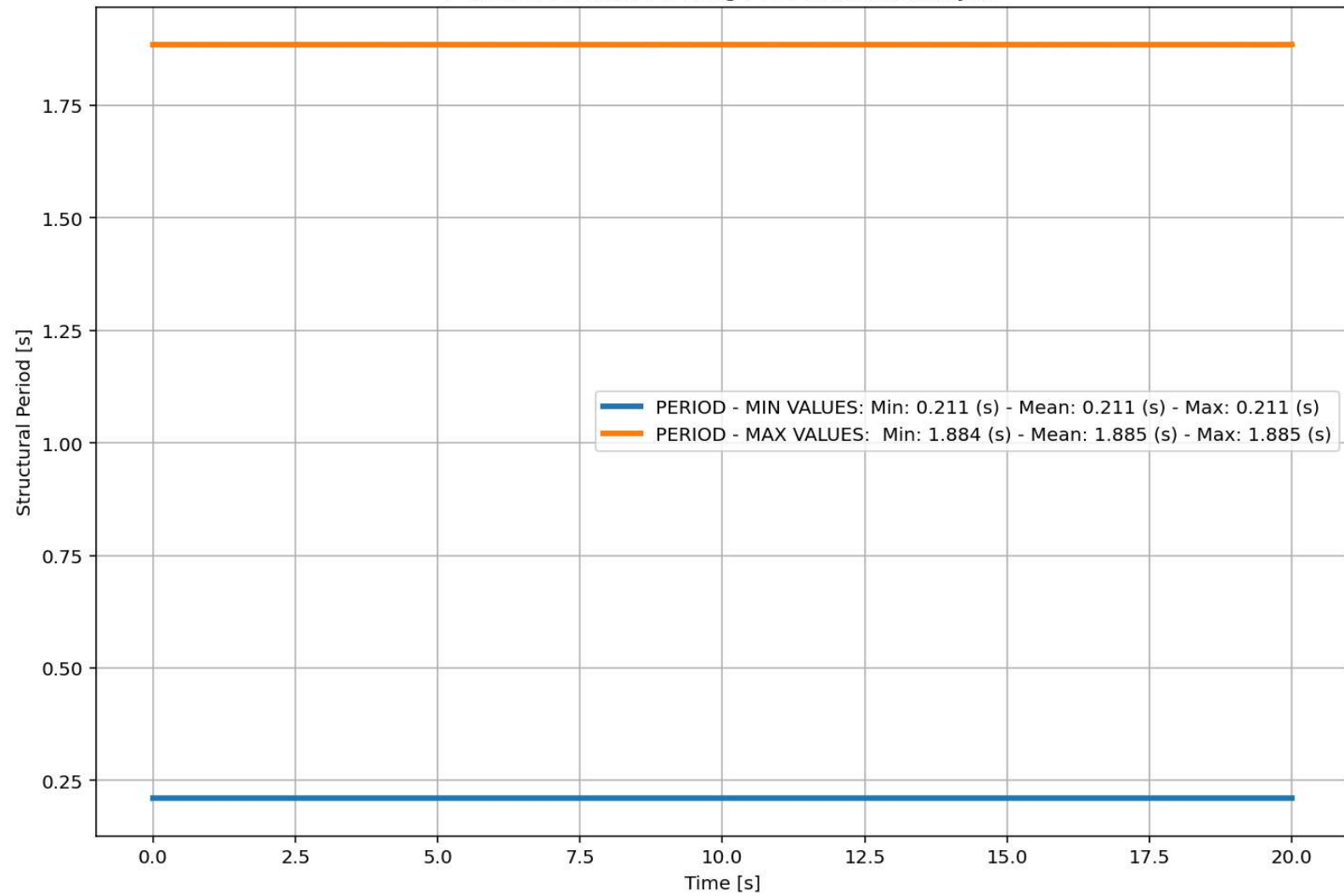




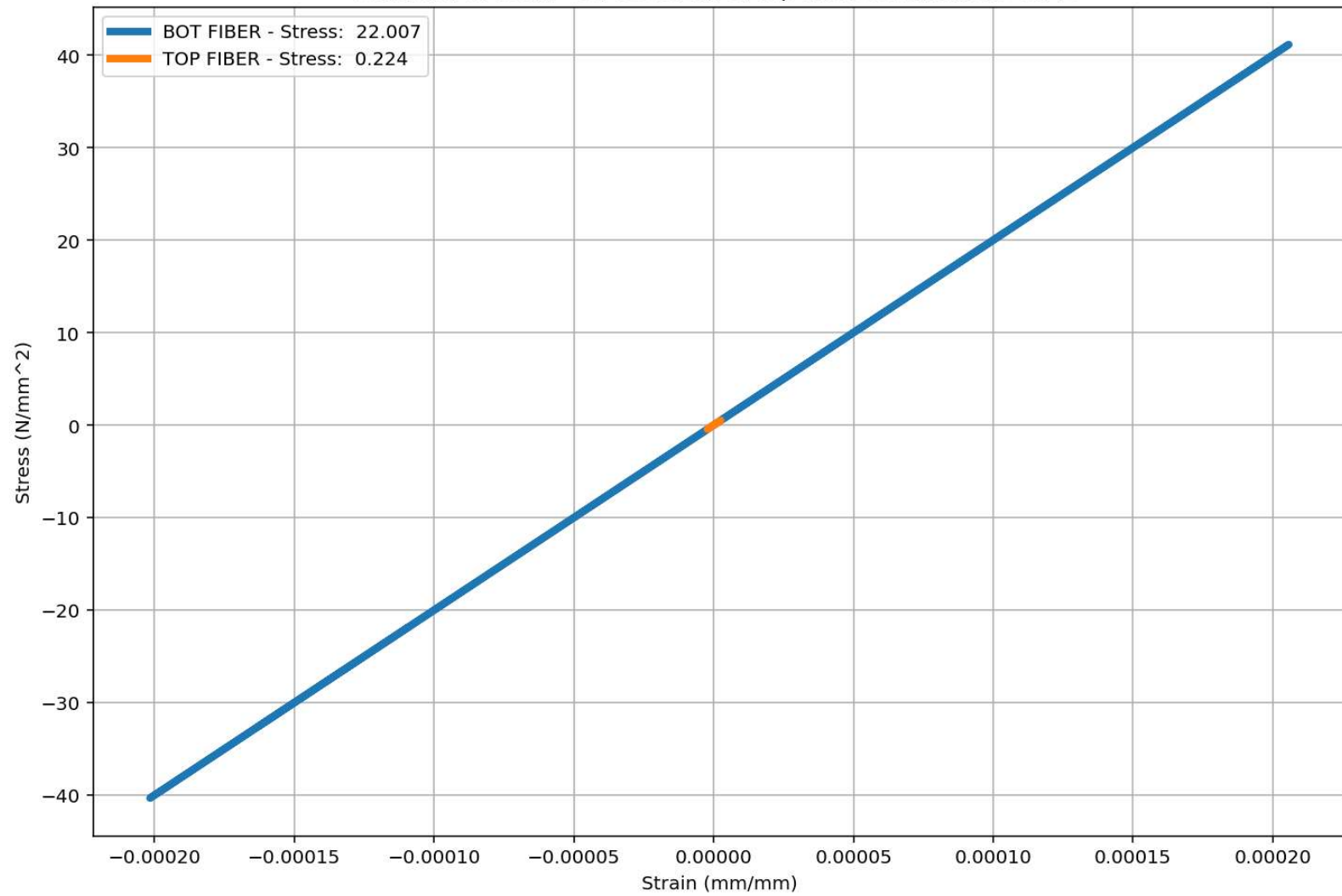
Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis



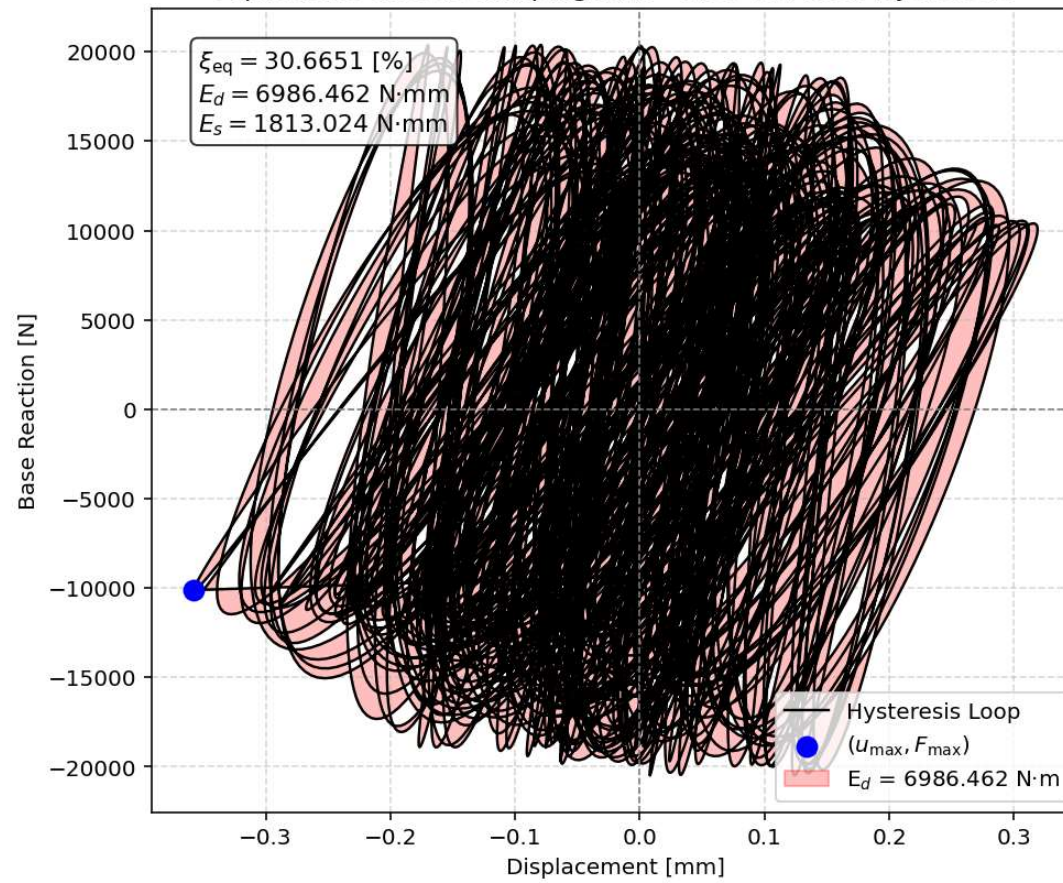
Period of Structure During Free-vibration Analysis

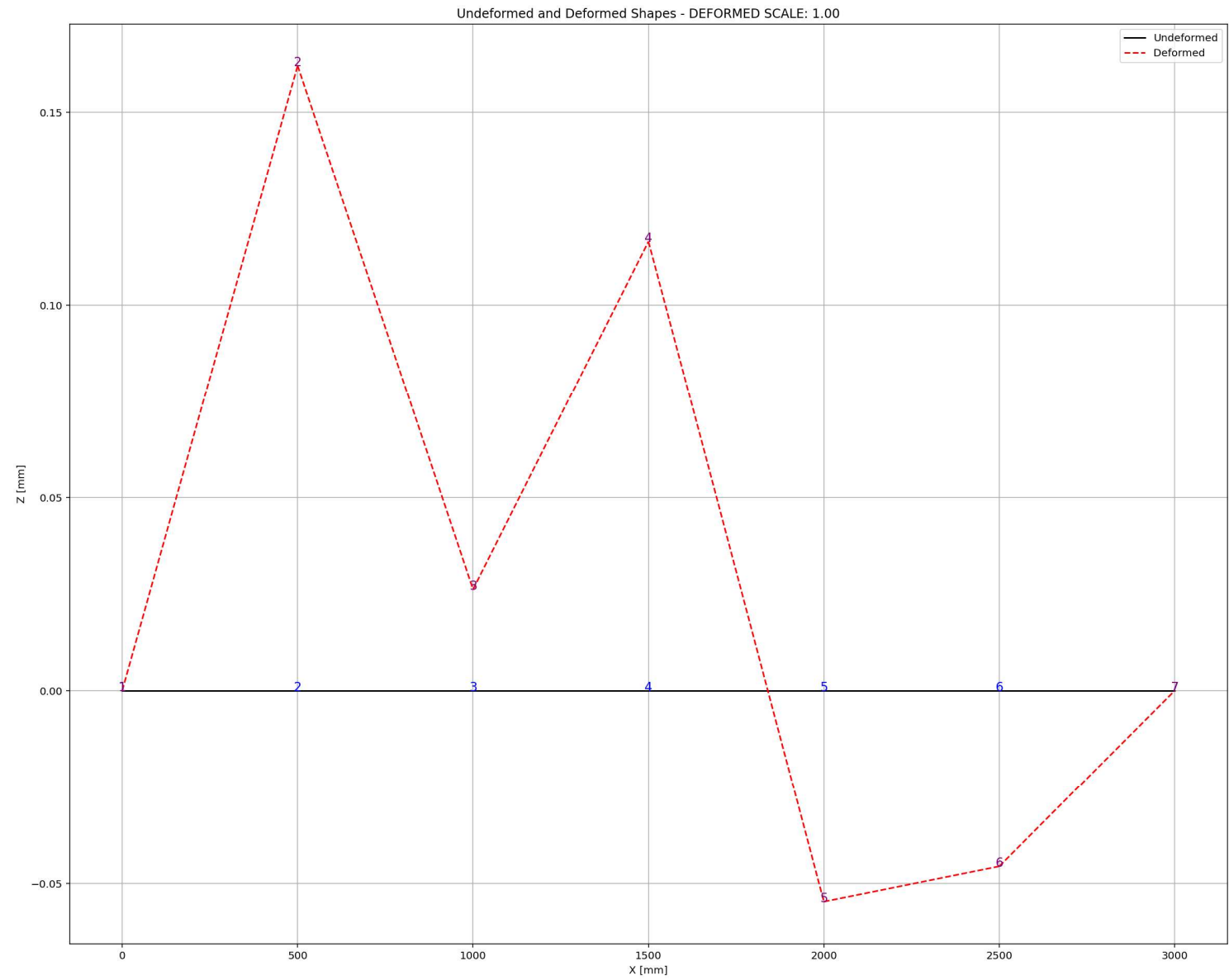


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve



Equivalent viscous damping ratio - Free-vibration Hysteresis





SEISMIC ANALYSIS

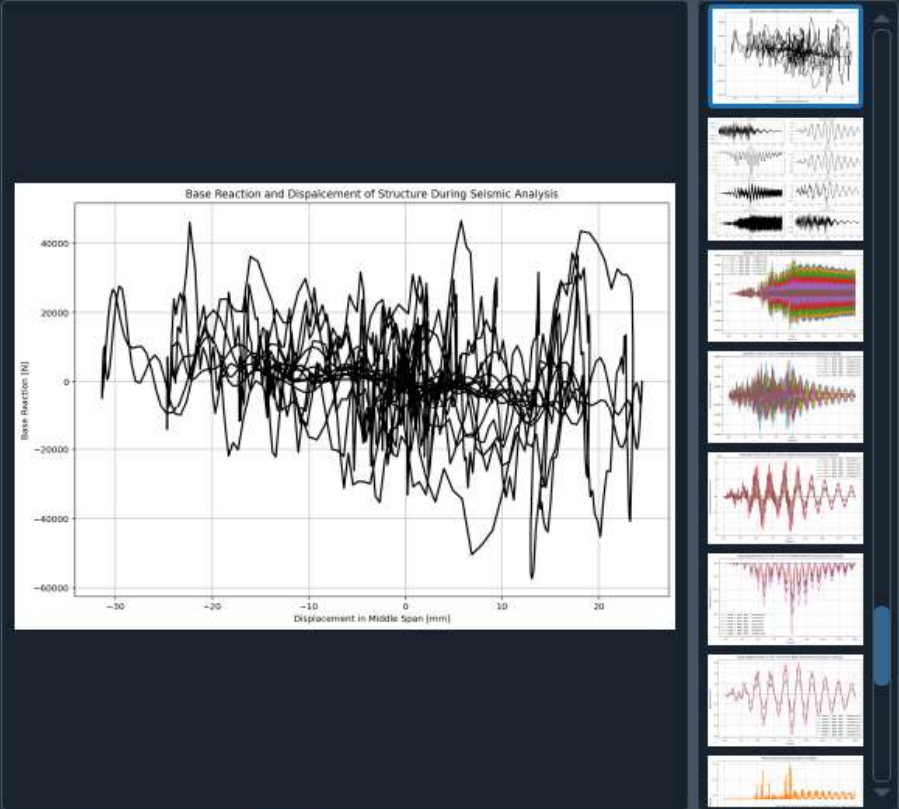
C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...PPORTED\P(t)_BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED.py

P(t)_BEAM_ONE_ELEM...IMPLY_SUPPORTED.py X

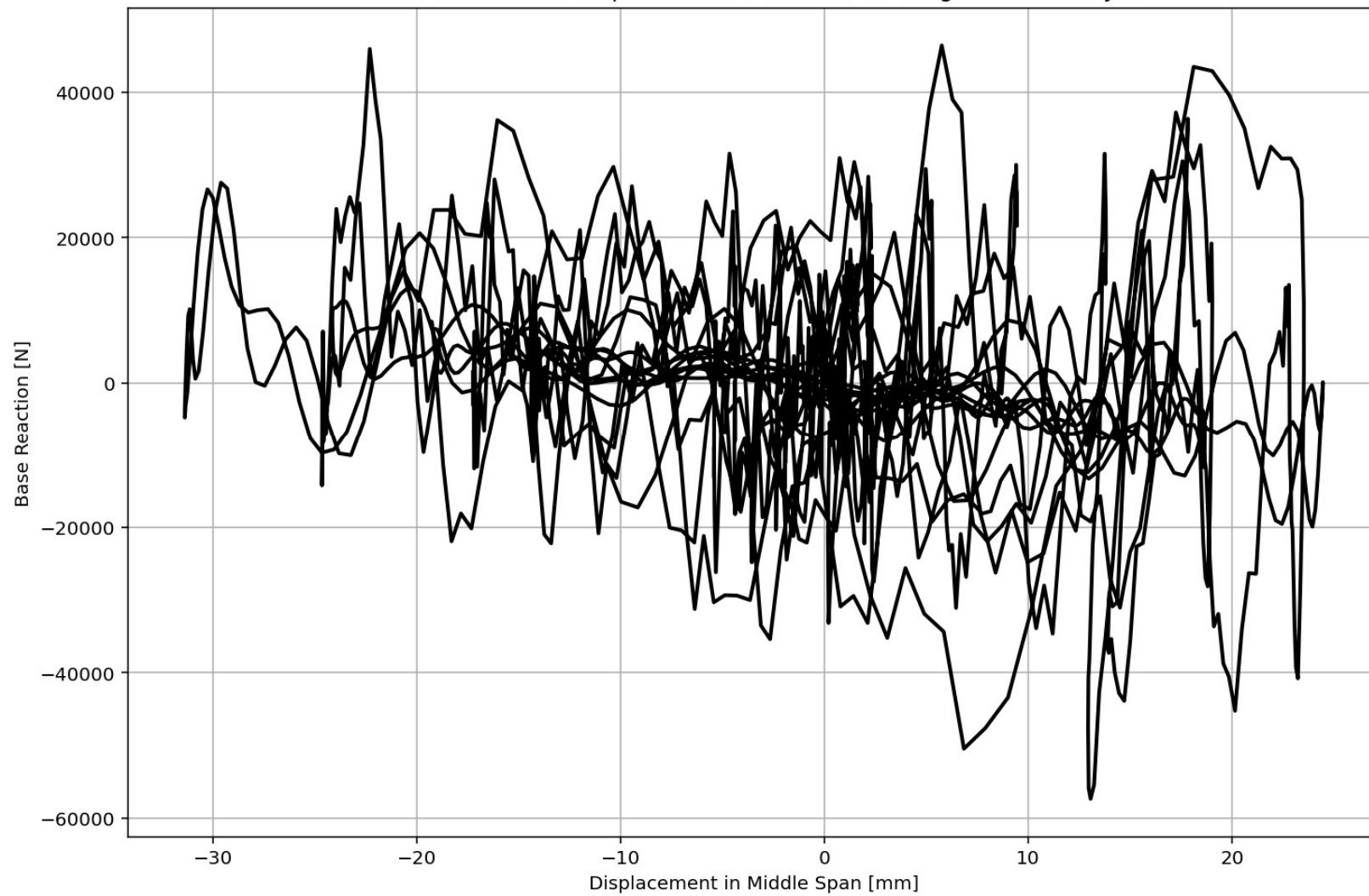
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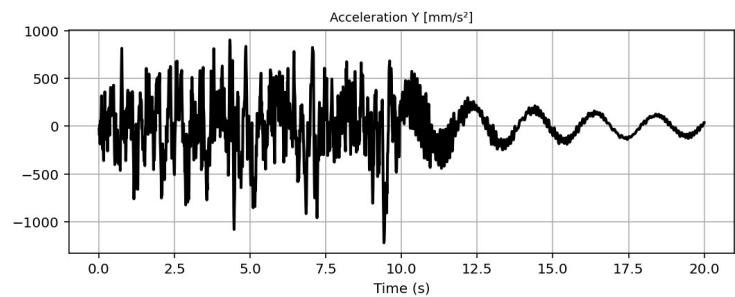
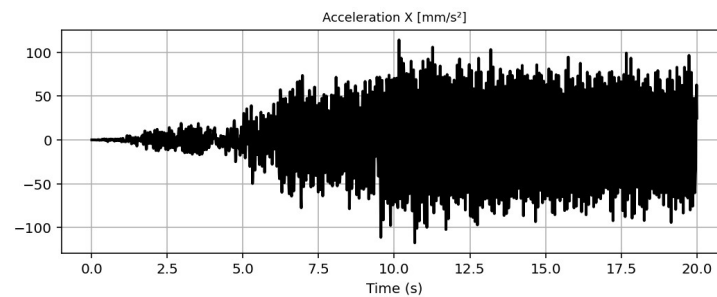
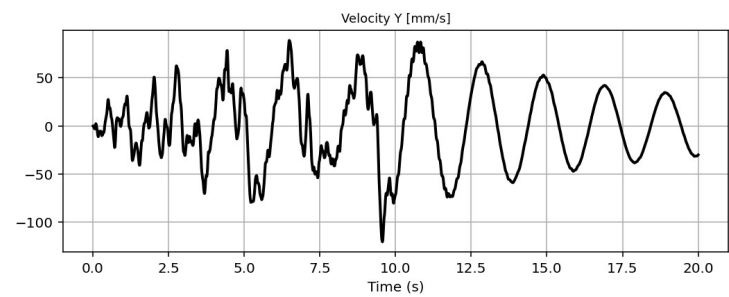
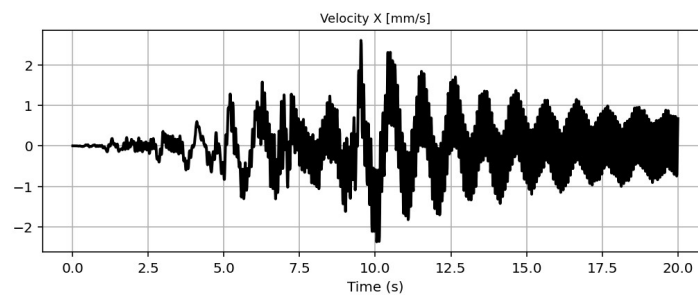
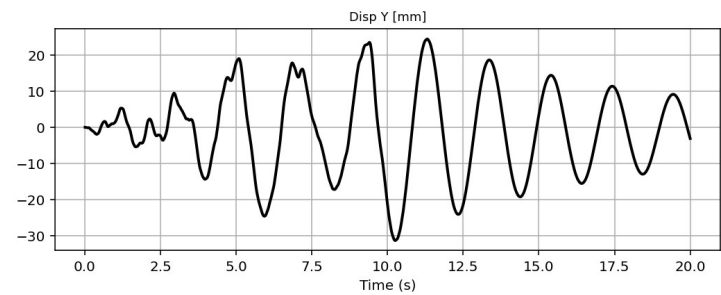
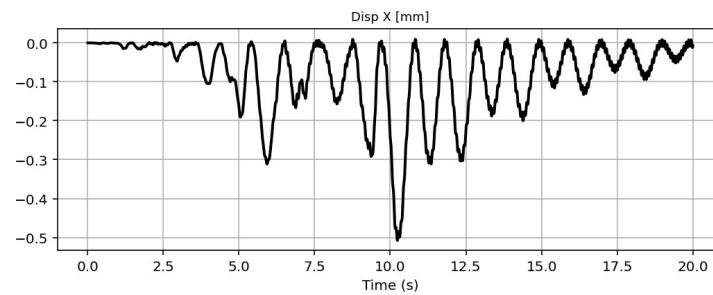
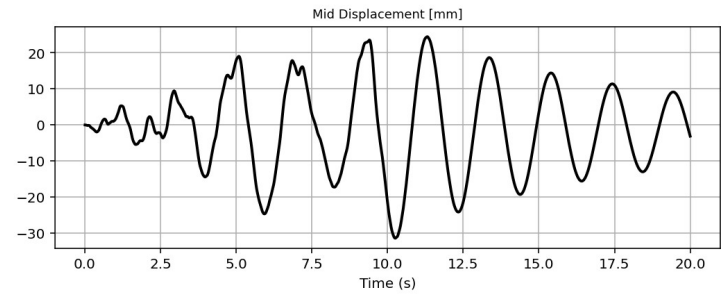
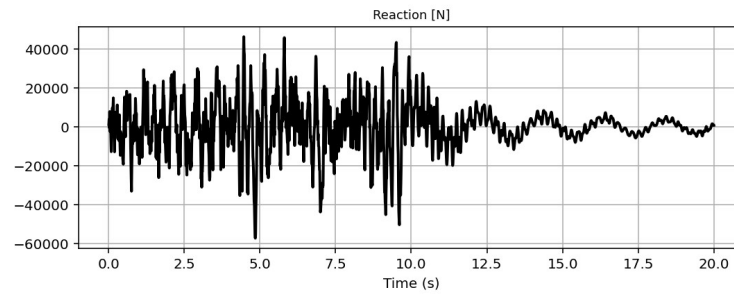
1408 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1409 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1410 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1411 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1412 ANAL_TYPE = 'SEISMIC'
1413
1414 DATA = BEAM_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1415
1416 (time_SEI, reaction_SEI, disp_mid_SEI,
1417  ele_axialforce_SEI, ele_shearforce_SEI, ele_momentforce_SEI,
1418  node_displacementsX_SEI, node_displacementsY_SEI,
1419  dispX_SEI, dispY_SEI,
1420  veloX_SEI, veloY_SEI,
1421  accX_SEI, accY_SEI,
1422  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1423  PERIOD_MIN_SEI, PERIOD_MAX_SEI,
1424  STRESSb_SEI, STRAINb_SEI, STRESSt_SEI, STRAINT_SEI, CURVATURE_SEI) = DATA
1425
1426
1427 XDATA = disp_mid_SEI
1428 YDATA = reaction_SEI
1429 XLABEL = 'Displacement in Middle Span [mm]'
1430 YLABEL = 'Base Reaction [N]'
1431 TITLE = 'Base Reaction and Displacement of Structure During Seismic Analysis'
1432 COLOR = 'black'
1433 SEMILOGY = False
1434 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1435
1436 PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
1437                   dispX_SEI, dispY_SEI,
1438                   veloX_SEI, veloY_SEI,
1439                   accX_SEI, accY_SEI)
1440
1441 # PLOT ELEMENTS AYTAN ENDE

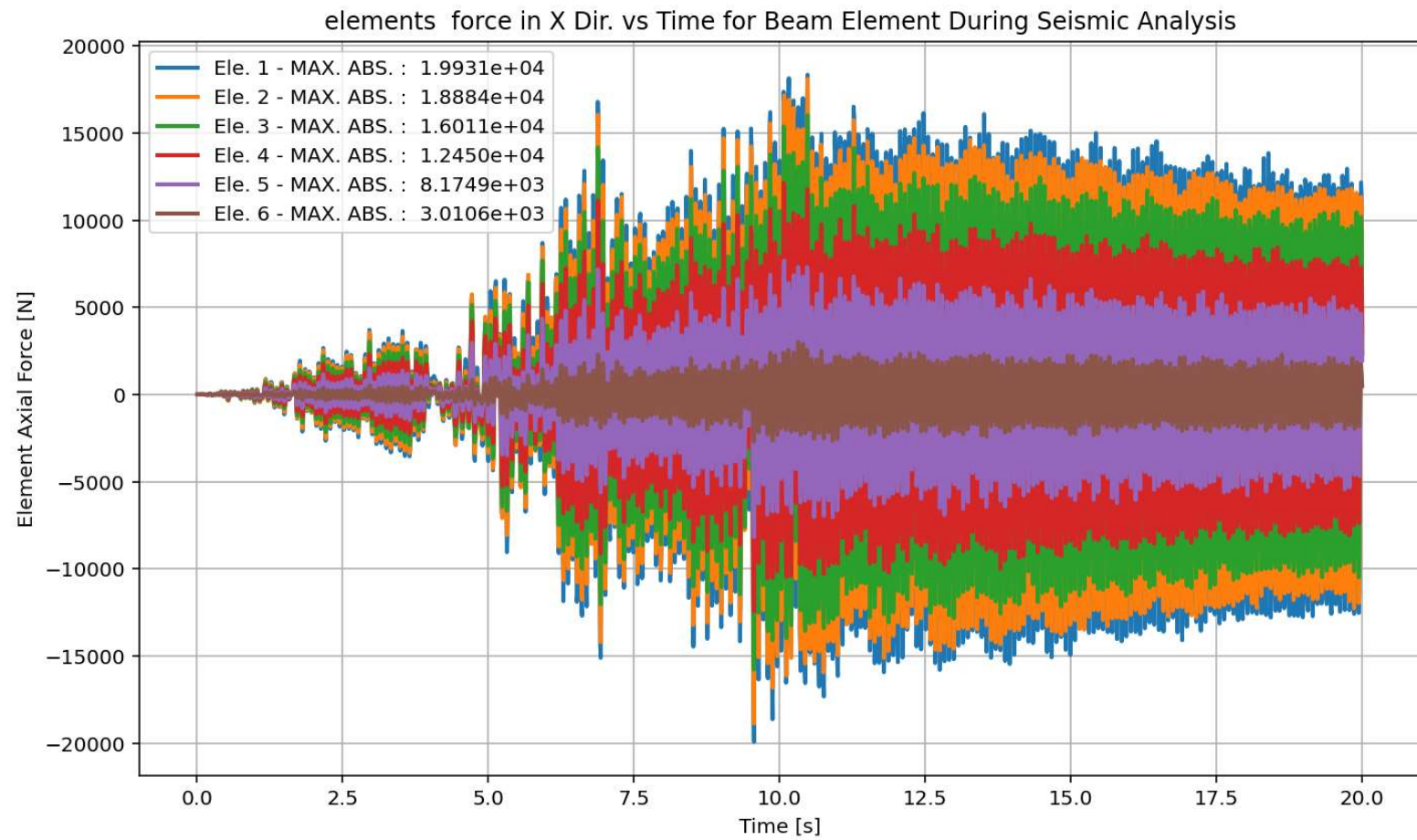
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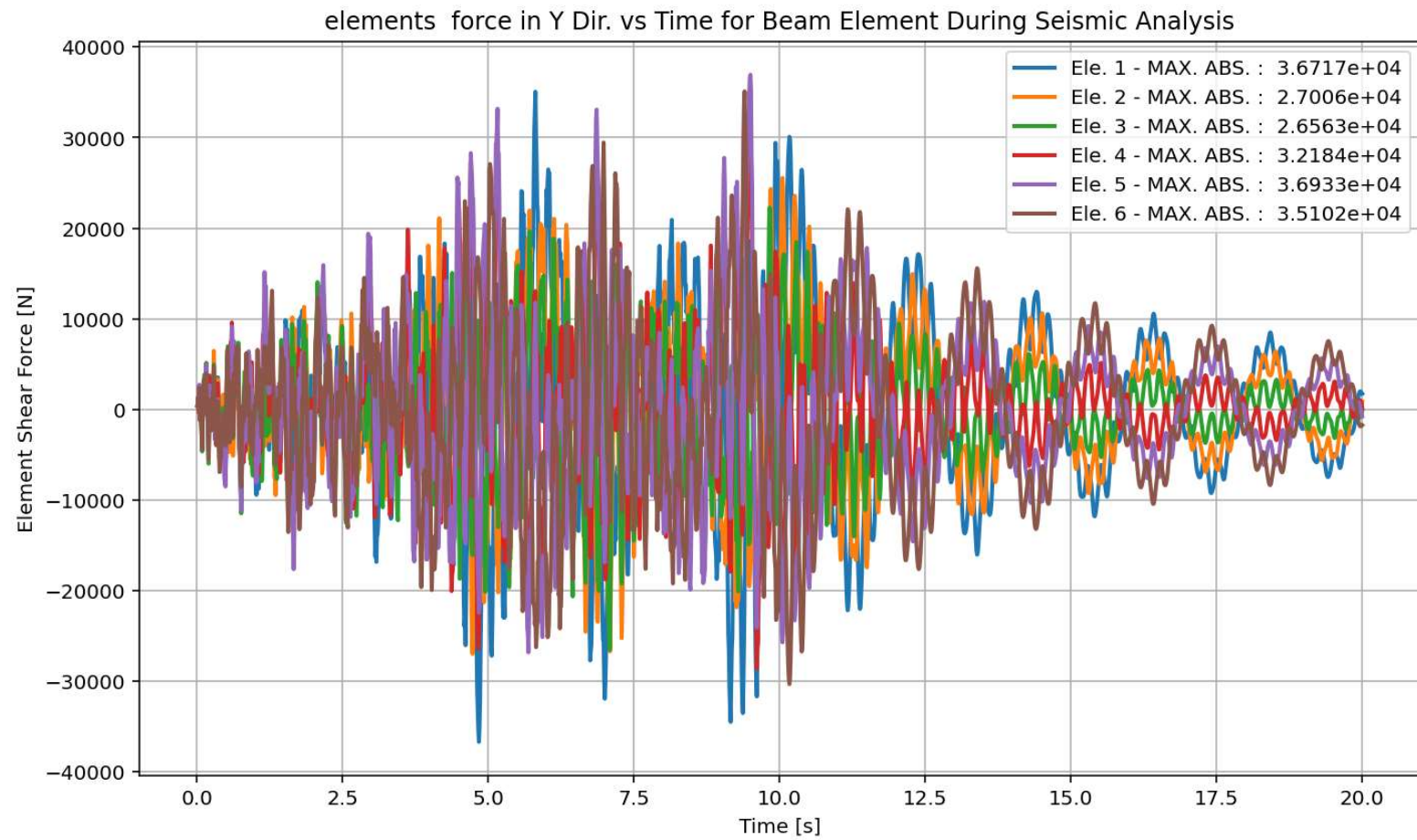


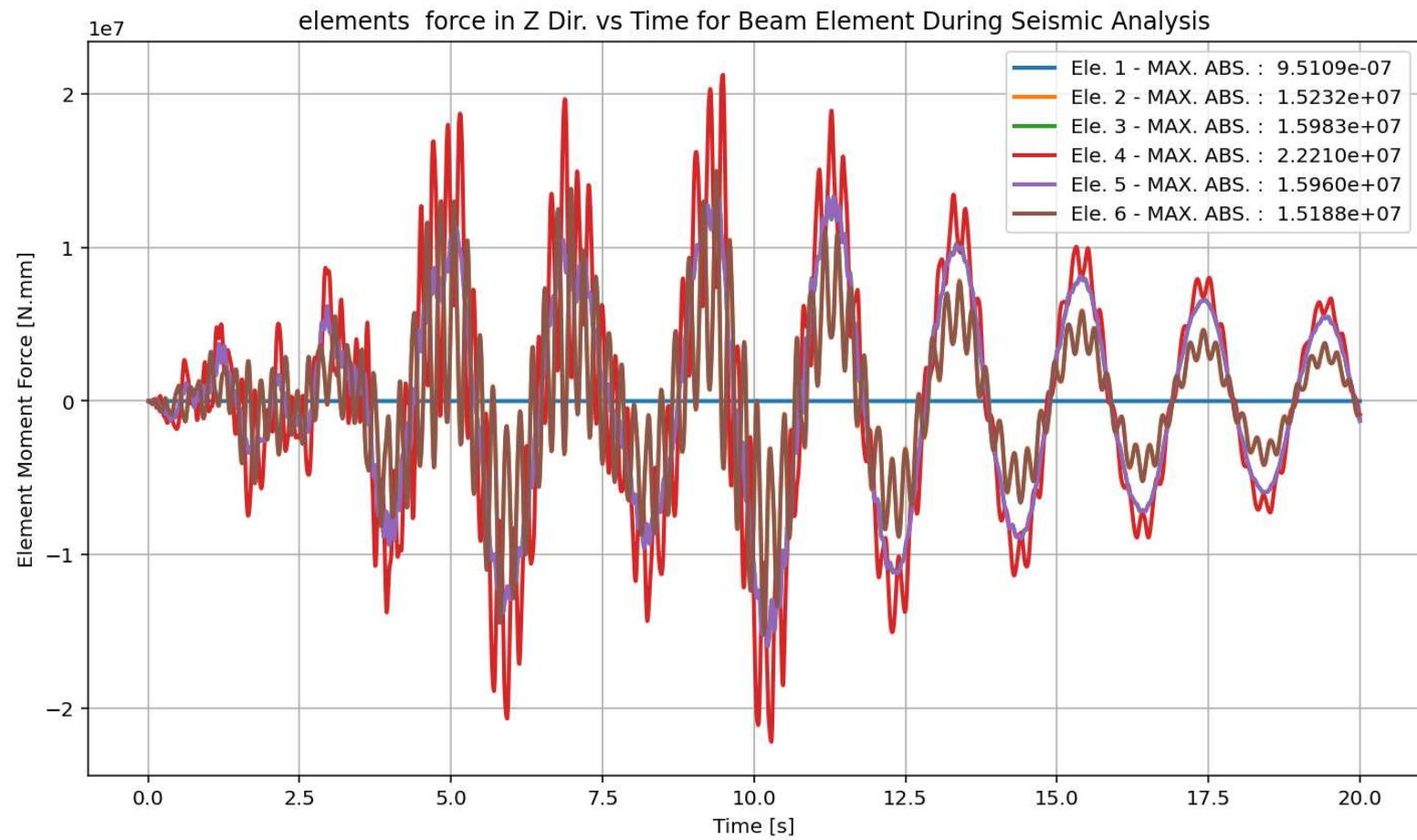
Base Reaction and Displacement of Structure During Seismic Analysis



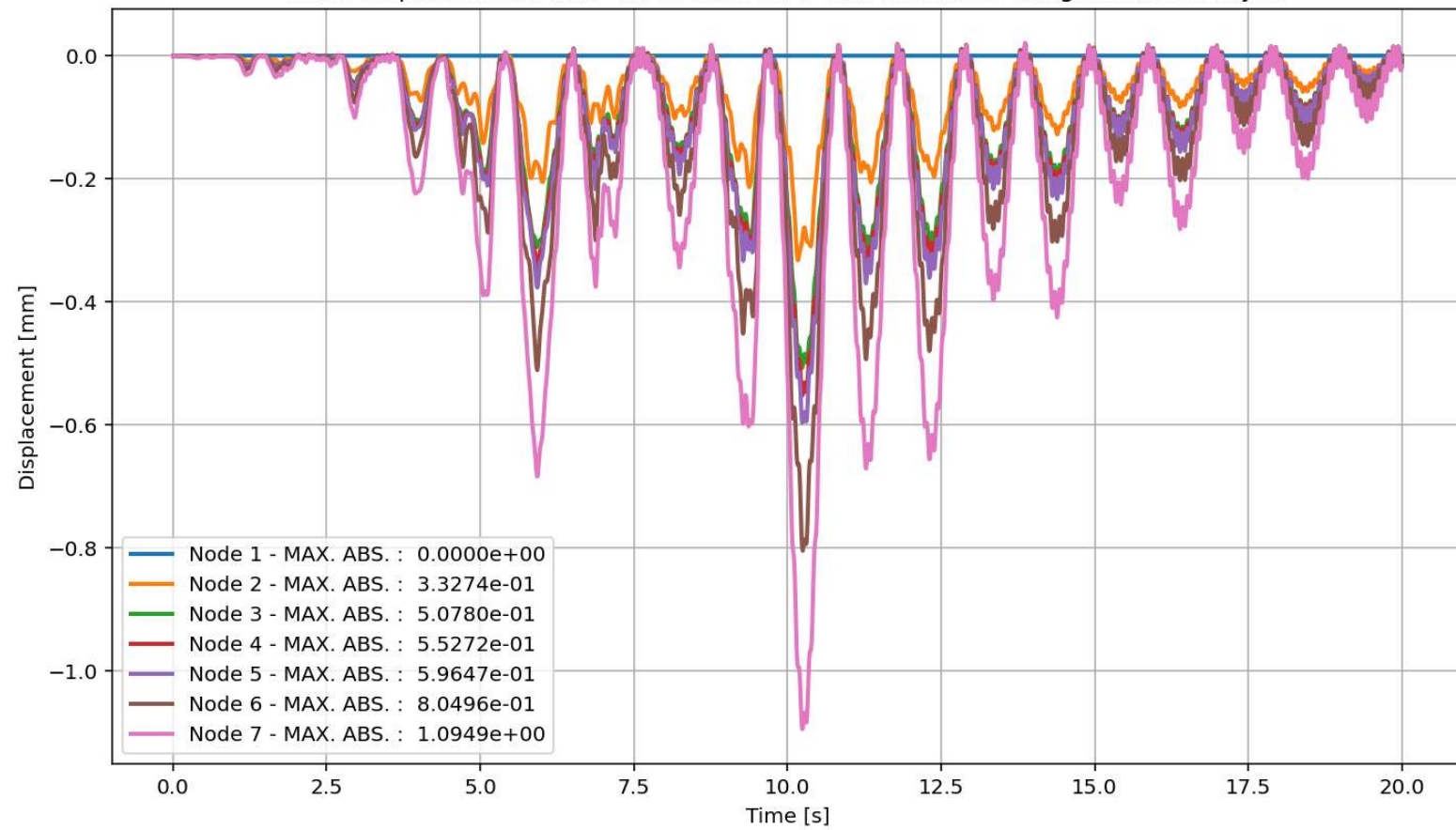




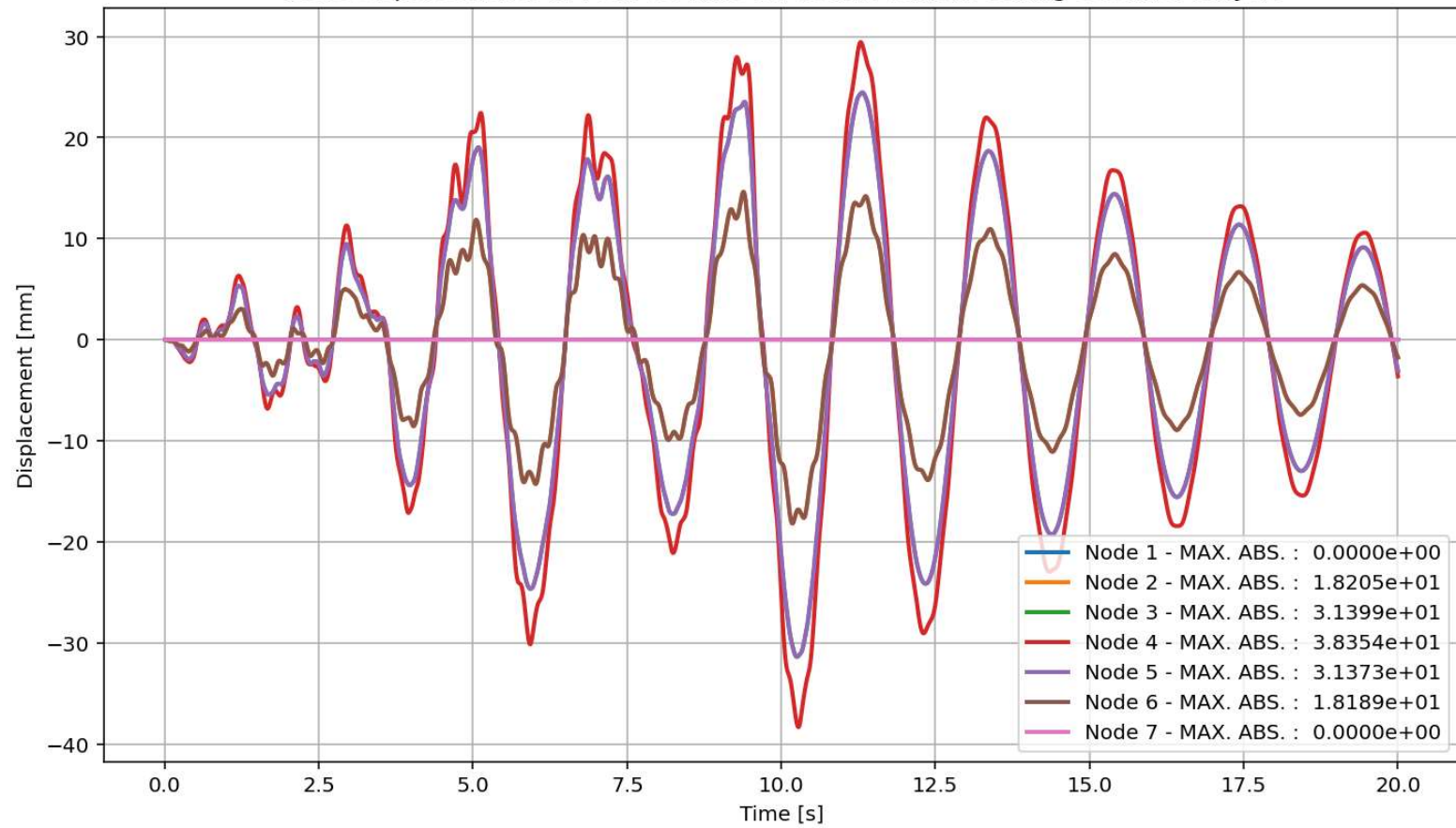




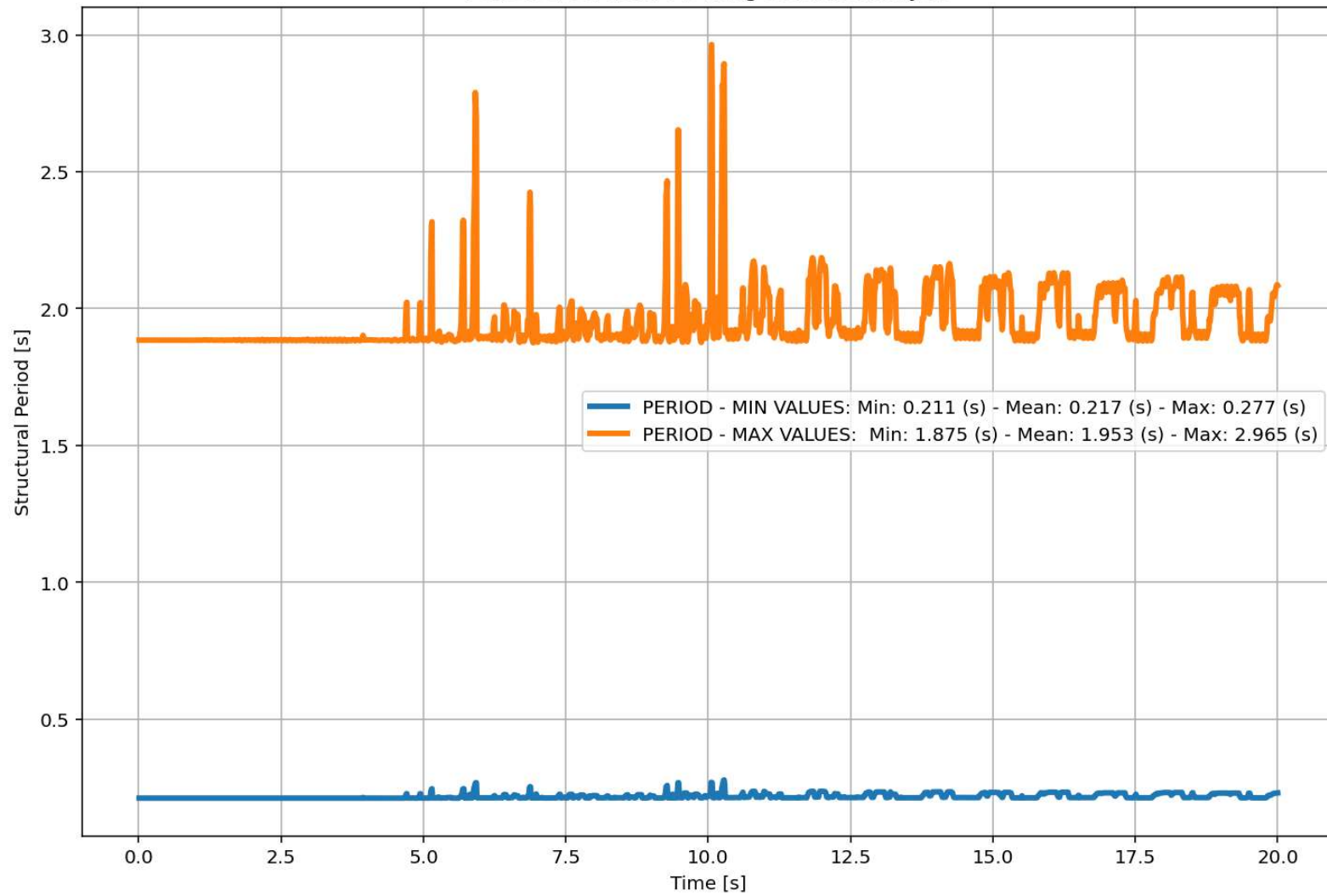
Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis



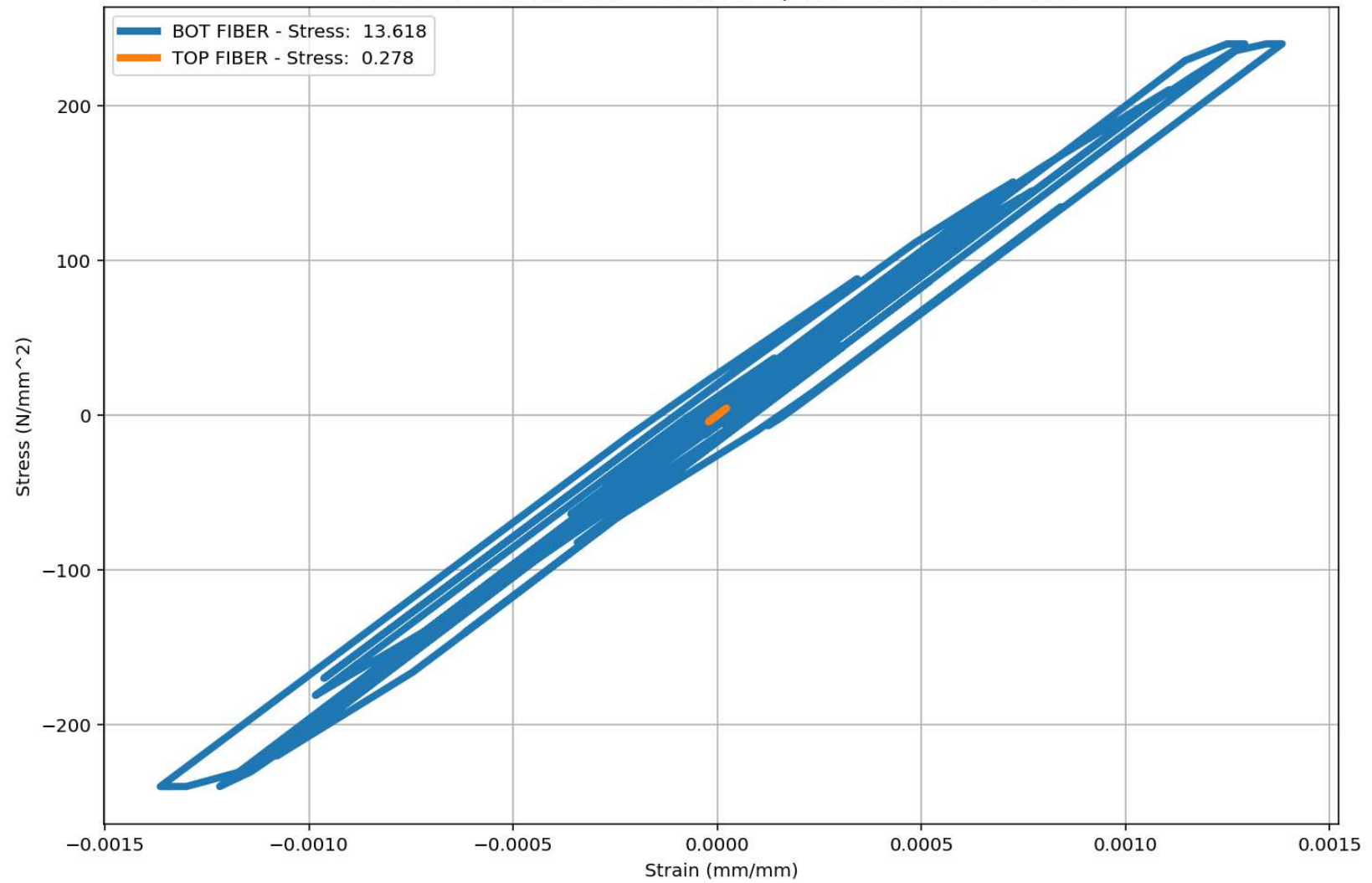
Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis



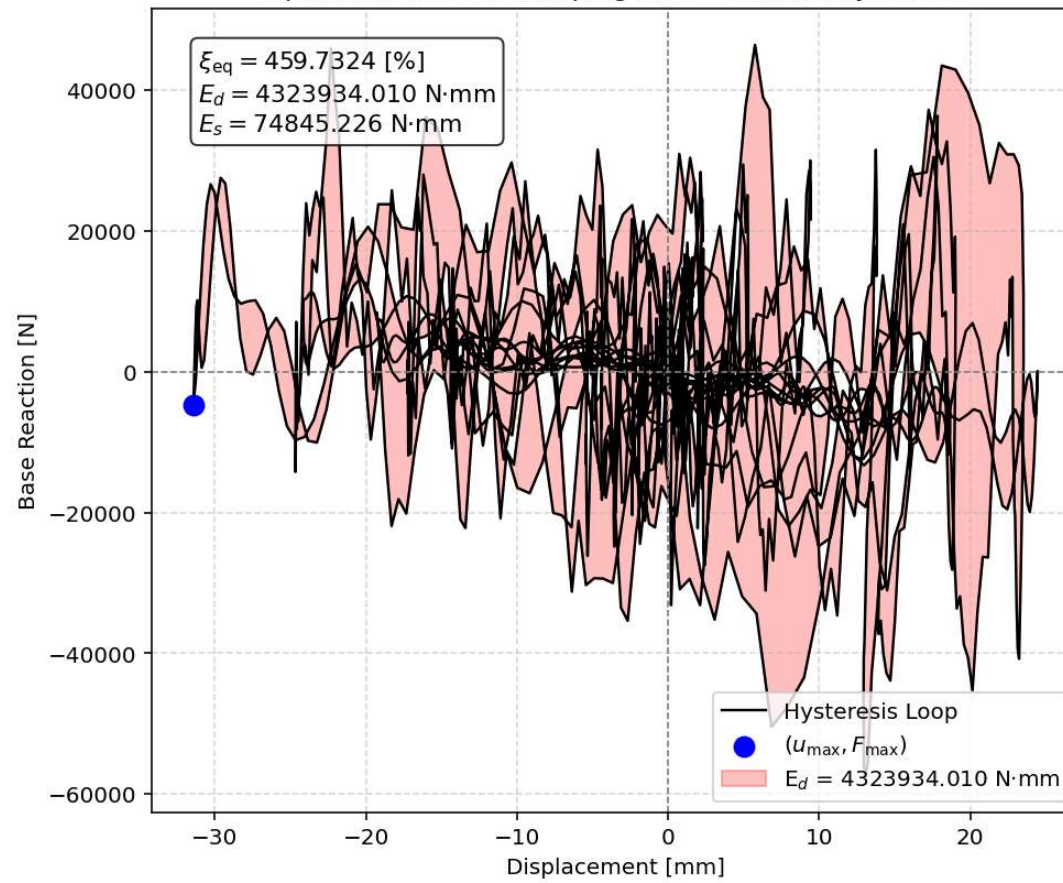
Period of Structure During Seismic Analysis

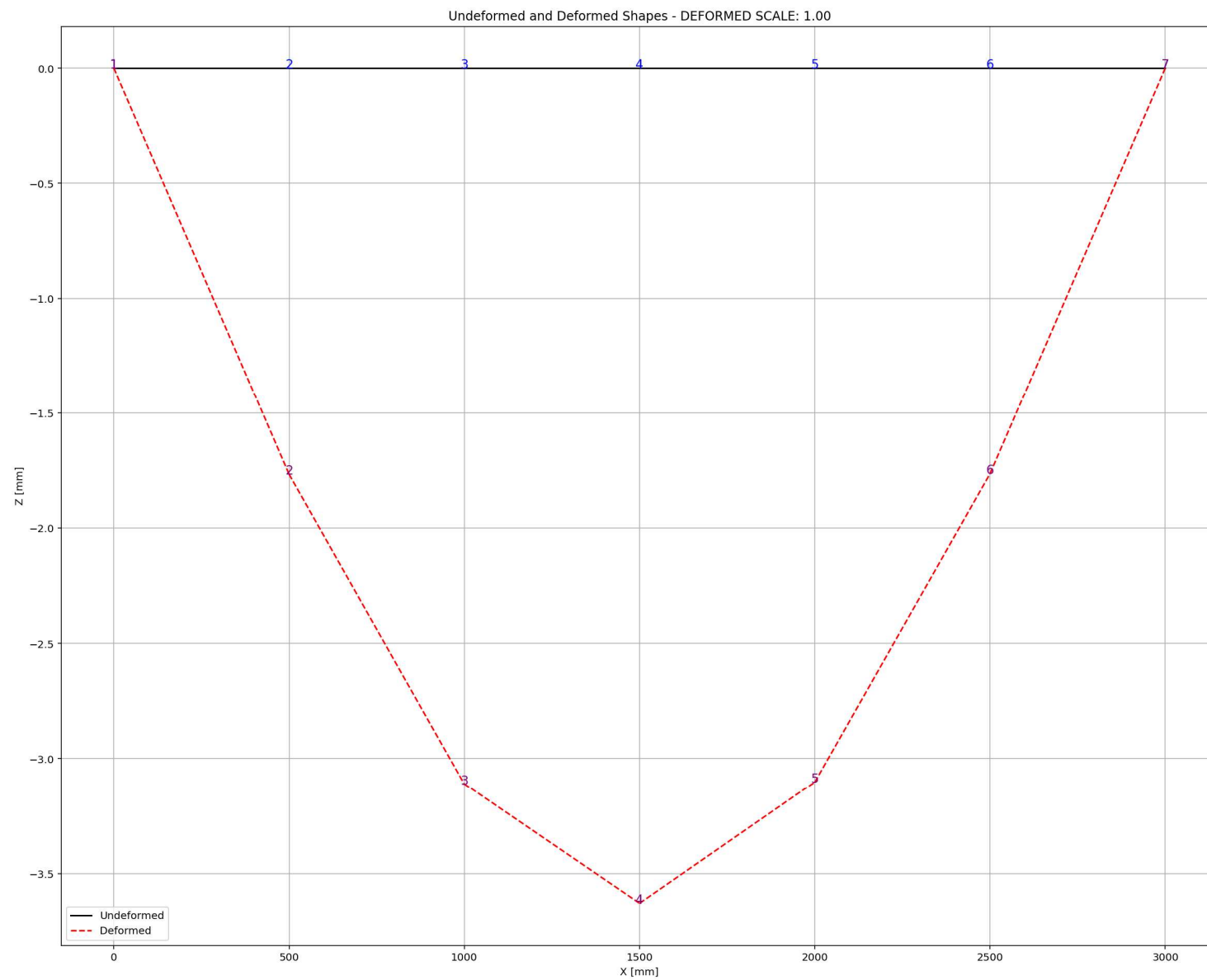


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve

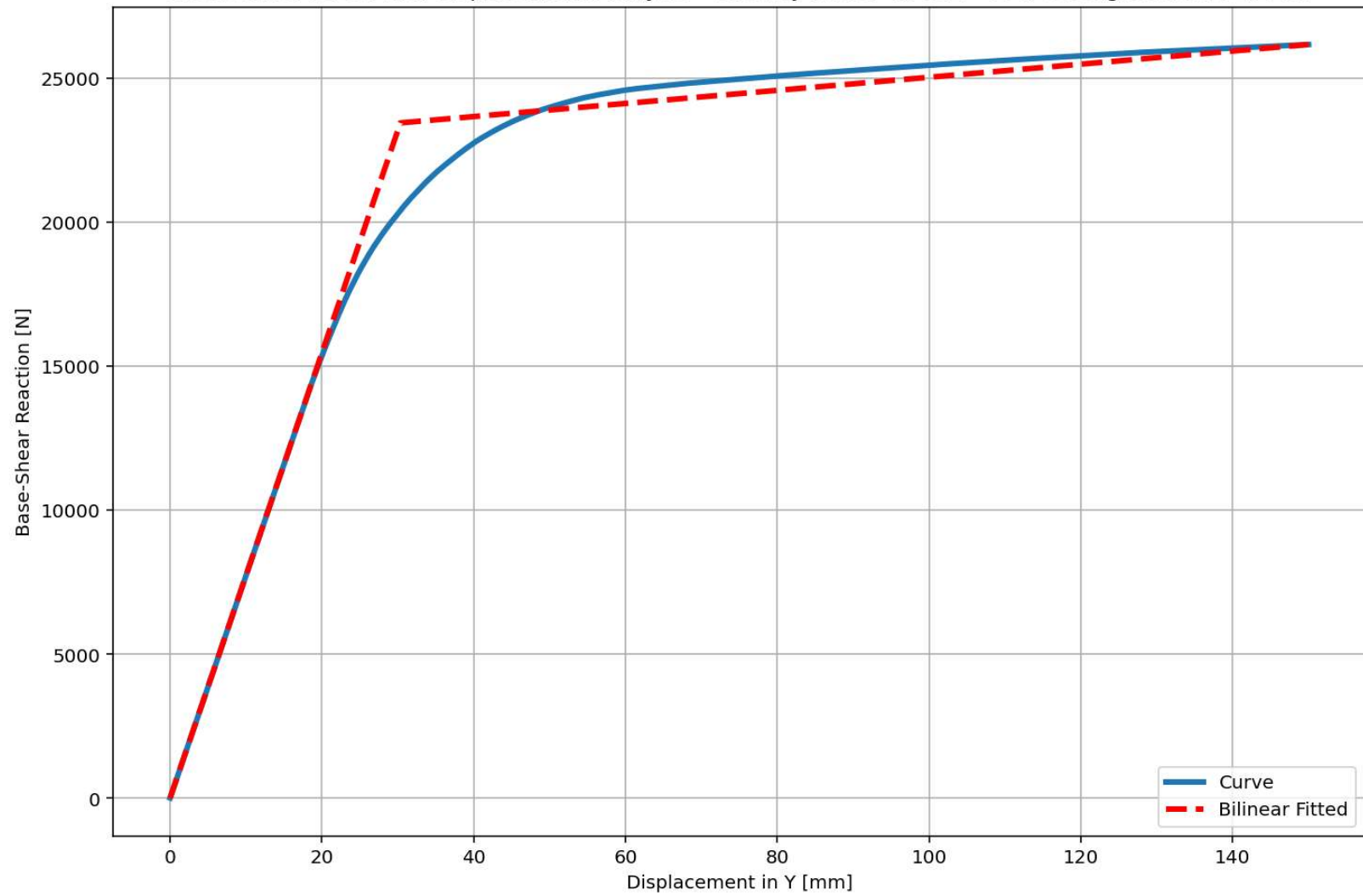


Equivalent viscous damping ratio - Seismic Hysteresis





Last Data of Base Axial-Displacement Analysis - Ductility Ratio: 4.9431 - Over Strength Factor: 1.1161



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0. 0.]

[30.34527209 23448.94886727]

[150. 26170.78767713]]

+=====+

+-----+

Structure Elastic Stiffness : 772.74

Structure Plastic Stiffness : 174.47

Structure Tangent Stiffness : 22.75

Structure Ductility Ratio : 4.94

Structure Over Strength Factor: 1.12

+-----+

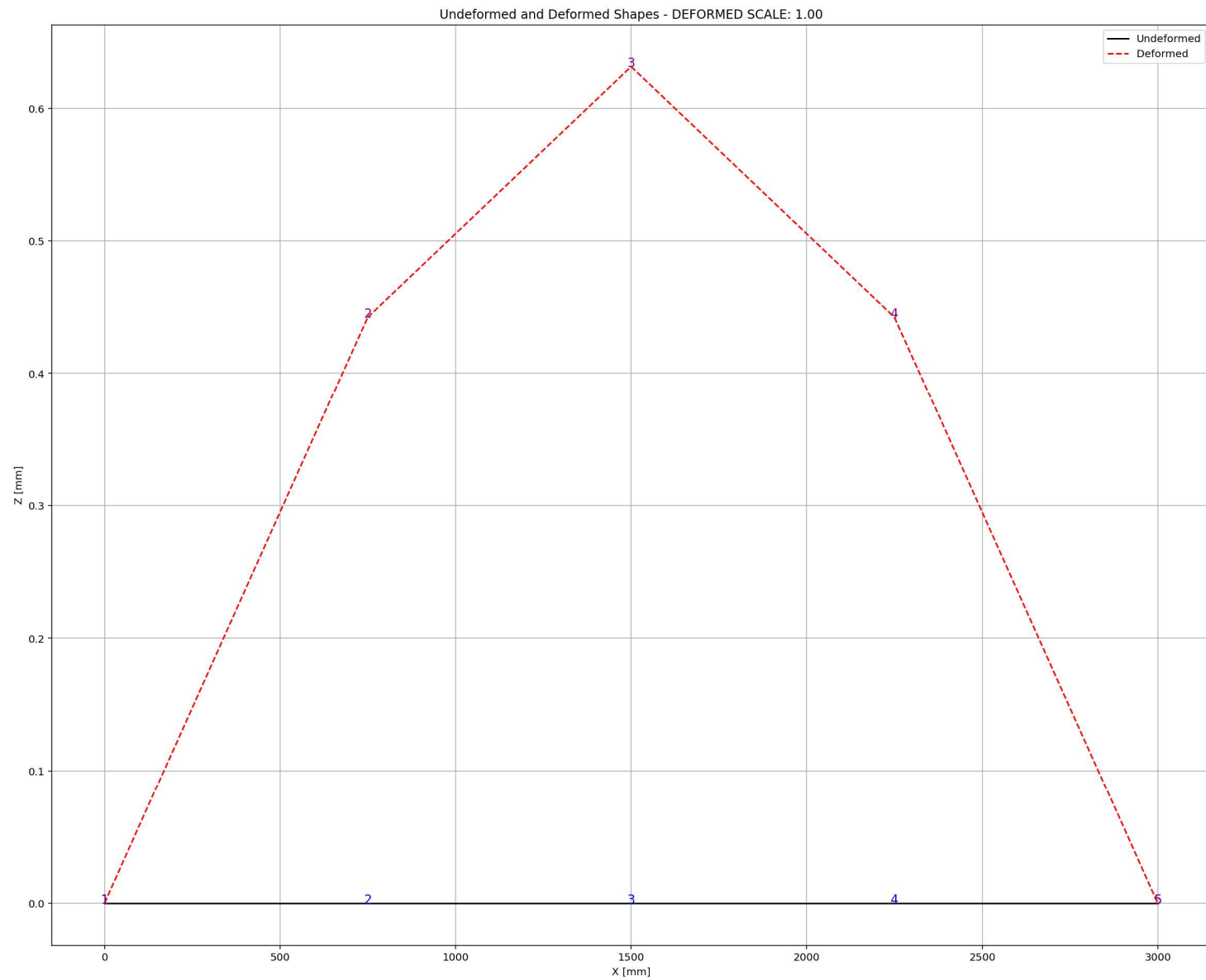
Over Strength Coefficient (Ω_0): 1.1161

Displacement Ductility Ratio (μ): 4.9431

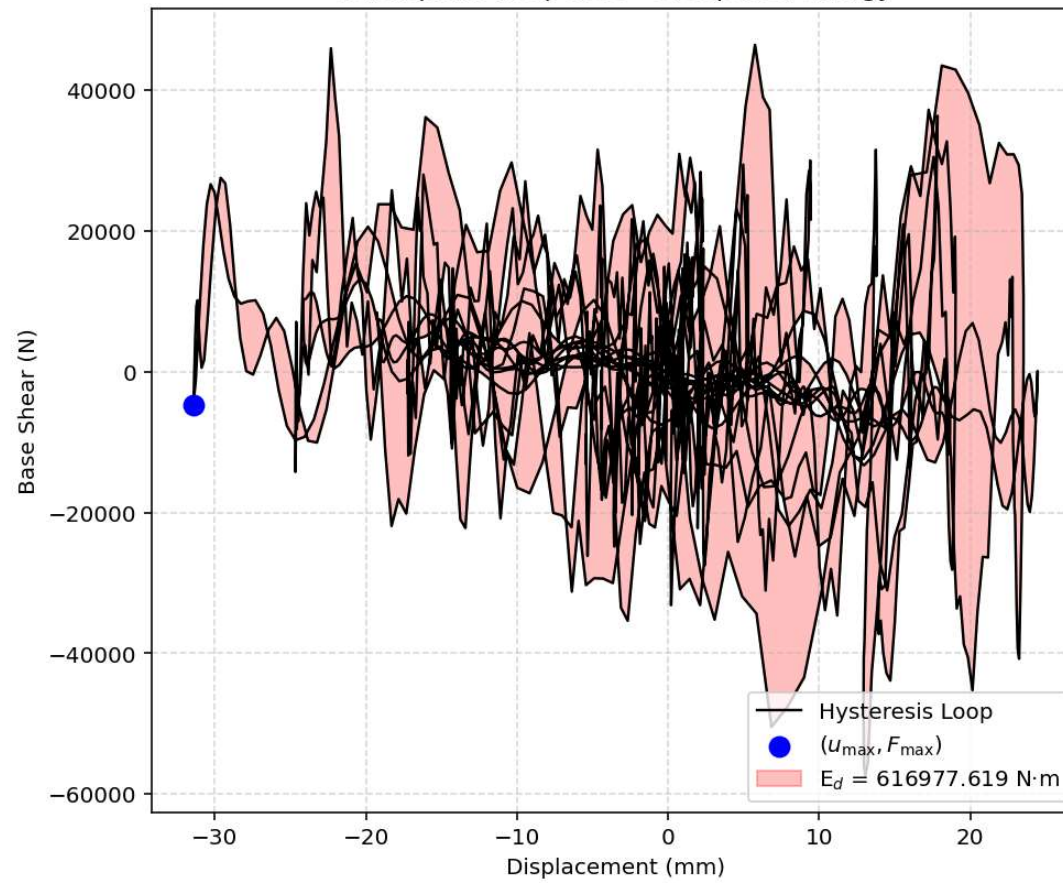
Ductility Coefficient (R_μ): 4.9431

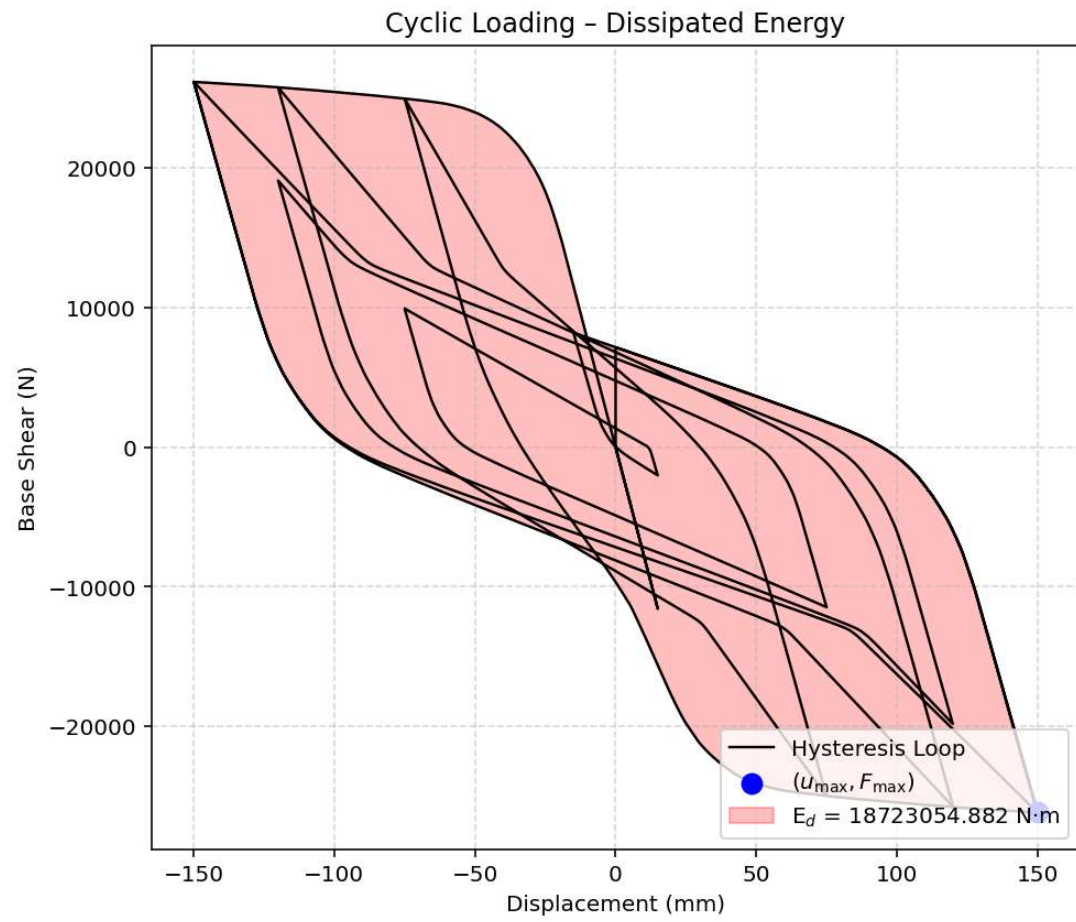
Structural Behavior Coefficient (R): 5.5169

Structural Ductility Damage Index in Y Direction: 0.8804 (%)



Earthquake Response - Dissipated Energy





Dissipated Energy from Earthquake= 616977.62 N·mm

Dissipated Energy from Cyclic Displacement= 18723054.88 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 3.295 [%]

ZONE 1: VERY MINOR DAMAGE

